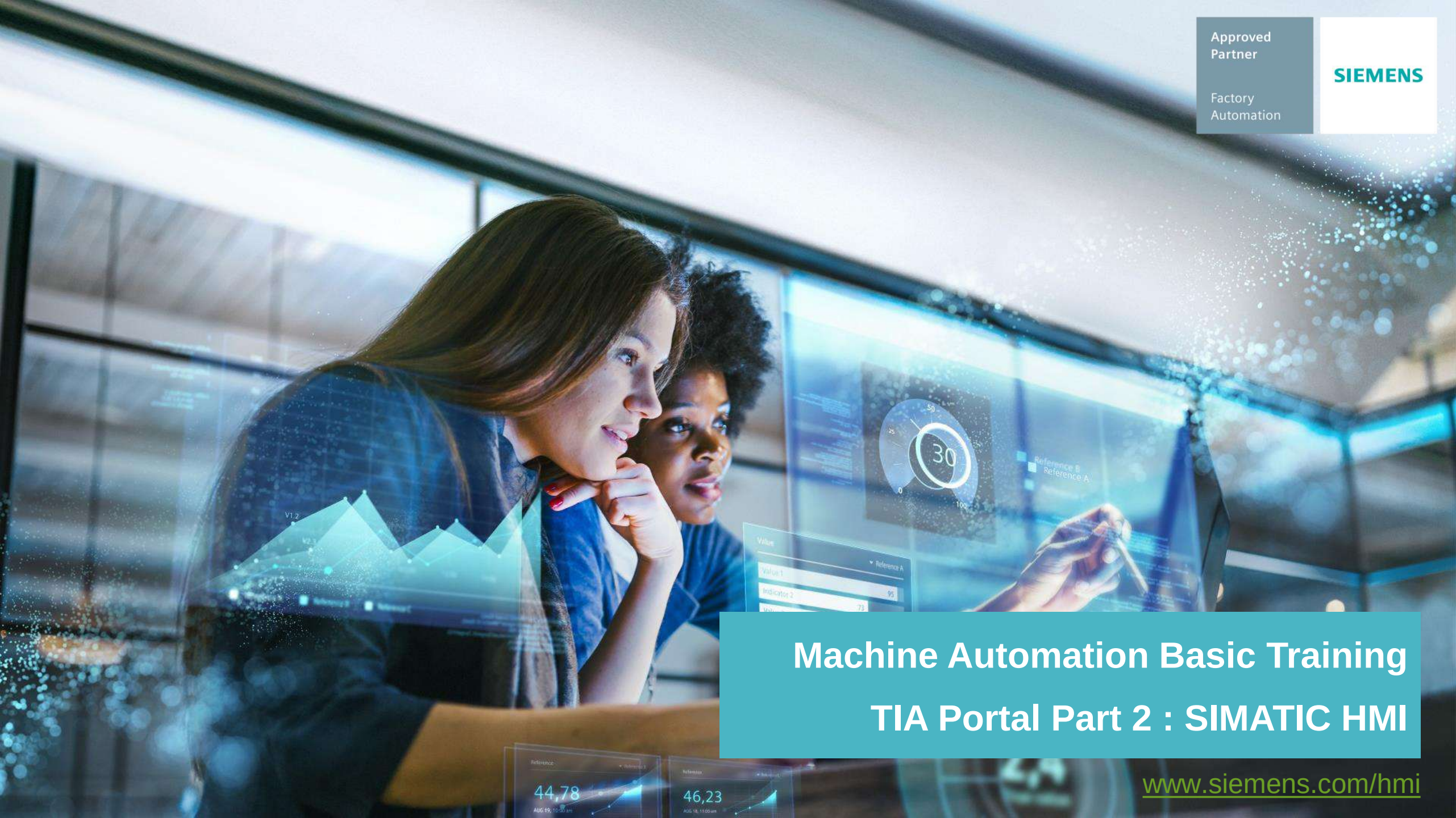


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The background image shows two women in a professional setting, looking at a large digital display. The display shows various industrial data visualizations, including a line graph with points labeled V1.2 and V2.3, a circular gauge with a needle pointing to 30, and a table with columns for Value, Reference A, and Reference B. The overall theme is industrial automation and data analysis.

Machine Automation Basic Training TIA Portal Part 2 : SIMATIC HMI

www.siemens.com/hmi

TIA Portal for Machinery Automation Solutions

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- **Part 1 : PLC - Step 7 Basic**
- **Part 2 : HMI - WinCC Basic**
- **Part 3 : Integrated Motion Control**
 - Speed control & Position control

SIMATIC S7-1200



SIMATIC HMI Panel



SINAMICS V20/V90



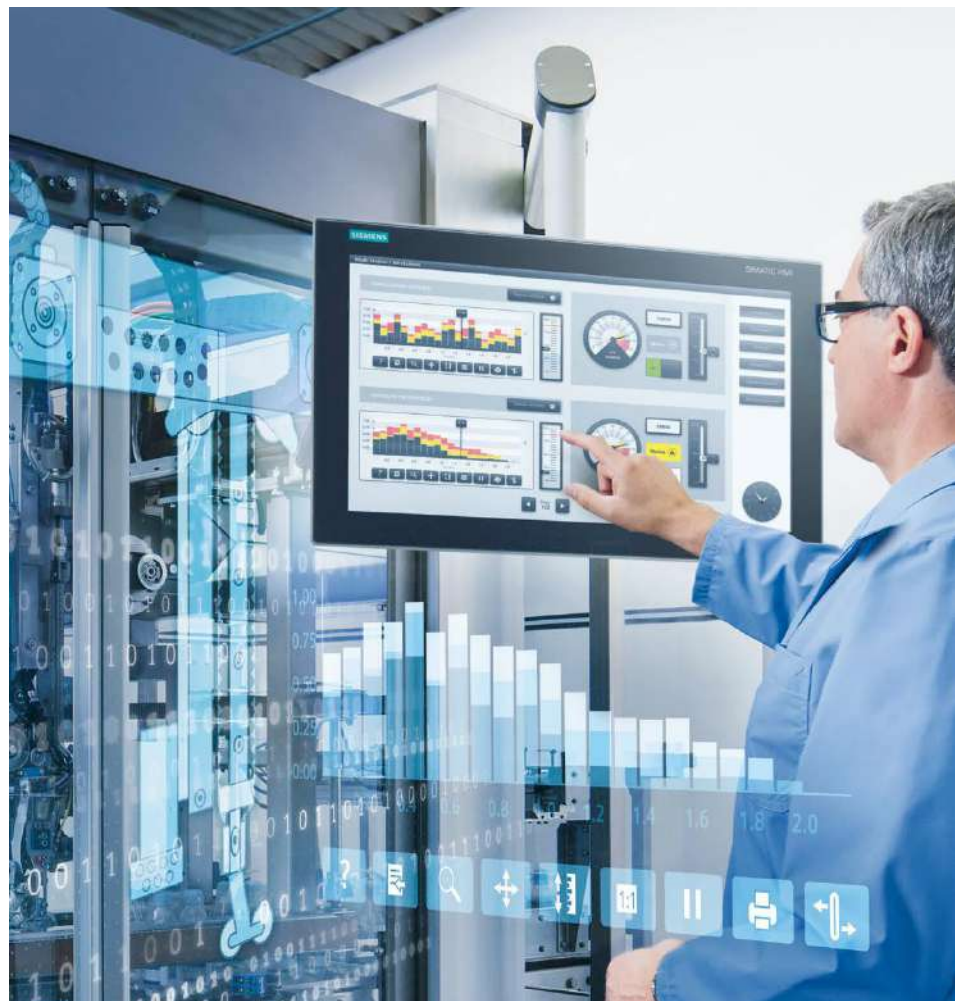
Basic training : TIA Portal WinCC Basic

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- An overview
- SIMATIC HMI
- Basic programming
 - HMI Setup 36
 - Graphic User Interface 55
 - Alarm Feature 118
 - Web browser 132
 - Others 145
- Technical note



Legal information (1/2)

Warning notice system

This manual contains notices you have to observe in order to ensure your personal safety, as well as to prevent damage to property. The notices referring to your personal safety are highlighted in the manual by a safety alert symbol, notices referring only to property damage have no safety alert symbol. These notices shown below are graded according to the degree of danger.

DANGER

indicates that death or severe personal injury **will** result if proper precautions are not taken.

WARNING

indicates that death or severe personal injury **may** result if proper precautions are not taken.

CAUTION

indicates that minor personal injury can result if proper precautions are not taken.

NOTICE

indicates that property damage can result if proper precautions are not taken.

If more than one degree of danger is present, the warning notice representing the highest degree of danger will be used. A notice warning of injury to persons with a safety alert symbol may also include a warning relating to property damage.

Legal information (2/2)

Qualified Personnel

The product/system described in this documentation may be operated only by **personnel qualified** for the specific task in accordance with the relevant documentation, in particular its warning notices and safety instructions. Qualified personnel are those who, based on their training and experience, are capable of identifying risks and avoiding potential hazards when working with these products/systems.

Proper use of Siemens products

Note the following:

WARNING

Siemens products may only be used for the applications described in the catalog and in the relevant technical documentation. If products and components from other manufacturers are used, these must be recommended or approved by Siemens. Proper transport, storage, installation, assembly, commissioning, operation and maintenance are required to ensure that the products operate safely and without any problems. The permissible ambient conditions must be complied with. The information in the relevant documentation must be observed.

Trademarks

All names identified by ® are registered trademarks of Siemens AG. The remaining trademarks in this publication may be trademarks whose use by third parties for their own purposes could violate the rights of the owner.

Disclaimer of Liability

We have reviewed the contents of this publication to ensure consistency with the hardware and software described. Since variance cannot be precluded entirely, we cannot guarantee full consistency. However, the information in this publication is reviewed regularly and any necessary corrections are included in subsequent editions.

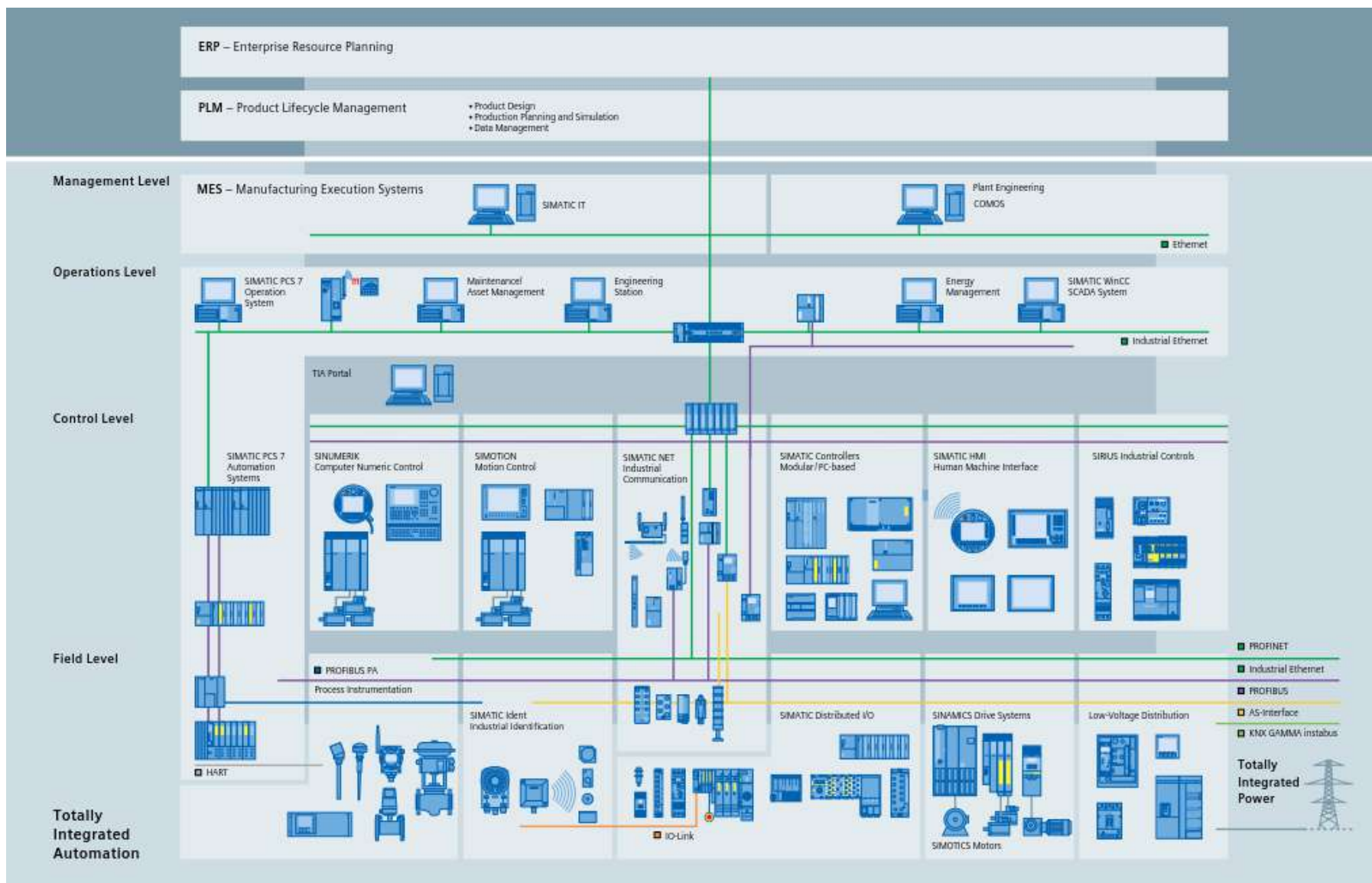


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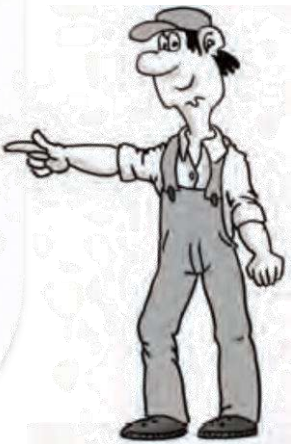


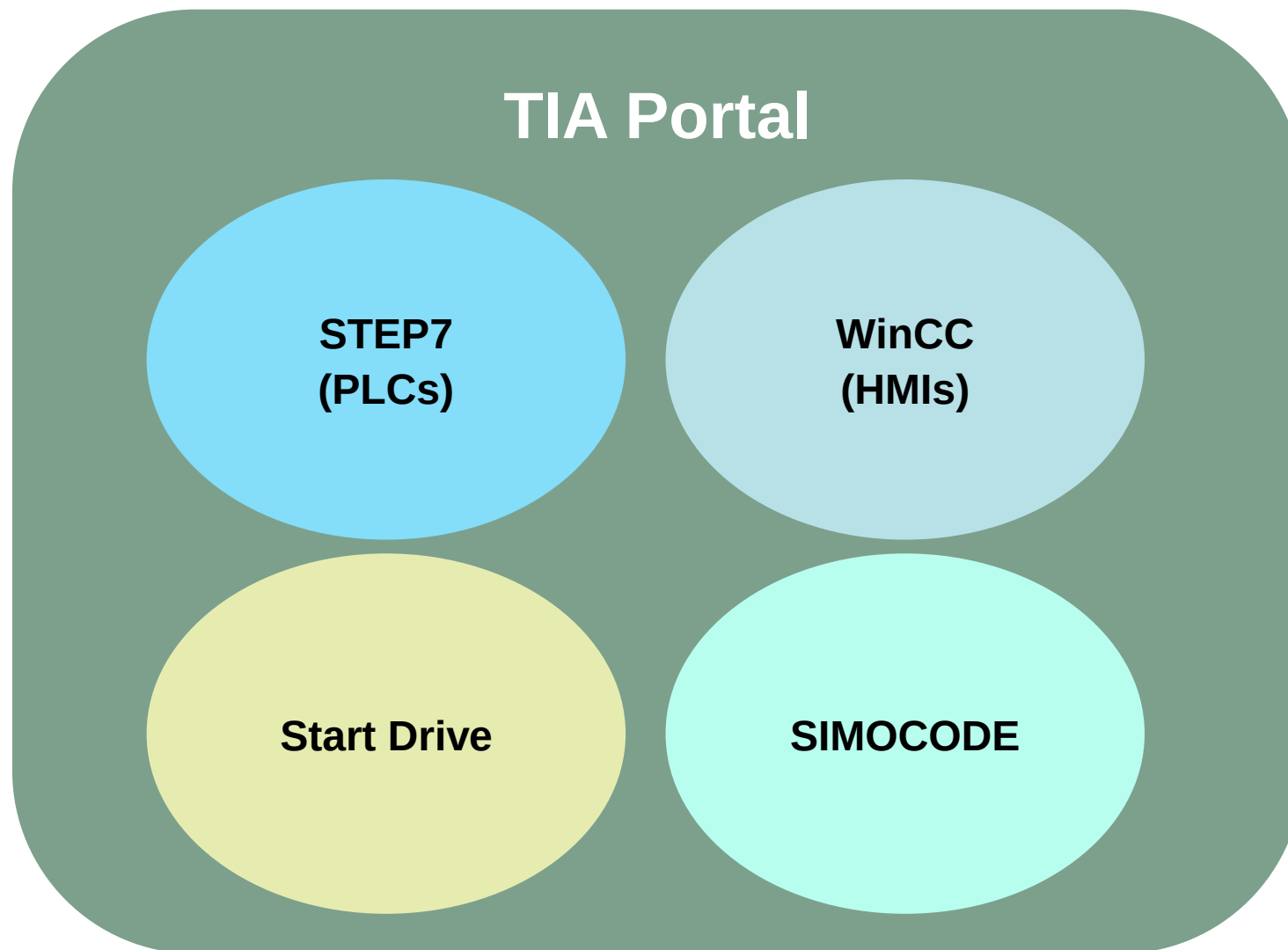
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Software packages

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	SIMATIC STEP 7/ STEP 7 Safety	SIMATIC WinCC	SINAMICS Startdrive	SIMOTION SCOUT TIA	Soft Starter ES/ SIMOCODE ES
Scope of services	Software Controller – SIMATIC S7-1500 (F/T/TF)	SCADA	Convenient, guided acceptance of the Safety Integrated functions		Additional access via PROFIBUS/PROFINET
	Advanced Controller – SIMATIC S7-1500 and SIMATIC S7-300/400 (F/T/TF)	Single PC workstation	Commissioning, optimization, and diagnostics for continuous and cyclic applications: S210, S120, G120, G130, G150, S150		Additional graphic parameterization and extended diagnostics
	Distributed Controller – SIMATIC ET 200 CPUs (F/T/TF)	Comfort Panels and Mobile Panels		SIMOTION D SIMOTION P SIMOTION C	Additional online functions via SIRIUS PtP
	Basic Controller – SIMATIC S7-1200 (F)	Basic Panels	V90 PROFINET		List parametri- zation via PN/PB; parametri- zation during start-up
	Basic	Basic	STEP 7 integrated	Professional	STEP 7 integrated
	Basic	Comfort	Basic (free)		Basic (free)
	Professional	Advanced	Advanced		Standard
	Advanced	Professional			Premium

Software options

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	SIMATIC STEP 7/ STEP 7 Safety	SIMATIC WinCC	SINAMICS Startdrive	SIMOTION SCOUT TIA	Soft Starter ES/ SIMOCODE ES
Engineering options	TIA Portal Multiuser Engineering				
	TIA Portal Teamcenter Gateway				
	TIA Portal Cloud Connector				
	SIMATIC Energy Suite ES				
	SIMATIC S7-PLCSIM Advanced				
	SIMATIC STEP 7 Safety				
	SIMATIC ODK 1500S				
	SIMATIC Target 1500S for SIMULINK				
	SIMATIC Visualization Architect		SINAMICS Drive Control Chart (DCC)		
Runtime options	SIMATIC ProDiag				
	SIMATIC Energy Suite RT				
	SIMATIC OPC UA	WinCC/WebUX			
	TIA User Management Component				

SIMATIC HMI (Basic Panel 2nd GEN)

- 4,7,9,12,15" Widescreen
- Built-in Ethernet
- Screen layout & Root screen
- Diagnostic functions
- Wide screen with 65,000 colors with 9 function keys



Engineering Efficiency

New engineering concept with TIA Portal V16

- Single Engineering framework (PLC, HMI, Network and Drive)
- Online detect and graphical configuration
- Common database with drag & drop
- Integrated Technology Object for PID / Motion

SIMATIC S7-1200 (PLC)

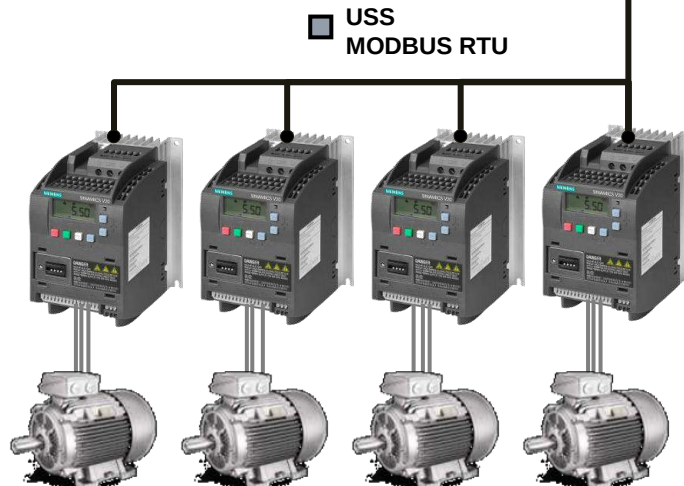
- Built-in Ethernet & Web embedded
- Diagnostic function
- Integrated High-speed input & Pulse output
- Expandable with signal module and/or communication module



Communication board (RS485)

SINAMICS V20 (Basic VSD)

- Easy to use Inverter for Fan / Pump and Conveyor application (0.12-30 kW)
- Integrated application and connection macro (less setting required)
- Built-in USS/MODBUS RTU
- Parameter cloning
- Without power supply
- ECO mode / Hibernation
- Mode for energy saving



USS
MODBUS RTU

PROFINET

SINAMICS V90 (Servo System)

- Single axis Servo system (0.4-7 kW)
- Simple plug & play
- Parameter cloning via SD-Card
- Integrated control mode
 - Pulse train input
 - Analog input
 - DI signal (with integrated positioning function)



Introduction HMI Panel

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Basic HMI



Panel-based

HMI devices with excellent price-performance ratio for simple visualization tasks

Advanced HMI



Panel-based

High-performance HMI devices with high convenience for demanding visualization tasks



PC-based

High-performance HMI devices for data-intensive and complex visualization tasks

Low

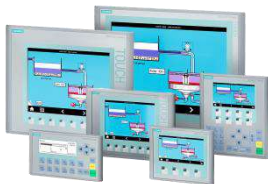
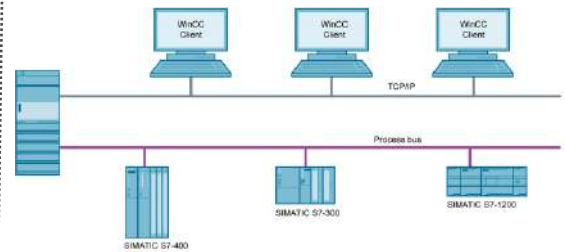
Application complexity & system performance

High



Brochure

Do I need different engineering systems for different HMI applications?

Engineering Edition	WinCC Professional			
	WinCC Advanced			
	WinCC Comfort			
	WinCC Basic			
Devices	Basic Panels	Comfort Panels, Mobile Panels	Panel PCs WinCC RT Advanced	SCADA WinCC RT Professional
				

Specification

Basic Panel 2nd Generation

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	KTP1200 Basic DP	KTP1200 Basic	KTP900 Basic	KTP700 Basic DP	KTP700 Basic	KTP400 Basic
Size	12"	12"	9"	7"	7"	4.3"
Color	65,536	65,536	65,536	65,536	65,536	65,536
Resolution	1280x800 pixel	1280x800 pixel	800x480 pixel	800x480 pixel	800x480 pixel	480x272 pixel
Backlight	20,000h (dimmbale)	20,000h (dimmbale)	20,000h (dimmbale)	20,000h (dimmbale)	20,000h (dimmbale)	20,000h (dimmbale)
Function keys	10	10	8	8	8	4
Memory	10MB	10MB	10MB	10MB	10MB	10MB
Clock	Yes (6 weeks backup)	Yes (6 weeks backup)	Yes (6 weeks backup)	Yes (6 weeks backup)	Yes (6 weeks backup)	Yes (6 weeks backup)
Ethernet	-	1	1	-	1	1
RS485	1	-	-	1	-	-
USB	1 (up to 16GB)	1 (up to 16GB)	1 (up to 16GB)	1 (up to 16GB)	1 (up to 16GB)	1 (up to 16GB)
SD Card	-	-	-	-	-	-
Printer	No	No	No	No	No	No
Maximum PLC connection	1 protocol (4 PLCs)	1 protocol (4 PLCs)	1 protocol (4 PLCs)	1 protocol (4 PLCs)	1 protocol (4 PLCs)	1 protocol (4 PLCs)
Marine Approval	OK					
Hazardous area	NO					
Ambient condition / %RH	0-50 °C / 90%, no condensation					
IP (front/rear)	IP65 / IP20					
Multilingual	10 languages					
Web-browser	Yes					
No.of recipe	50					
No. of user group/user	50/50					
Alarm	Yes (Bit message =1,000 / Analog message = 25 / Message buffer = 256)					
VB Script	No					

Specification

Comfort Panel - Standard

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	TP2200 Comfort	TP1900 Comfort	TP1500 Comfort KP1500 Comfort	TP1200 Comfort KP1200 Comfort	TP900 Comfort KP900 Comfort	TP700 Comfort KP700 Comfort	KTP400 Comfort KP400 Comfort
Size	21.5"	18.5"	15.4"	12.1"	9"	7"	4.3"
Color	16M	16M	16M	16M	16M	16M	16M
Resolution	1920x1080 pixel	1366x768 pixel	1280x800 pixel	1280x800 pixel	800x480 pixel	800x480 pixel	480x272 pixel
Backlight	30,000h (dimmbale)	50,000h (dimmbale)	80,000h (dimmbale)	80,000h (dimmbale)	80,000h (dimmbale)	80,000h (dimmbale)	80,000h (dimmbale)
Function keys	0	0	0 / 36	0 / 34	0 / 26	0 / 24	4 / 8
Memory	24MB	24MB	24MB	12MB	12MB	12MB	4MB
Clock	Yes (6weeks backup)	Yes (6weeks backup)	Yes (6weeks backup)	Yes (6weeks backup)	Yes (6weeks backup)	Yes (6weeks backup)	Yes (6weeks backup)
Ethernet	2 (2 switch + 1 free)	2 (2 switch + 1 free)	2 (2 switch + 1 free)	1 (2 switch)	1 (2 switch)	1 (2 switch)	1
RS485	1	1	1	1	1	1	1
USB / USB mini B	2 / 1	2 / 1	2 / 1	2 / 1	2 / 1	2 / 1	1 / 1
SD Card	2	2	2	2	2	2	2
Speaker	Yes	Yes	Yes	Yes	Yes	Yes	No
Max. PLC connection	>=8 PLCs (No. of devices of each protocol varies per protocol, please use TIA to configure & check)						4 PLCs
PROFINET	Yes	Yes	Yes	Yes	Yes	Yes	Yes
IRT	Yes	Yes	Yes	Yes	Yes	Yes	No
MRP	Yes	Yes	Yes	Yes	Yes	Yes	No
PROFIBUS	Yes	Yes	Yes	Yes	Yes	Yes	Yes
MPI	Yes	Yes	Yes	Yes	Yes	Yes	Yes
OPC Server / Client	Yes / Yes (can be used in parallel with the process links to SIMATIC S7 or non-Siemens PLCs.)						
HTTP Server / Client	Yes / Yes (can be used in parallel with the process links to SIMATIC S7 or non-Siemens PLCs.)						

Mounting Cutout

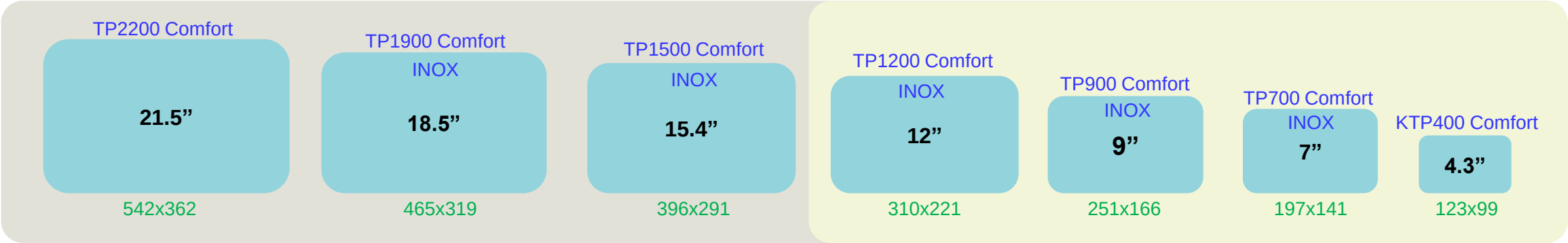
Unit: mm

Basic



Unit: mm

Comfort



Compatible to Basic Panel

Additional parts in Comfort Panel

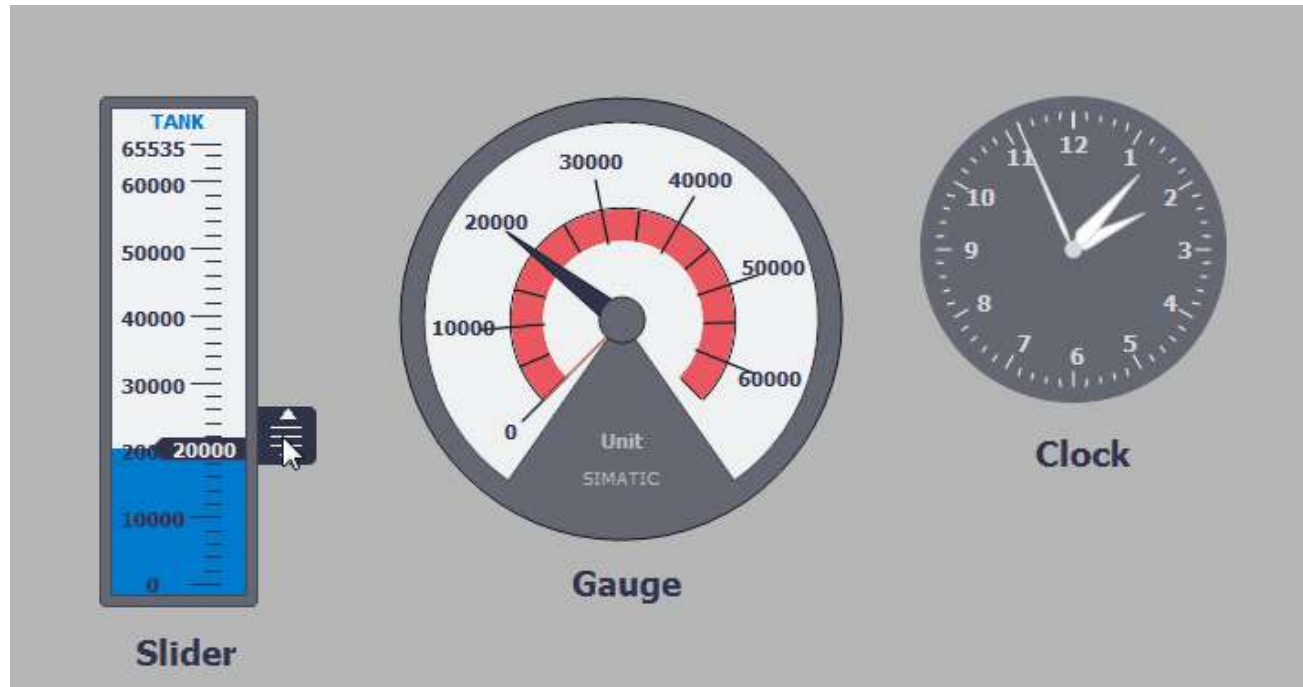
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Basic parts in Basic Panel are still maintain in Comfort Panel.

Advance parts are added e.g., Slider, Gauge and Clock.

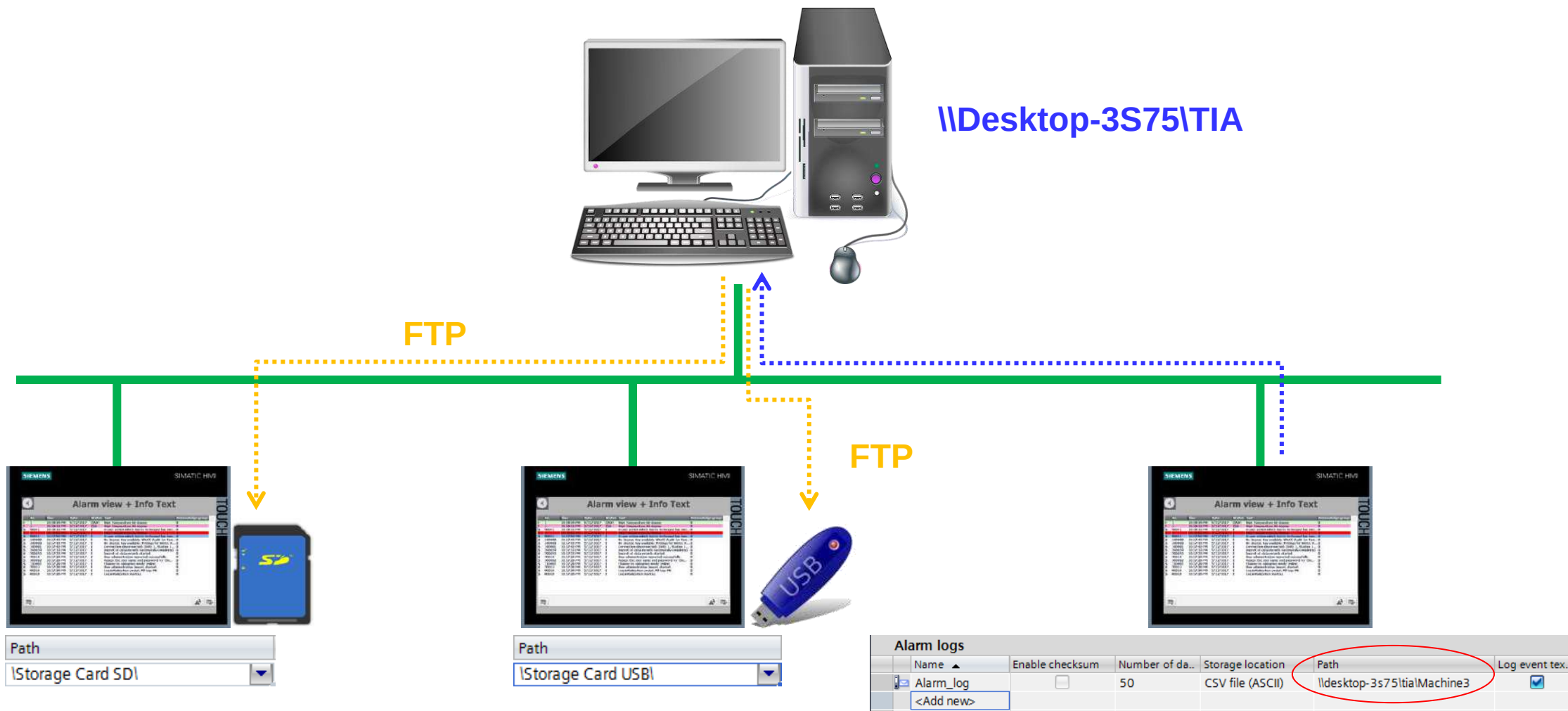


Additional features in Comfort Panel ALARM & Data Log

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Additional features in Comfort Panel Animation

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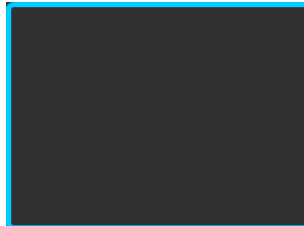
Appearance



Visible

Visible

Visibility

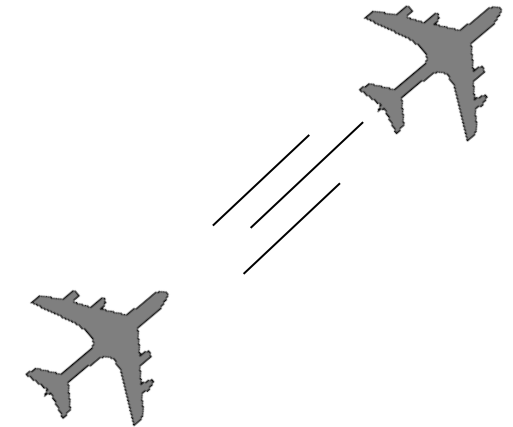


Control Enable



Flashing

Movements



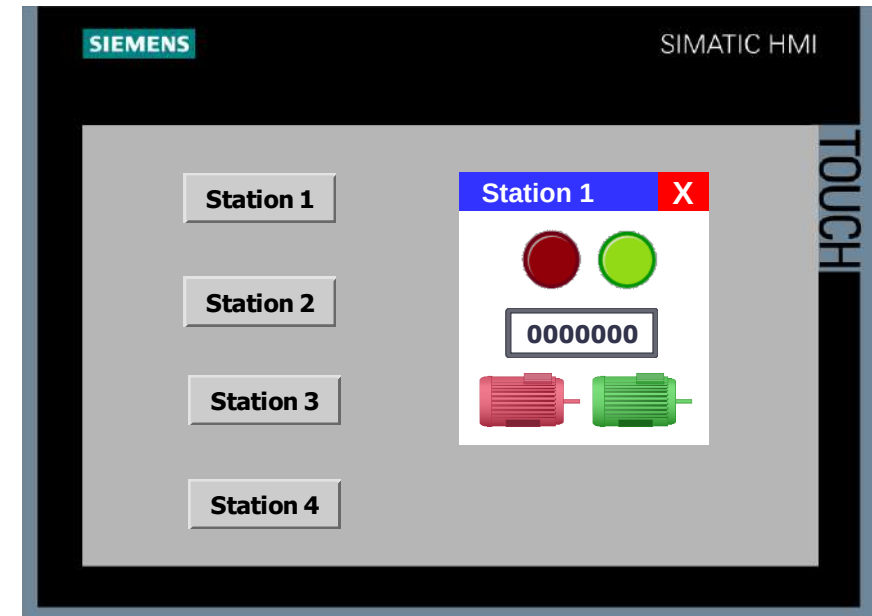
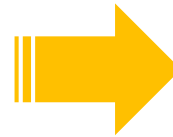
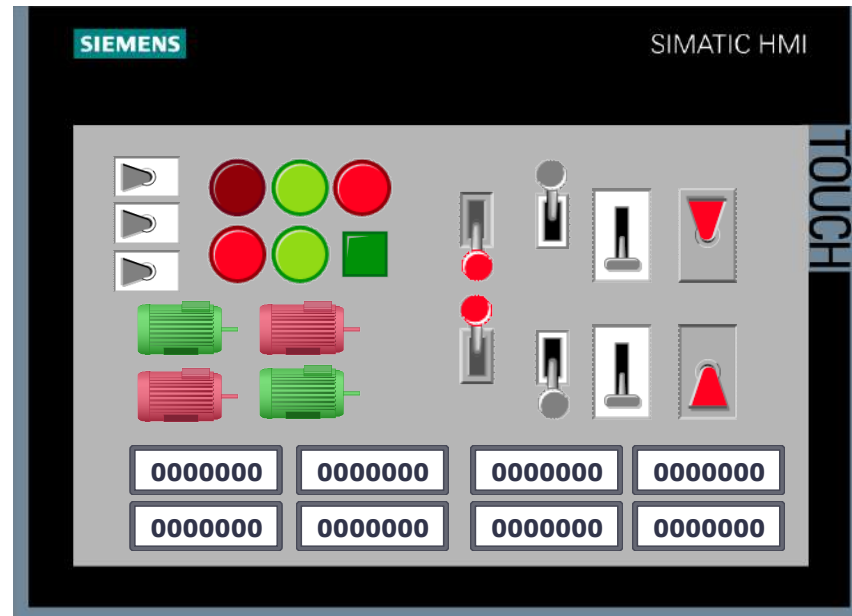
Additional features in Comfort Panel Popup Screen & Slide-in Screen

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Too many information is not easy to operate and understand



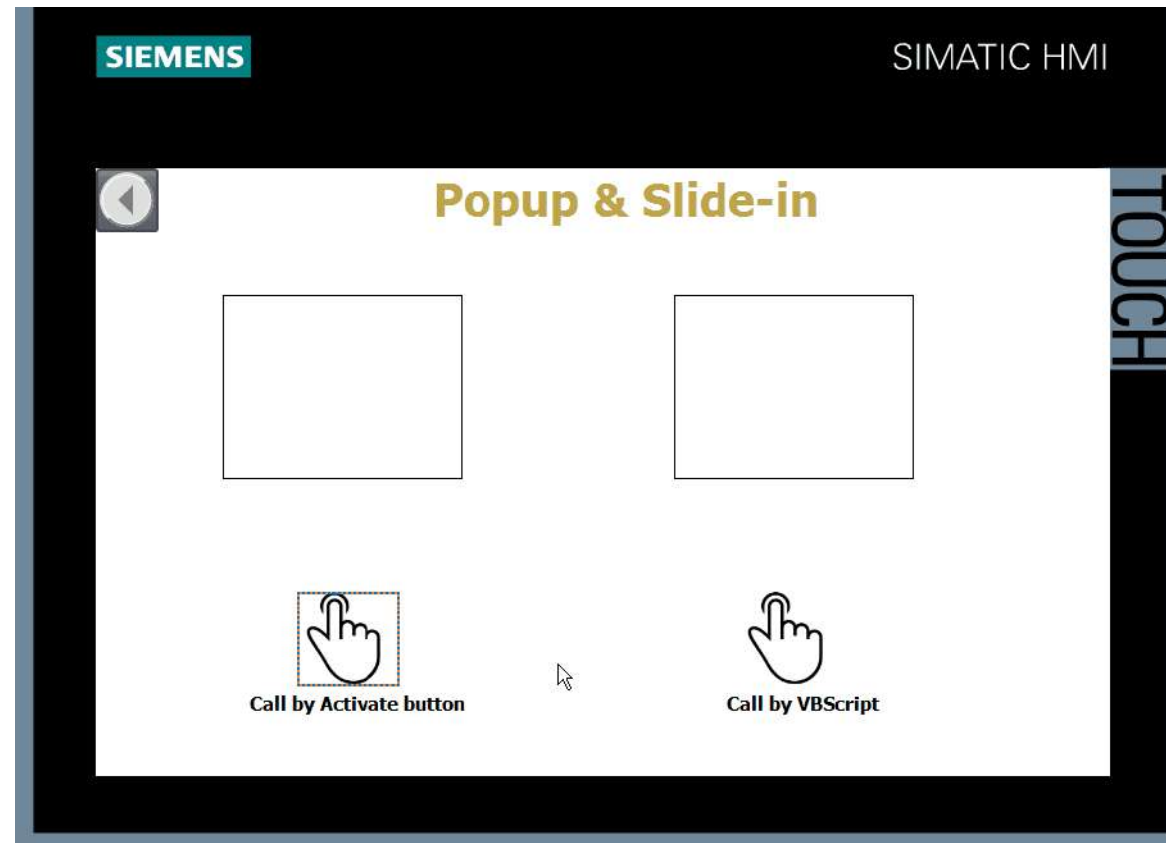
Additional features in Comfort Panel Popup Screen & Slide-in Screen

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Save space by using popup & slide-in



Only one pop-up screen at a time is displayed in a screen.

Additional features in Comfort Panel VBScript

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Flexible operation by VB programming style
Complex programming can be adapted in HMI



Powerful
Flexible

Additional features in Comfort Panel PDF view

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Essential information can be kept in HMI in pdf form.



Where's user
manual?



Additional features in Comfort Panel

Movie player

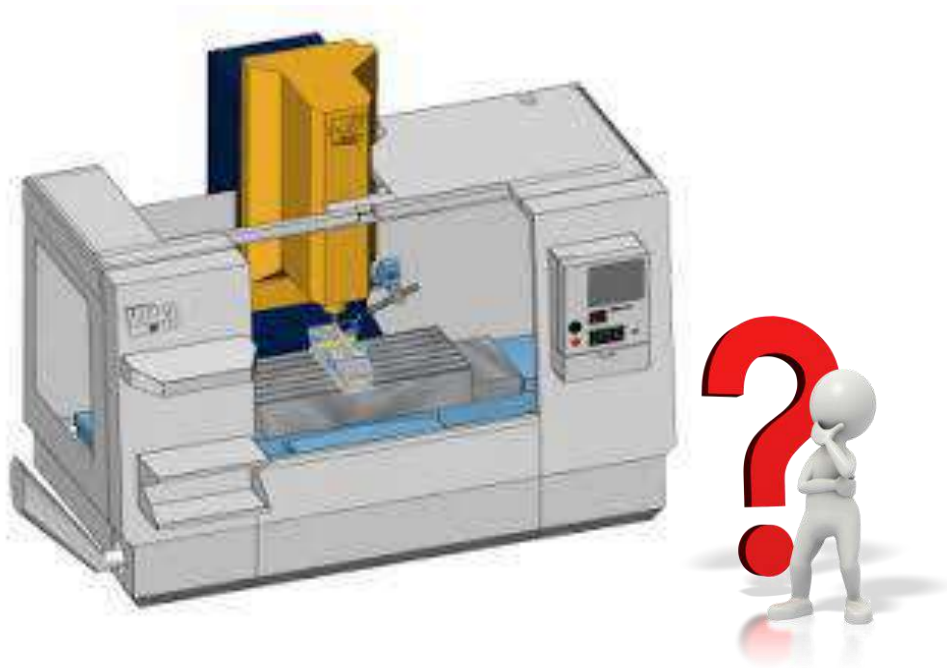
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Movie file can be viewed directly in Comfort Panel.

It can be used as work instruction. Movie file can be kept in HMI, SD or USB.



The following audio formats are supported by Media Player:

- Moving Picture Experts Group standard 1, Layer 1, 2, 3 (.mpa, .mp2, .mp3)
- Waveform Audio (.wav)

The following video formats are supported by Media Player:

- Moving Picture Experts Group standard 1 (.mpg, .mpeg, .mpv, .mpe)
- Advanced Streaming Format (.asf)
- Windows Media Video Format (.wmv)
- Audio-Video Interleaved (.avi)

Note: There's free space in flash memory around 700MB

Additional features in Comfort Panel

Camera view

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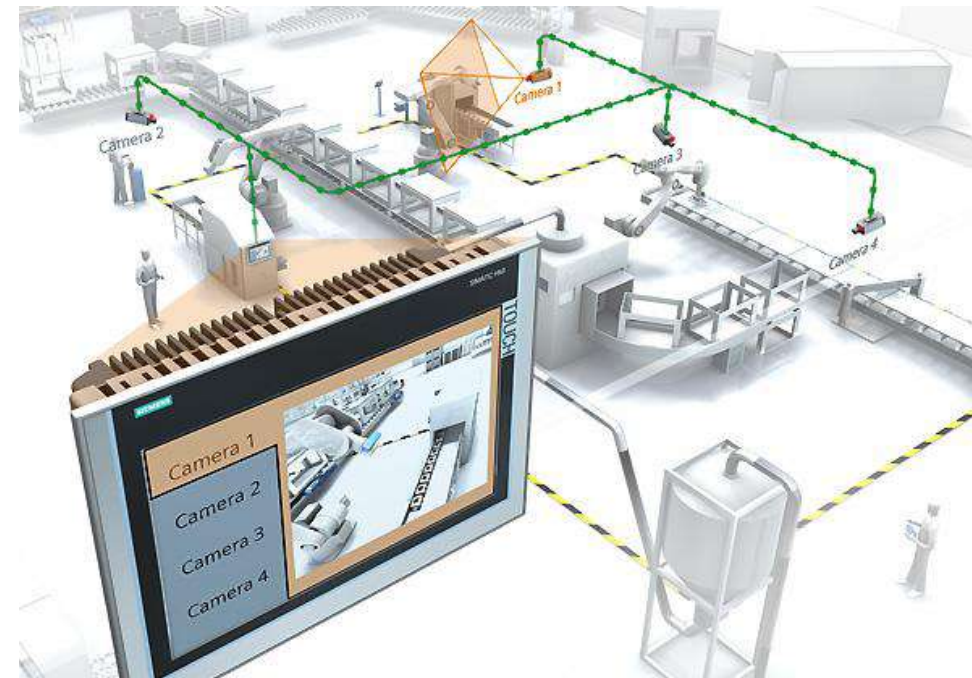
Tested Camera

OPC Data Access is an open standard for exchanging both local and remote variables between various applications via Industrial Ethernet.

- Siemens CCMS2025
- AXIS M1011
- AXIS M1013
- D-Link DCS-942L
- D-Link 4701E
- D-Link DCS-4201
- TL-SC2020

Doesn't work

- D-Link DCS-930L (no RTP/RTPS protocol)



In the "Camera view" object you display screens of a connected network camera on your HMI device. Use the "Camera view" object with the network cameras which support the streaming protocol **RTP/RTPS** and Codecs **MJPEG, H.264 and MPEG4**.

Additional features in Comfort Panel Report

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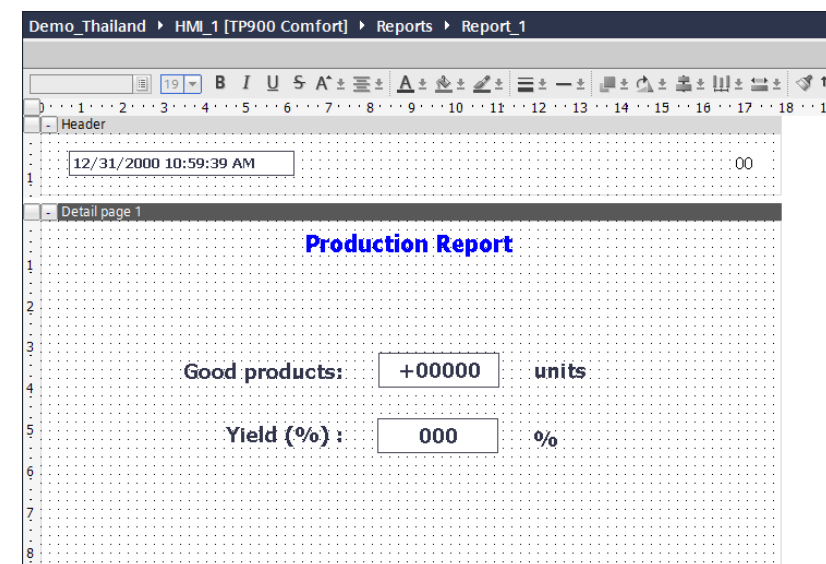
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Output report to printer, or to pdf file.

Report template can be created in advance.



PostScript



Output in Runtime

On the control panel of the HMI device, the report is output to the printer that is specified as the default printer.

The default printer enabled for a HMI device can be found in the "Printer list". For further information about the "Printer list", refer to the Internet page of Siemens Customer Support and the Article ID "11376409".

<https://support.industry.siemens.com/cs/document/11376409/printers-for-simatic-hmi-panels?dti=0&lc=en-WW>

Additional features in Comfort Panel

Automatic backup

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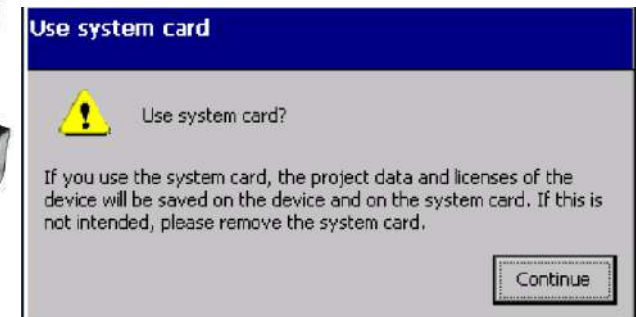
SIEMENS

Automatic Backup

System SD card will be detected automatically.



SIMATIC HMI SD storage card



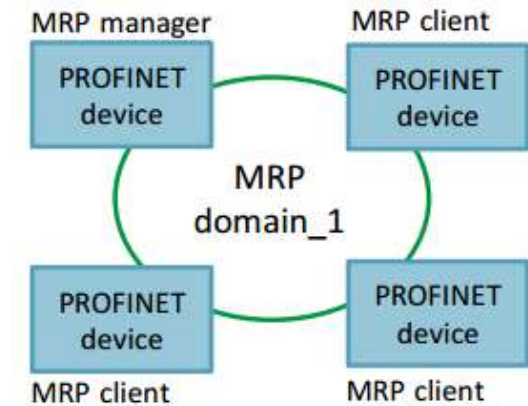
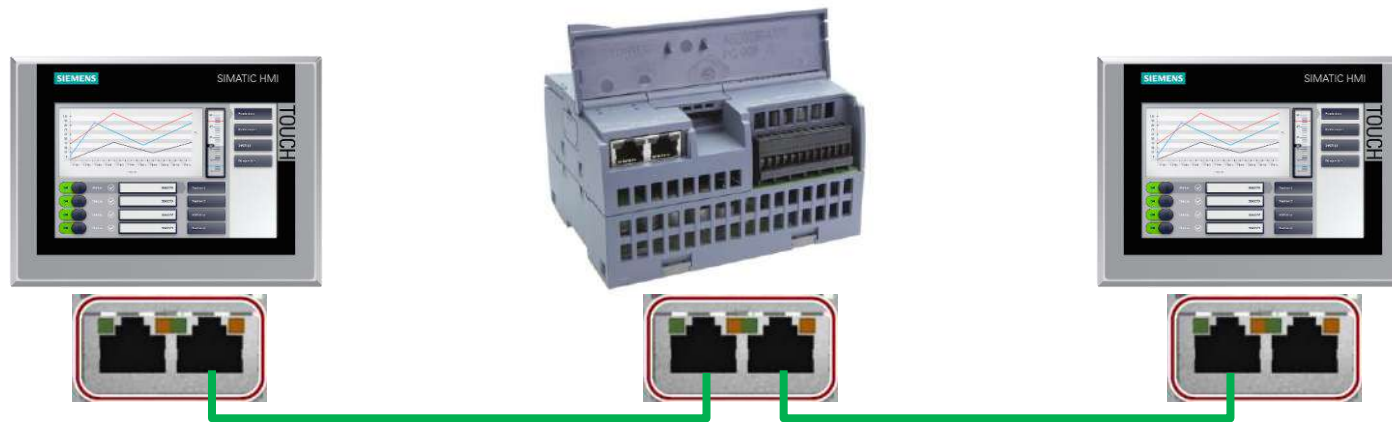
Additional features in Comfort Panel Embedded Ethernet Switch

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Flexible connection, cost saving & high reliability with MRP.



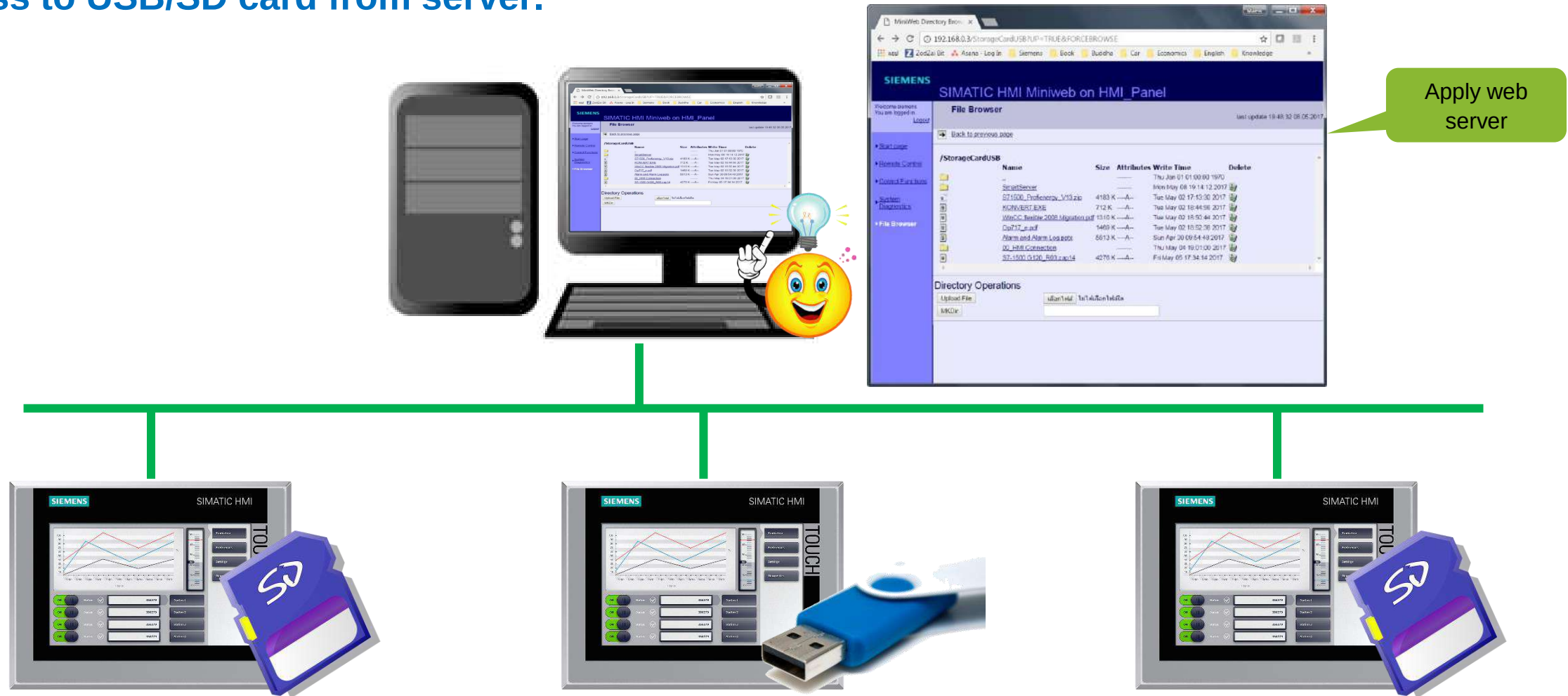
Additional features in Comfort Panel FTP Server

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Access to USB/SD card from server.



Additional features in Comfort Panel Email

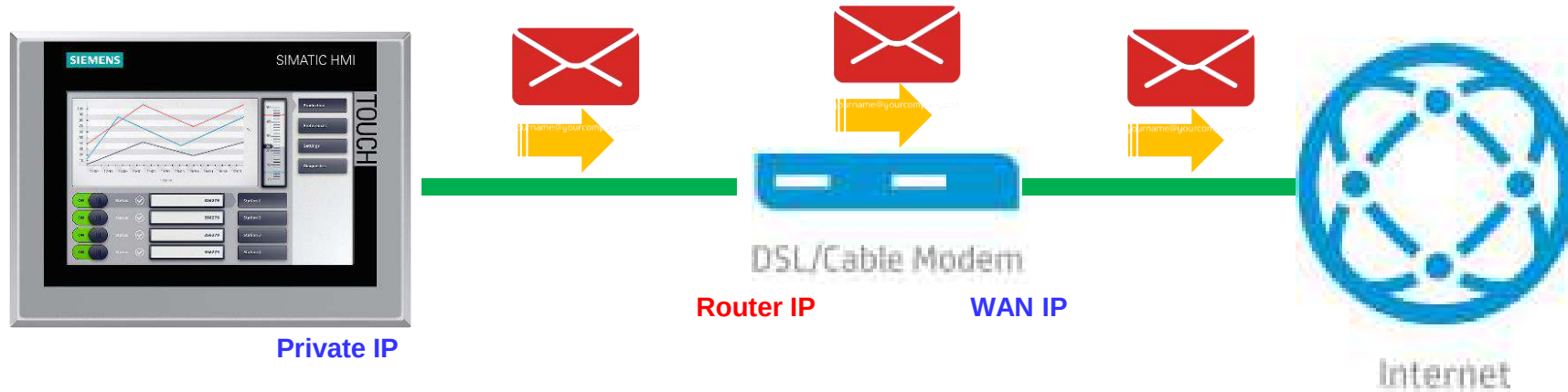
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Email can be sent out from HMI directly.

No need any external software or license.

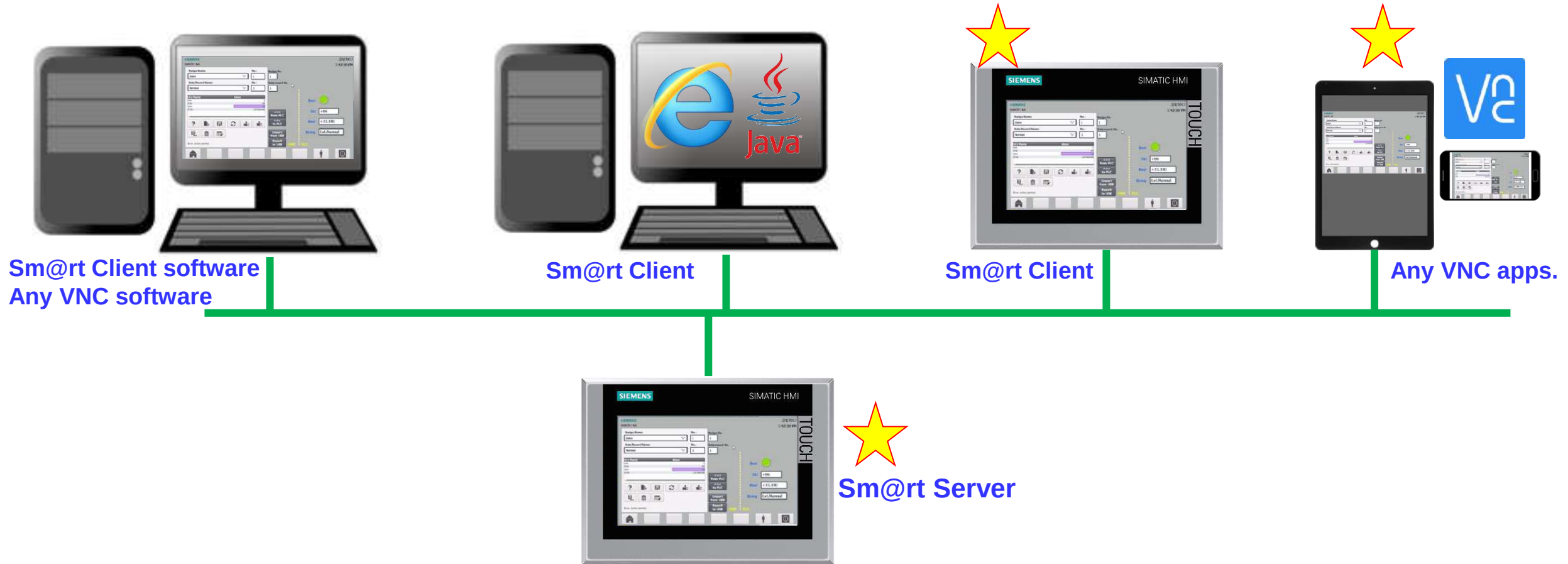


Additional features in Comfort Panel Sm@rtServer/Client

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Additional features in Comfort Panel HTTP Client / Server

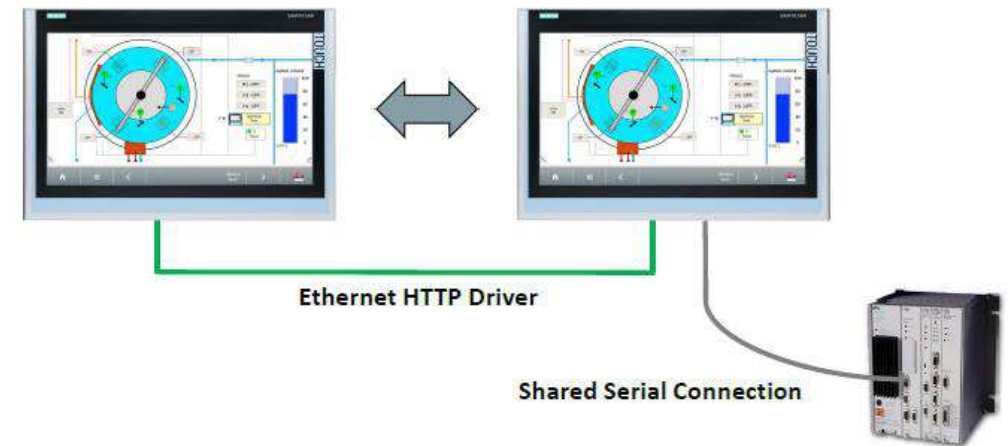
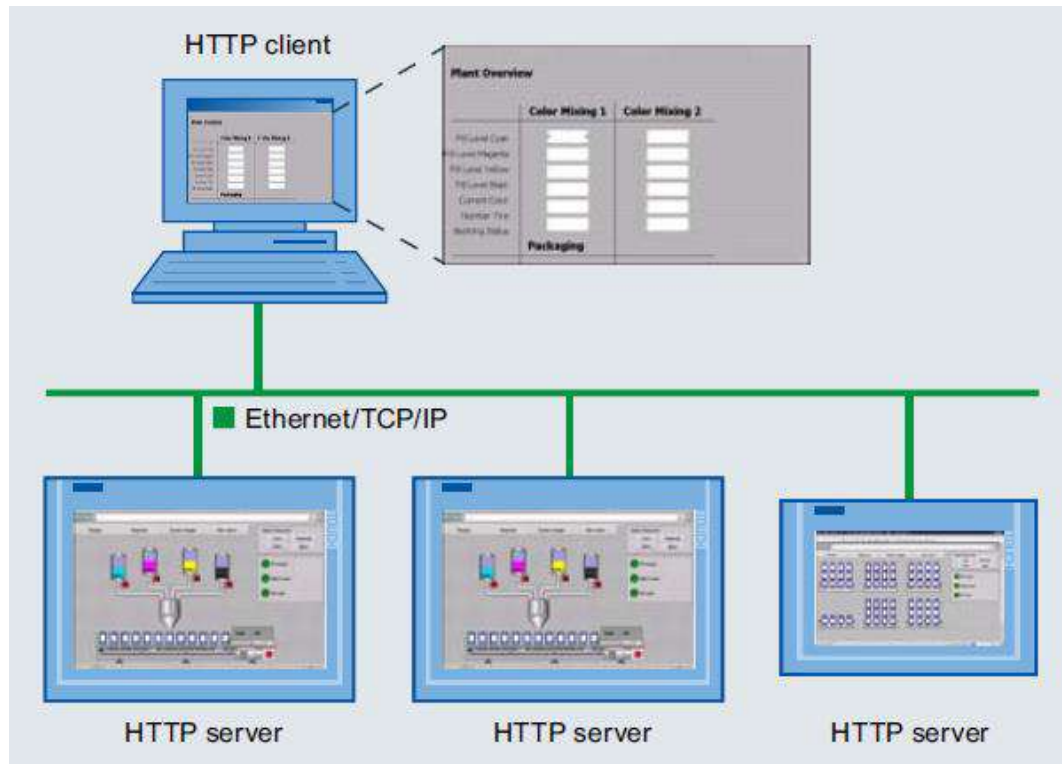
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HMI-HMI tag sharing

Communication based on HTTP message frames enables variables to be exchanged between SIMATIC HMI systems.



Additional features in Comfort Panel OPC Server / Client

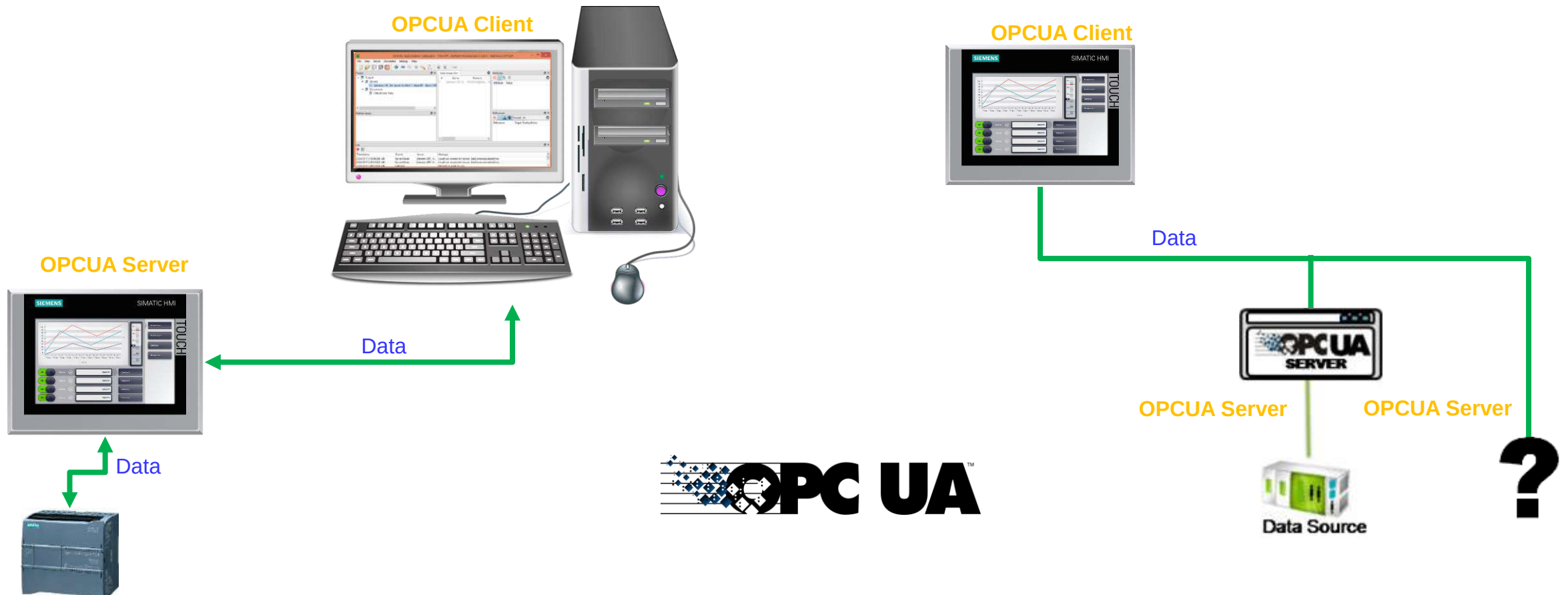
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Comfort Panel can act as OPC Server and OPC Client.

OPC Data Access is an open standard for exchanging both local and remote variables between various applications via Industrial Ethernet.



Additional features in Comfort Panel ProDiag

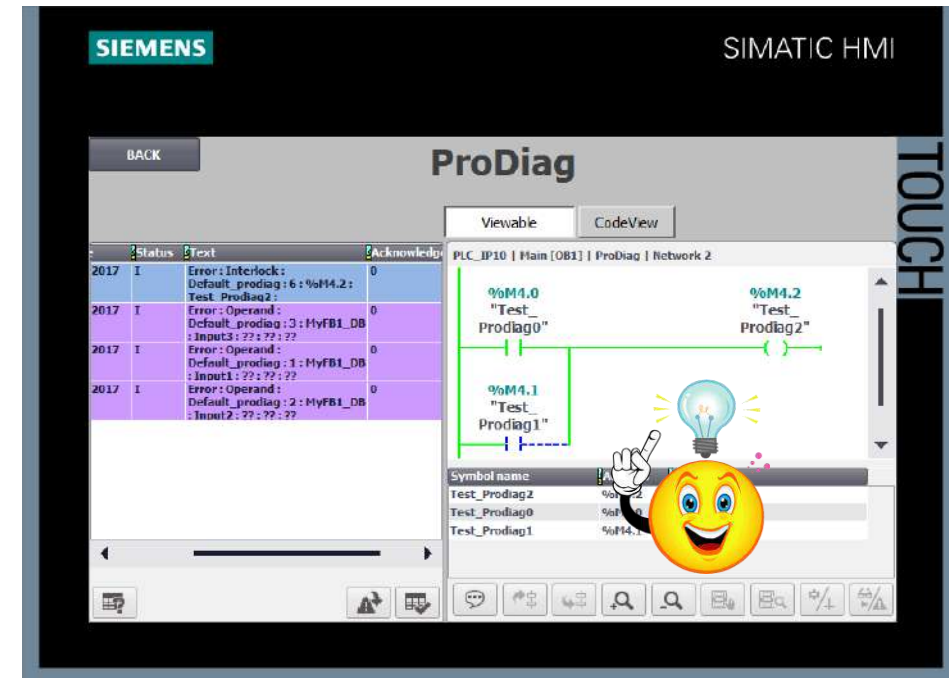
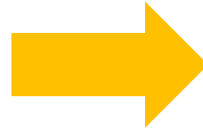
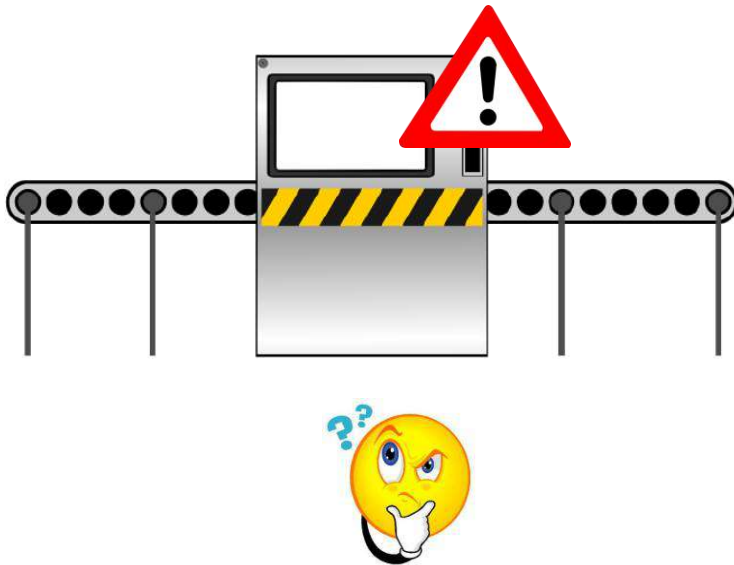
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Code generation diagnostic

SIMATIC ProDiag offers detailed plant and machine monitoring with minimal configuration and visualization costs and provides support thanks to automated code generation and synchronization of HMI devices.



Note: Only with S7-1500 (no license required if using <= 25 supervisions)
Comfort panel needs license "SIMATIC ProDiag for SIMATIC Comfort/Mobile Panels" (6AV2107-0UP00-0BB0)

Additional features in Comfort Panel GMP

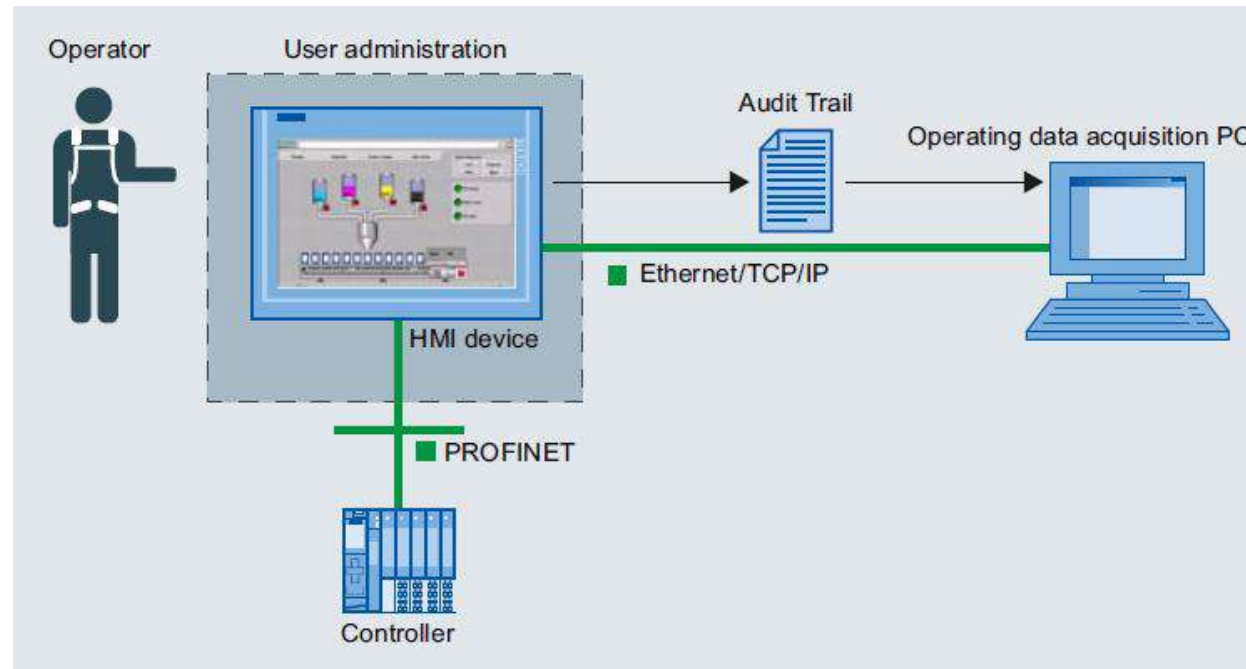
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GMP Standard

WinCC Audit covers essential requirements outlined by GMP (Good Manufacturing Practice) and the FDA (Food and Drug Administration) in accordance with 21 CFR Part 11.



Basic programming

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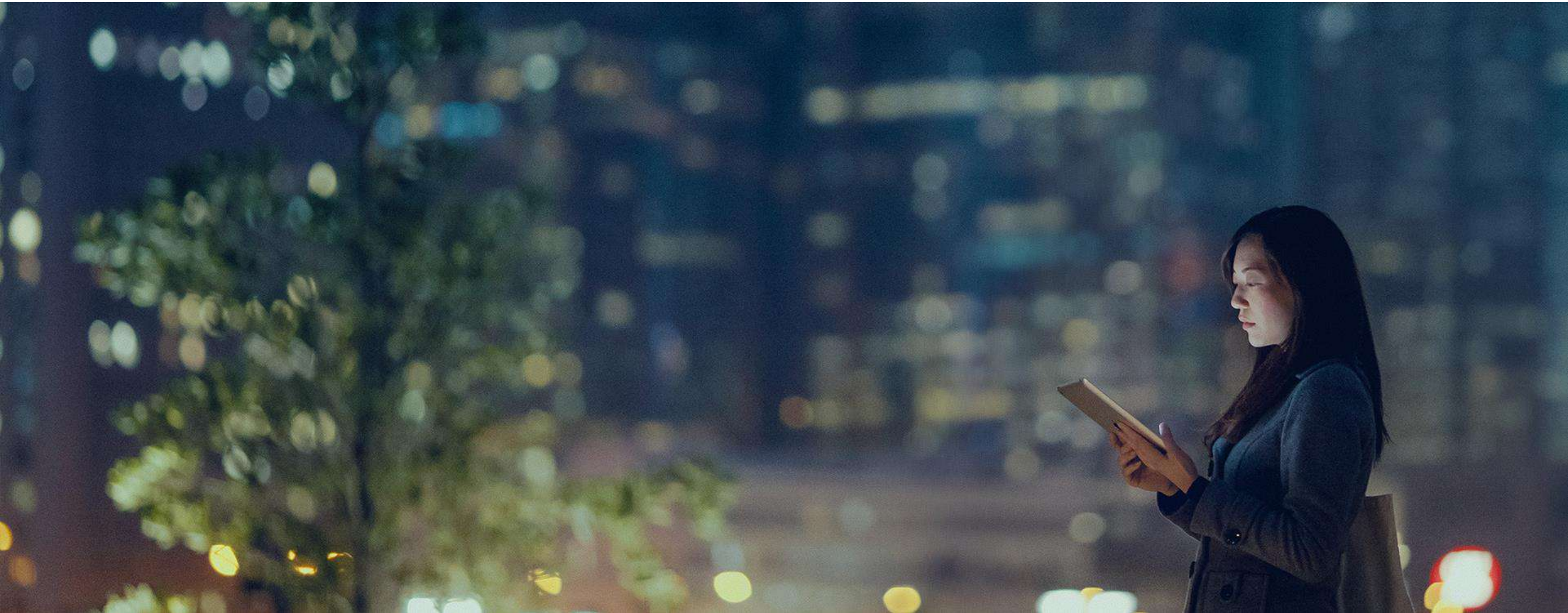
- HMI Setup
- Graphic User Interface
- Alarm Feature
- Web browser
- Others

01 - HMI Setup

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Add HMI Device: Method1

Drag & Drop in network view

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Siemens - C:\Users\User\Documents\Automation\Basic Workshop\Basic Workshop

Project Edit View Insert Online Options Tools Window Help

Save project Go online Go offline <Search in project>

Basic Workshop > Devices & networks

Topology view Network view Device view

Network Connections: Add connection Relations

1

2

3

4 Drag & Drop

Devices & networks

Hardware catalog

Options

Catalog

Filter Profile: <All>

Controllers

HMI

SIMATIC Basic Panel

3" Display

4" Display

6" Display

7" Display

KTP700 Basic

6AV2 123-2GA03-0AX0

6AV2 123-2GB03-0AX0

KTP700 Basic Portrait

9" Display

10" Display

11" Display

15" Display

SIMATIC Panel

SIMATIC Comfort Panel

SIMATIC Multi Panel

SIMATIC Mobile Panel

SIMATIC Key Panel

SIMATIC Push Button Panel

SIMATIC WinAC for Multi Panel

Information

Device:

KTP700 Basic PN

Article no.: 6AV2 123-2GB03-0AX0

Version: 14.0.1.0

Partner interface does not exist

เลือก รุ่นของ HMI ตามต้องการ

Add HMI Device: Method1

Drag & Drop in network view

Approved
Partner

Factory
Automation

SIEMENS

Siemens - C:\Users\User\Documents\Automation\Basic Workshop\Basic Workshop

Project Edit View Insert Online Options Tools Window Help

Save project Go online Go offline <Search in project>

Basic Workshop > Devices & networks

Topology view Network view Device view

Hardware catalog

Options

Catalog

Filter Profile: <All>

Controllers

HMI

SIMATIC Basic Panel

3" Display

4" Display

6" Display

7" Display

KTP700 Basic

6AV2 123-2GA03-0AX0

6AV2 123-2GB03-0AX0

Devices & networks

Name

Basic Workshop

Add new device

Devices & networks

MASS_PL_CIP90 [CPU 1214C DC/DC/DC]

HMI_1 [KTP700 Basic PN]

Ungrouped devices

Common data

Documentation settings

Languages & resources

Online access

Card Reader/USB memory

Details view

PROFINET interface_1 [X1 : PN(LAN)]

Properties

General IO tags System constants Texts

General

Ethernet addresses

Time synchronization

Operating mode

Advanced options

Web server access

Hardware identifier

Name: PROFINET interface_1

Author: User

Comment:

1

ลากเส้นระหว่าง Port ของทั้ง 2 อุปกรณ์เพื่อเชื่อมต่อกัน

Basic Workshop > Devices & networks

Topology view Network view Device view

Connections HMI connection

2

MASS_PL_CIP90 CPU 1214C

HMI 1 KTP700 Basic PN

PN/IE_1

รูปแบบ เมื่ออุปกรณ์เชื่อมต่อในวงแลนเดียวกันแล้ว

Add HMI Device: Method1

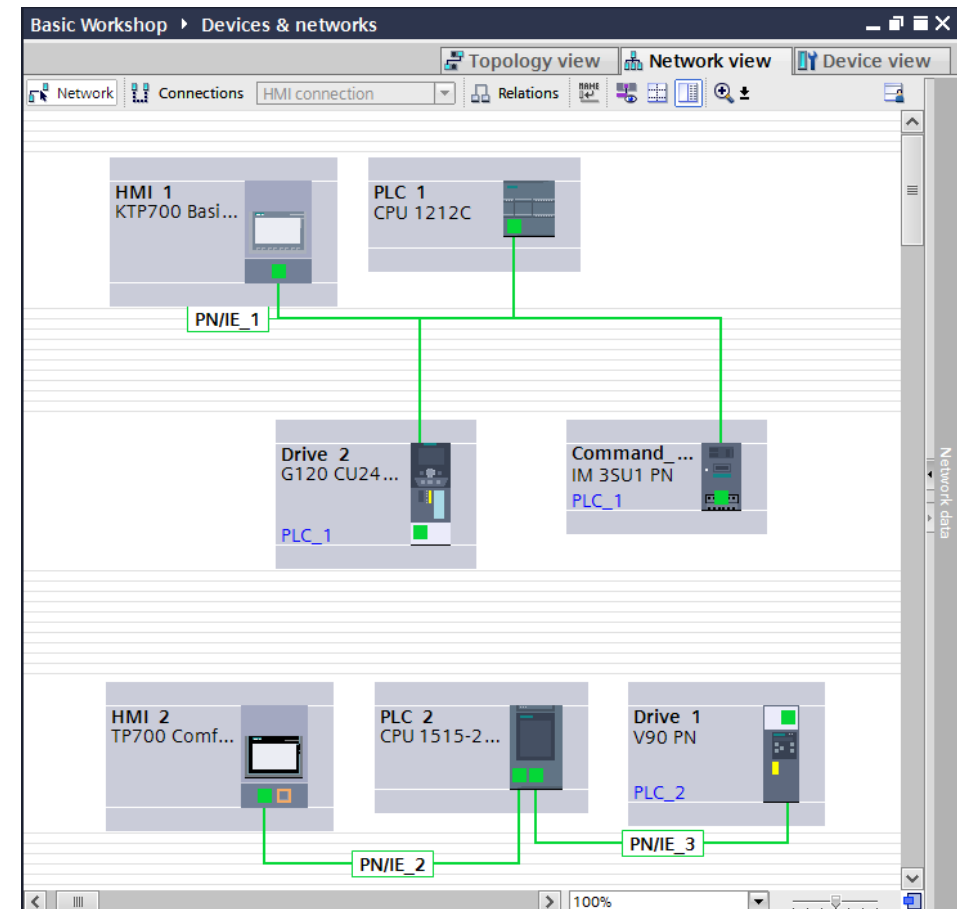
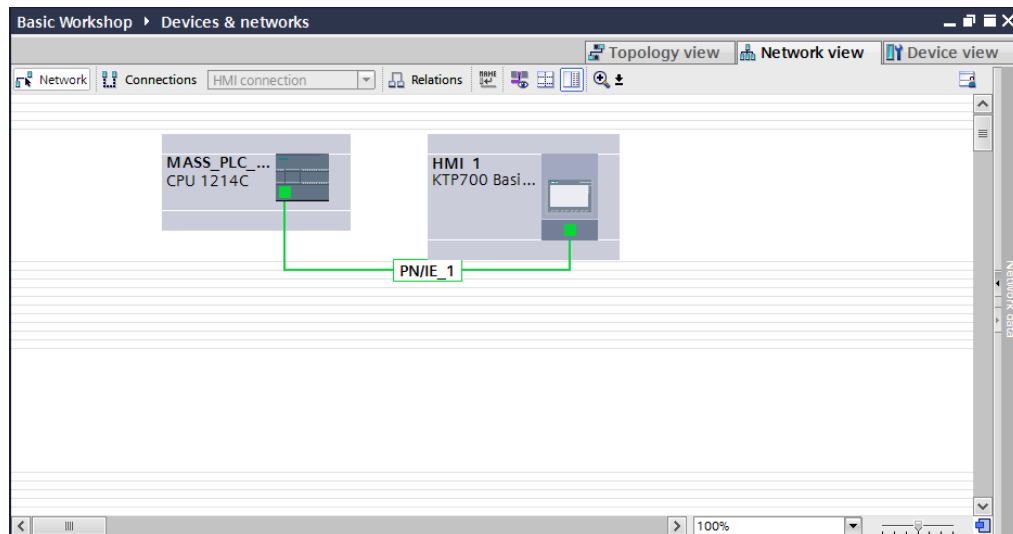
Drag & Drop in network view

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- สิ่งที่ควรระวังคือ การลากเส้นเชื่อมระหว่างพอร์ตระหว่างอุปกรณ์นี้ เพียงแค่บอกว่าอุปกรณ์ 2 ตัวนี้อยู่ใน network เดียวกันเท่านั้น ไม่ได้หมายความว่าจะสามารถสื่อสารกับ PLC ได้แล้ว



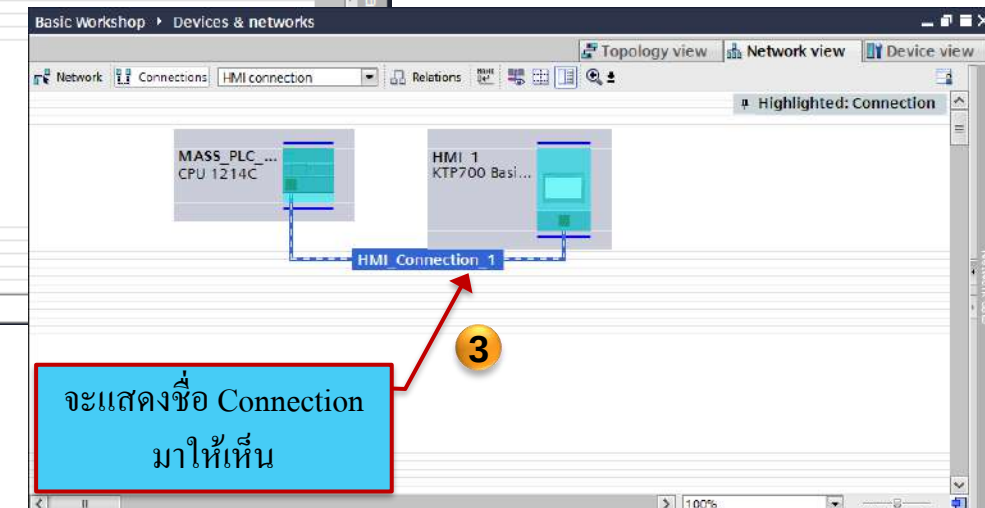
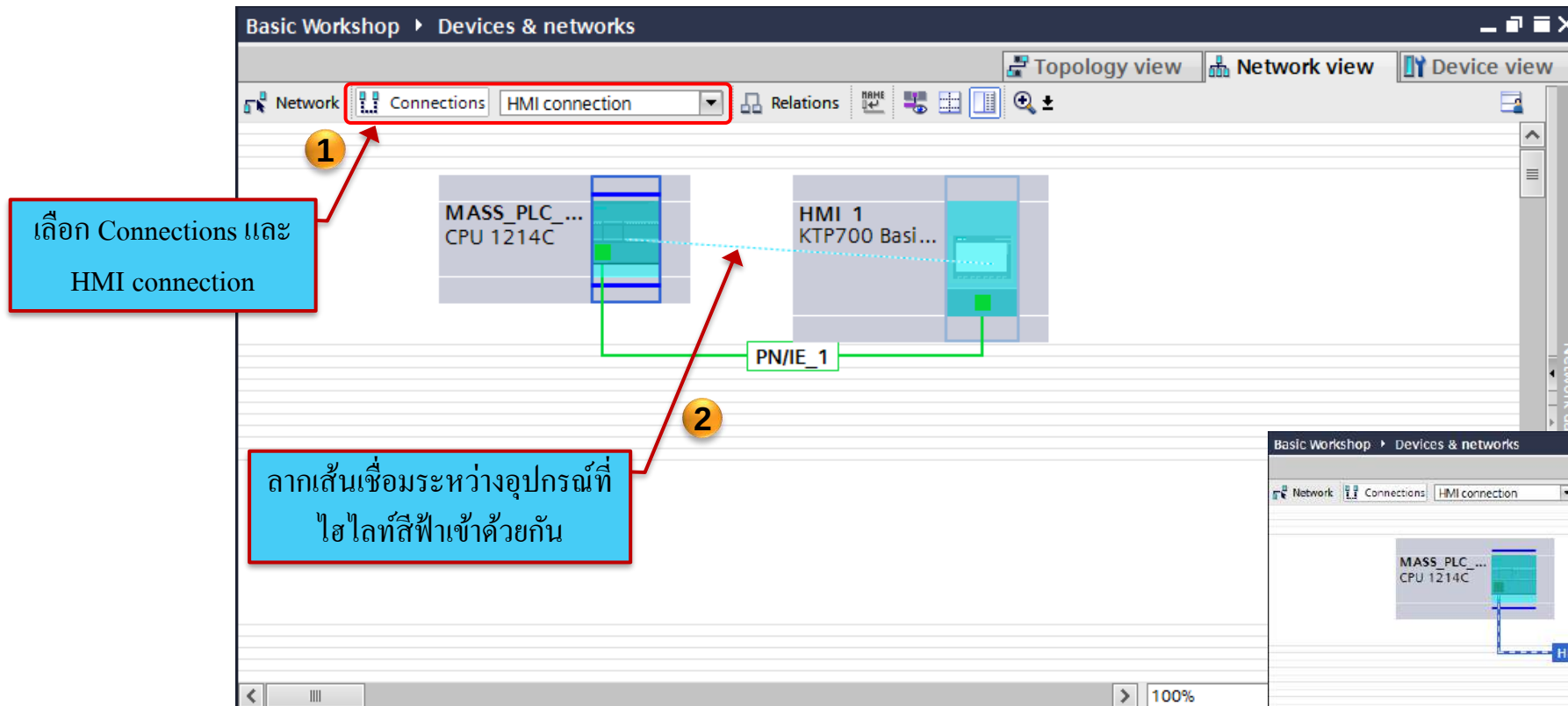
Add HMI Device: Method1 Drag & Drop in network view

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- การทำให้จอคุยกับ PLC ได้คือให้ทำ Connections -> HMI connection



Add HMI Device: Method1

Drag & Drop in network view

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- เราสามารถตรวจสอบการสร้าง Connection ได้ในส่วนของ Connections

The screenshot displays the Siemens TIA Portal interface. On the left, the 'Project tree' shows the 'Connections' folder under 'HMI_1 [KTP700 Basic PN]' highlighted with a red box. The main area shows a table titled 'Connections to S7 PLCs in Devices & Networks' with a red box around the first row. A red arrow points from a blue text box 'Connections ได้ถูกสร้างแล้ว' to the 'HMI time synchronization mode' column of this row.

Name	Communication driver	HMI time synchronization mode	Station	Partner	Node	Online	Comment
HMI_Connection_1	SIMATIC S7 1200	None	57-1200 station 2	MASS_PLC_IP90	CPU 1214C DC/DCI...	☒	
<Add new>							

Below the table, the 'Parameter' tab is active, showing the 'KTP700 Basic PN' HMI device configuration. The 'Interface' is set to 'PROFINET (X1)'. The 'HMI device' address is '192.168.0.91' and the 'Access point' is 'S7ONLINE'. The 'PLC' address is '192.168.0.90' and the 'Access password' is empty. The 'Station' tab is also visible on the right.

At the bottom, the 'General' tab is selected, displaying the message: 'No 'properties' available. No 'properties' can be shown at the moment. There is either no object selected or the selected object does not have any displayable properties.'

Add HMI Device: Method2 HMI Wizard

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Siemens - C:\Users\User\Documents\Automation\Basic Workshop\Basic Workshop

Project Edit View Insert Online Options Tools Window Help

Save project

Go online Go offline

Totally Integrated Automation PORTAL

Tasks

Options

Find and replace

Find:

Whole words only

Match case

Find in substructures

Find in hidden texts

Use wildcards

Use regular expressions

Down

Replace

Replace all

Languages & resources

Editing language:

Reference language:

Portal view Overview

The project Basic Workshop was saved ...

1

2

3

4

5

6

7

ตั้งชื่อของจอ HMI

เลือก Version ที่ตรงกับ HMI

เลือกการ Setting HMI แบบ Wizard

Device name:

HMI_basic

Device:

KTP700 Basic PN

Article no.:

6AV2 123-2GB03-0AX0

Version:

13.0.1.0

Description:

7" TFT display, 800 x 480 pixel, 64K color and Touch operation, 8 function keys; 1 x PROFINET, 1 x USB

Start device wizard

OK

Cancel

HMI Wizard

PLC Connection (1)

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PLC connections

Screen layout

Alarms

Screens

System screens

Buttons

HMI Device Wizard: KTP700 Basic PN

PLC connections

Configure the PLC connection(s).

Communication driver: SIMATIC S7 1200

Interface: PROFINET (X1)

Select PLC: PLC_1

Click

2

3

4

Next

Details view

Portal view

Wizard: successfully configured KTP700...

เลือก PLC ที่จะเชื่อมต่อกับจอ

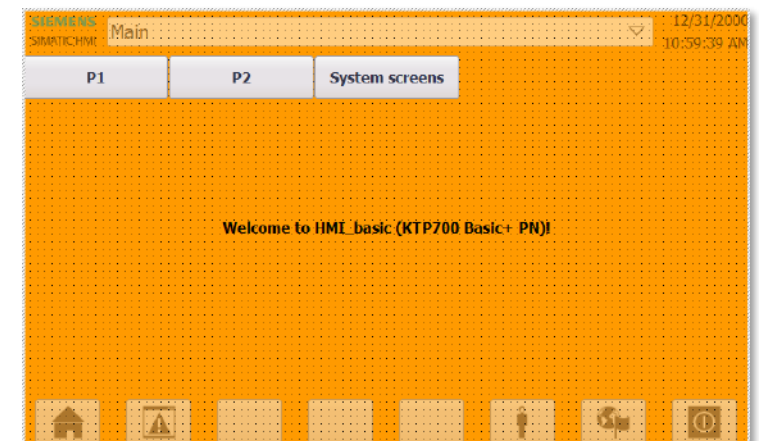
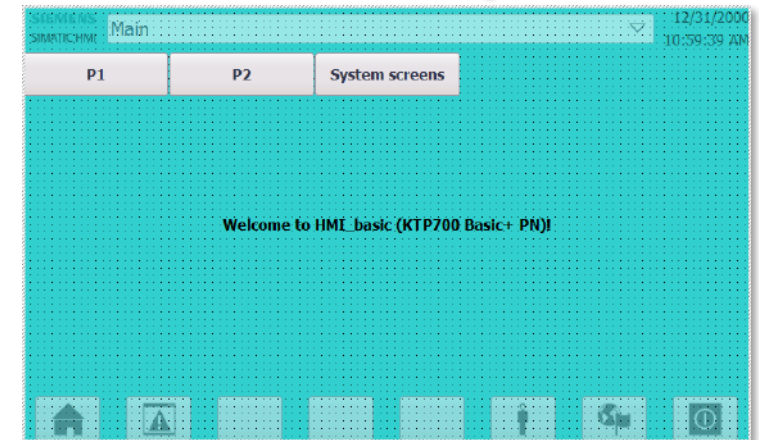
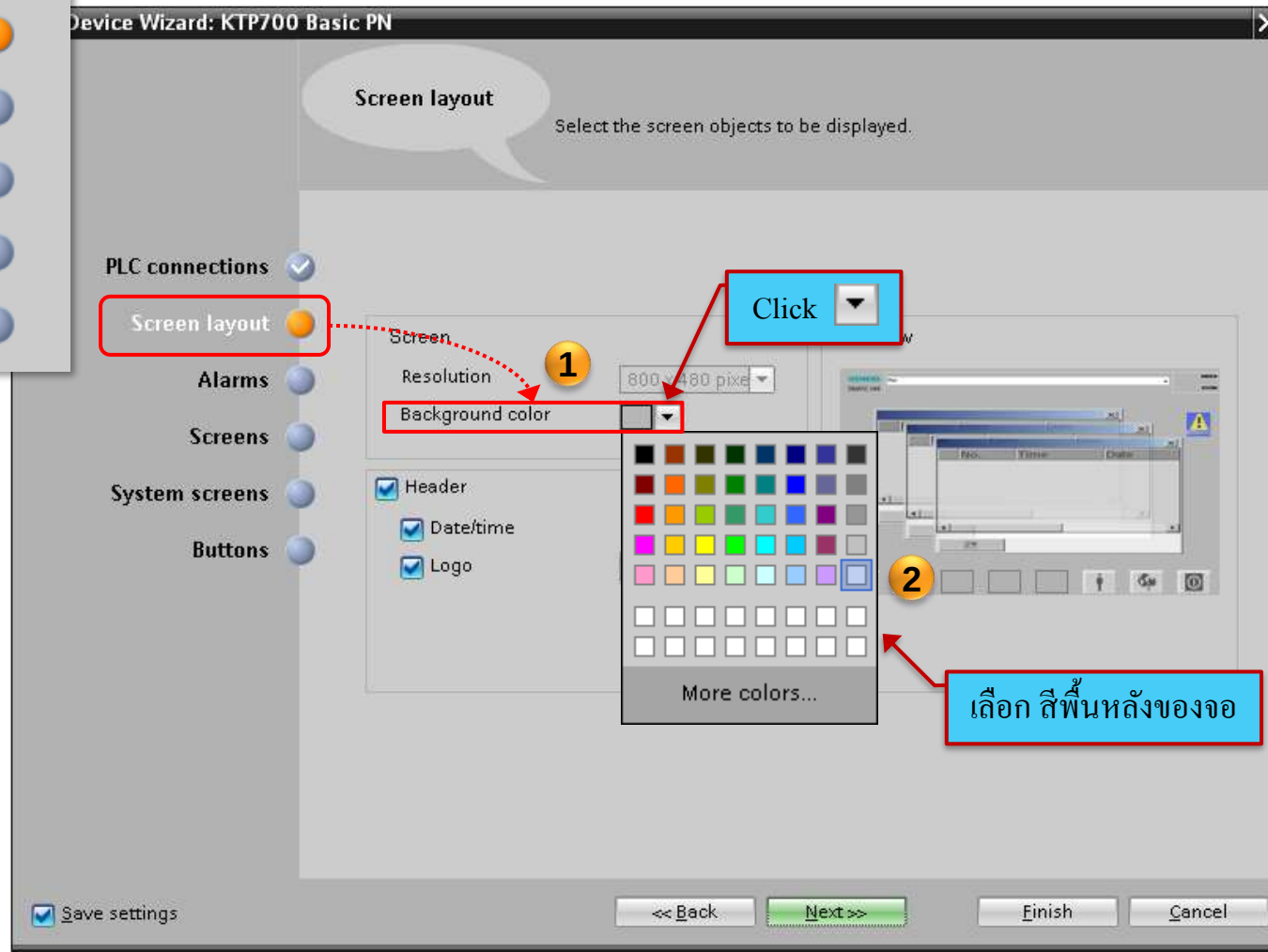
HMI Wizard Screen Layout (2)

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ตัวอย่างพื้นหลังที่เปลี่ยนสี



HMI Wizard Screen Layout (2)

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PLC connections ☒

Screen layout ☒

Alarms ☐

Screens ☐

System screens ☐

Buttons ☐

Device Wizard: KTP700 Basic PN

Screen layout

Select the screen objects to be displayed

PLC connections ☒

Screen layout ☒

Alarms ☐

Screens ☐

System screens ☐

Buttons ☐

Screen

Resolution

800 x 480 pixels

Background color

☒ Header

☒ Date/time

☒ Logo

3

Browse...

เลือก Header ของจอโดยการเลือก

Check Box (☒)

o วันและเวลา

o Logo เช่น Logo SIEMENS

☒ Save settings

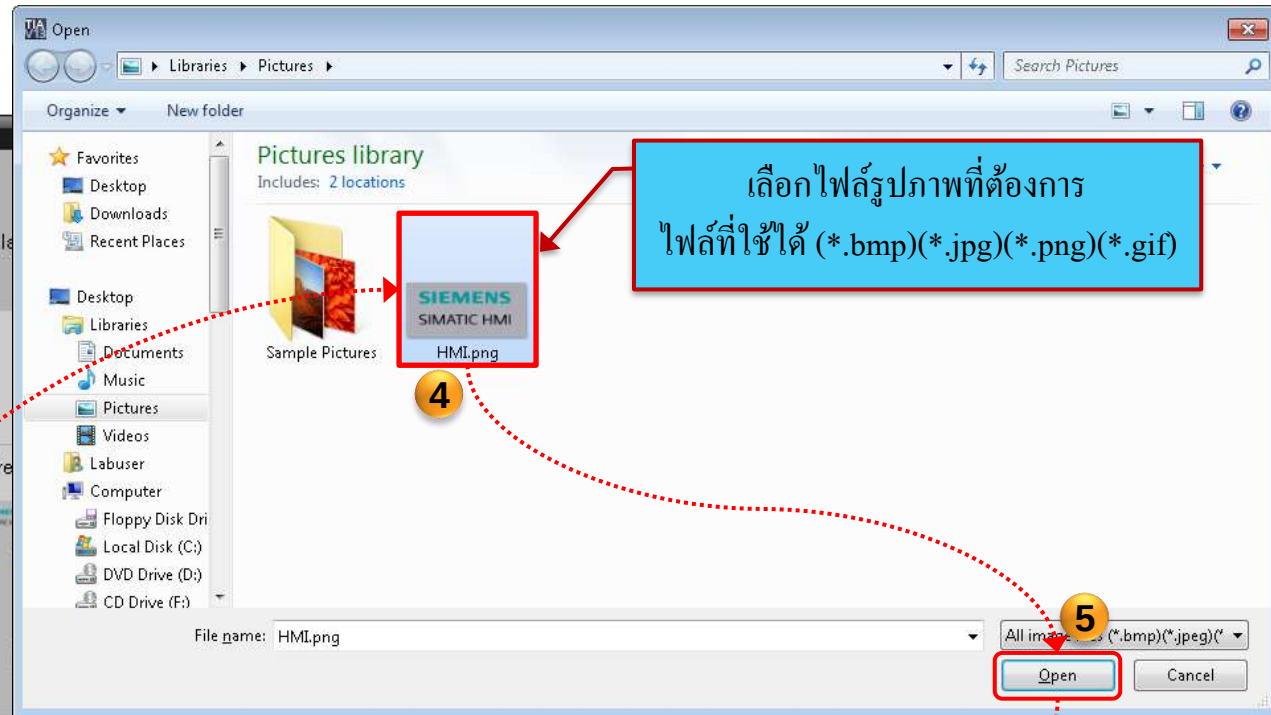
<< Back

Next >>

6

Finish

Cancel



เลือกไฟล์รูปภาพที่ต้องการ
ไฟล์ที่ใช้ได้ (*.bmp)(*.jpg)(*.png)(*.gif)

4

5

6

HMI Wizard Alarm (3)

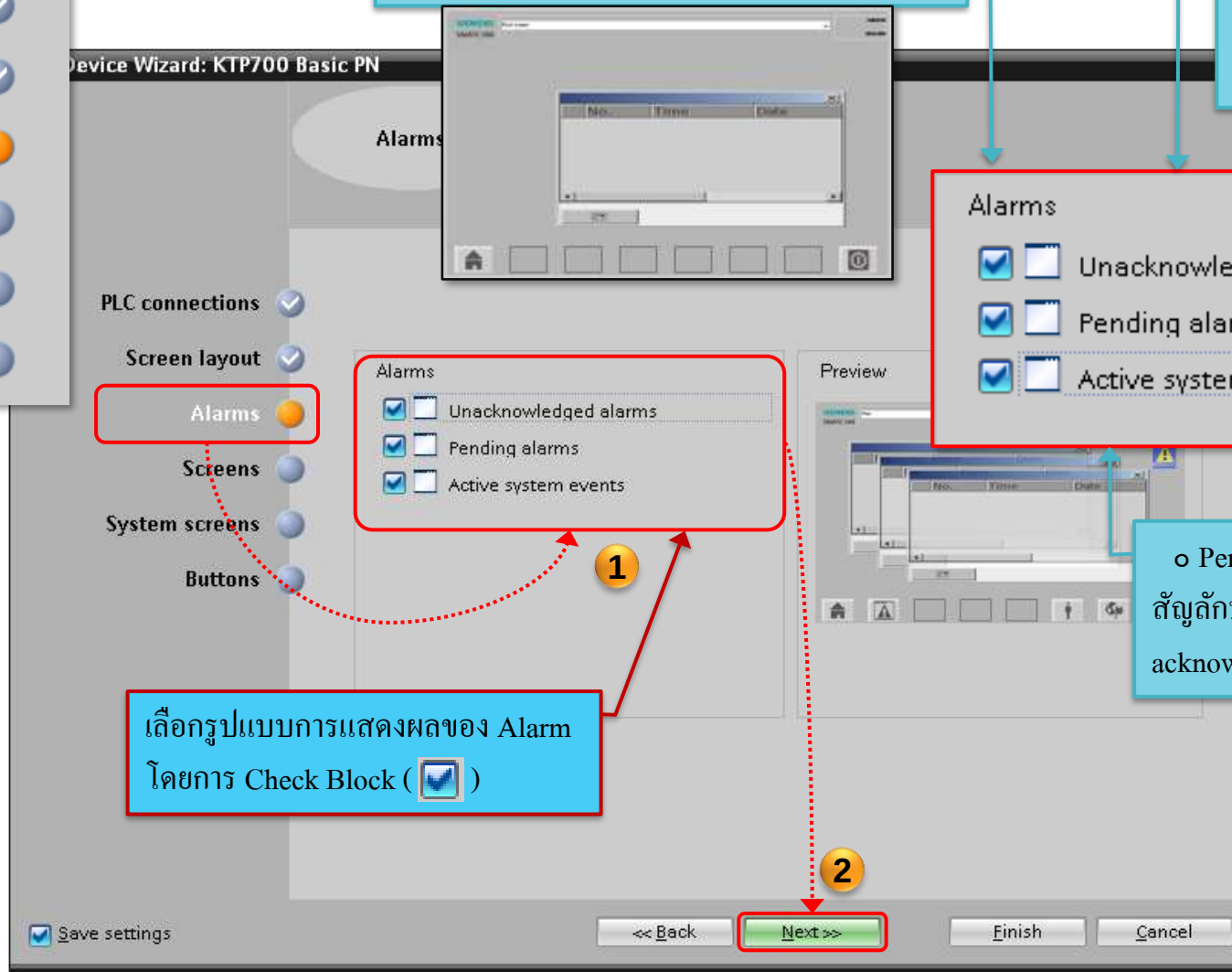
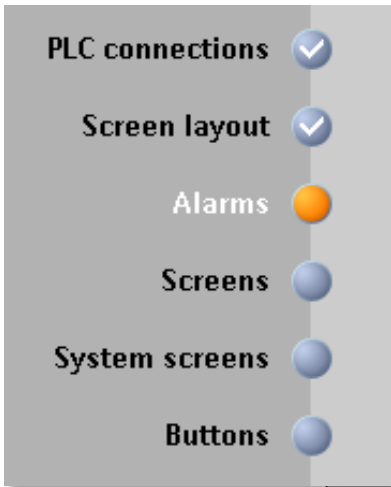
Approved
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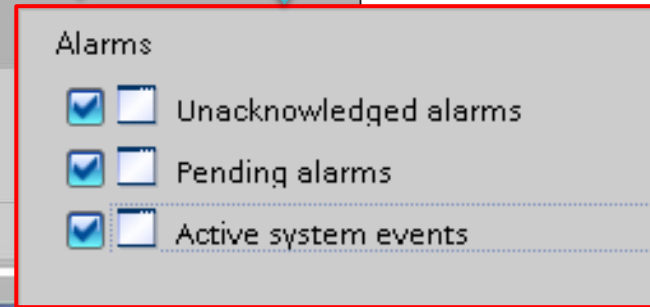
Factory
Automation

o Active system events คือ บอกระบบที่ใช้งาน

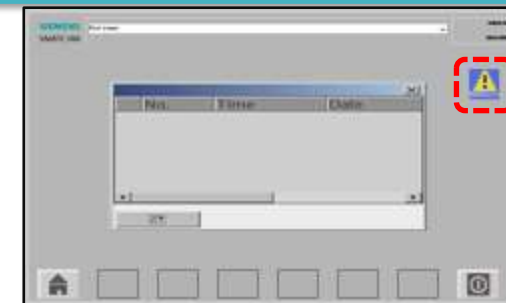
o Unacknowledged คือ เมื่อเกิด Alarm ขึ้นจะมี Pop up ขึ้นมาแจ้งเตือน และเมื่อกด acknowledge หน้าต่าง Pop up ก็จะหายไป แต่จะมีการแจ้งเตือนอยู่ตรงช่องปุ่มและจะหายไปเมื่อกำหนดการเกิด Alarm



เลือกรูปแบบการแสดงผลของ Alarm
โดยการ Check Block (☒)



o Pending คือ เมื่อเกิด Alarm ขึ้นจะมี Pop up ขึ้นมาพร้อมสัญลักษณ์ Alarm และ Pop up จะคงอยู่ไม่หายไปจนกว่าจะกด acknowledge และค่านั้นไม่อยู่ในขอบเขตการเกิด Alarm



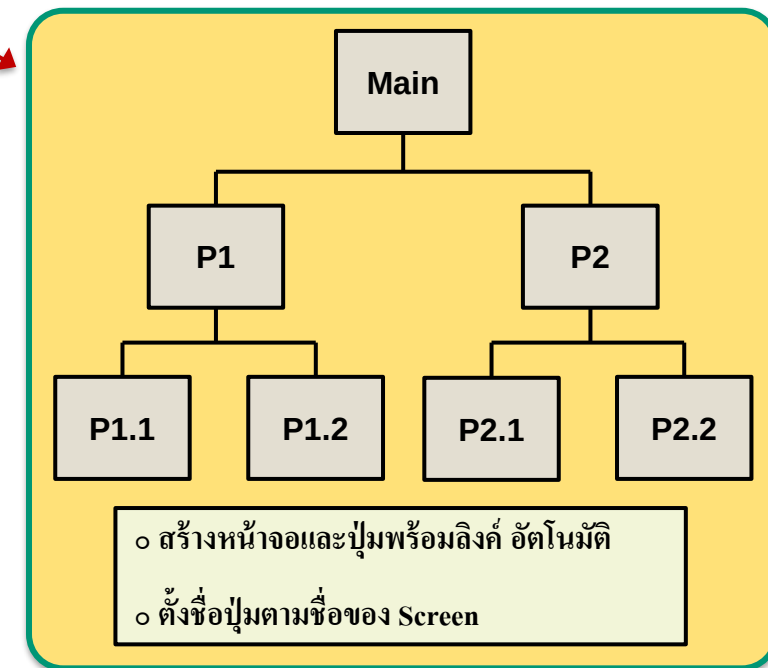
HMI Wizard Screen (4)

Approved
Partner

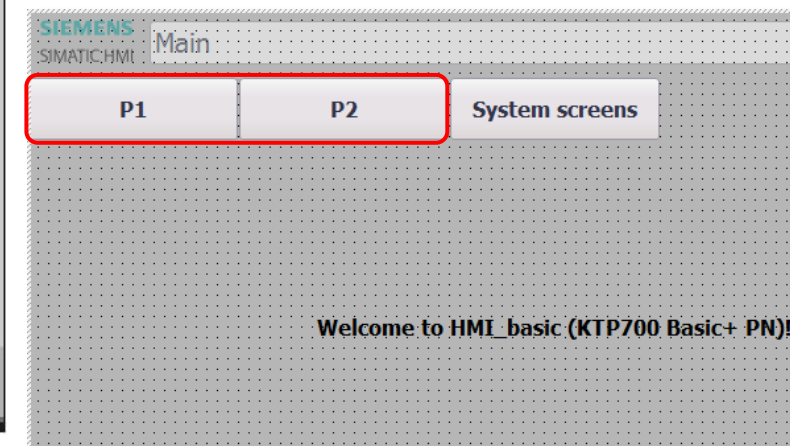
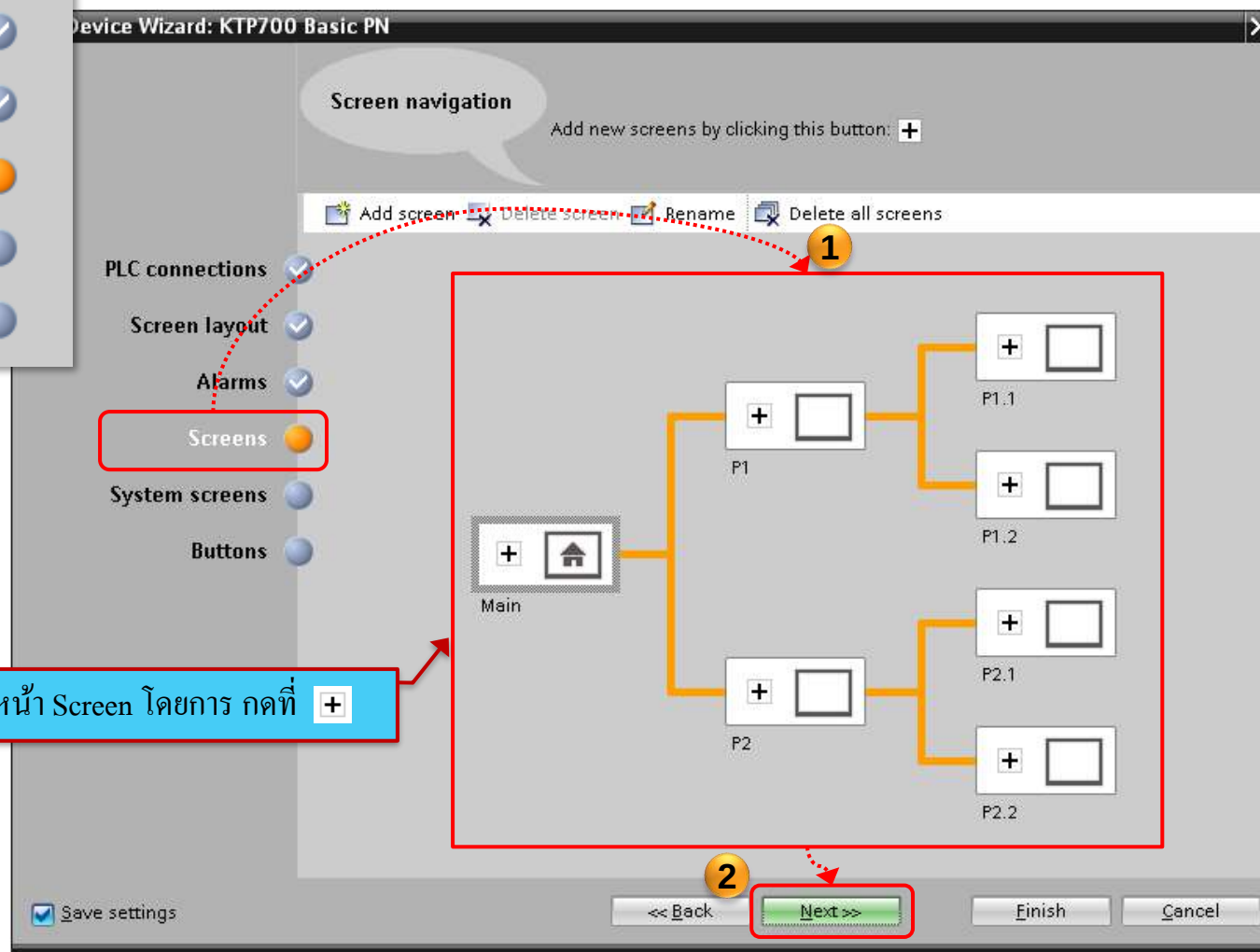
Factory
Automation

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ตัวอย่าง Flowchart ของหน้าจอ



เพิ่มหน้า Screen โดยการ กดที่ +



HMI Wizard System Screen (5)

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PLC connections ✓

Screen layout ✓

Alarms ✓

Screens ✓

System screens ○

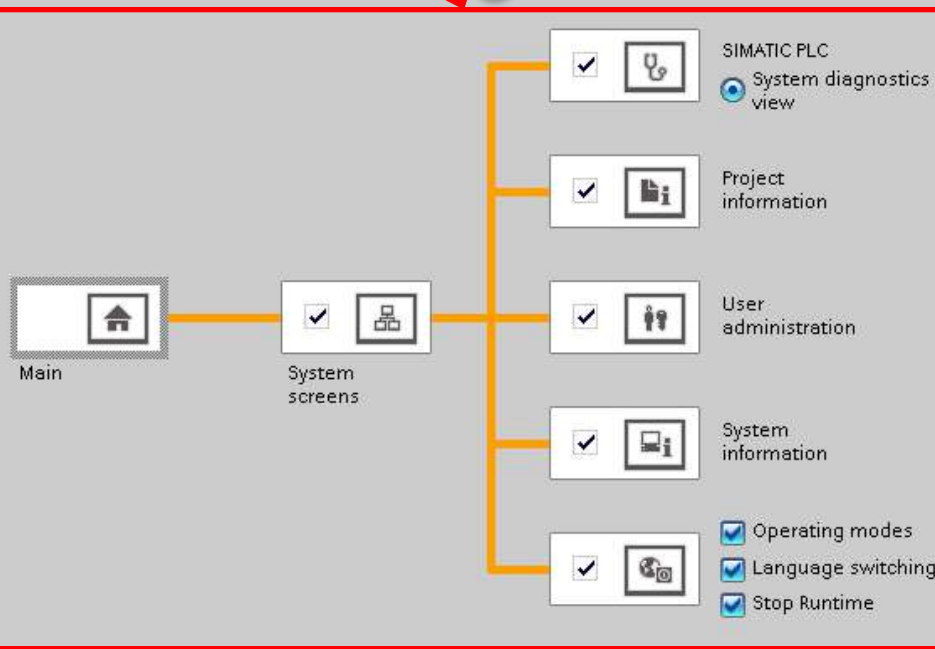
Buttons ○

Device Wizard: KTP700 Basic PN

System screens

Select the system screens.

1



เลือก System Screen โดยการ กดที่ ☒

☒ Select all

☒ Save settings

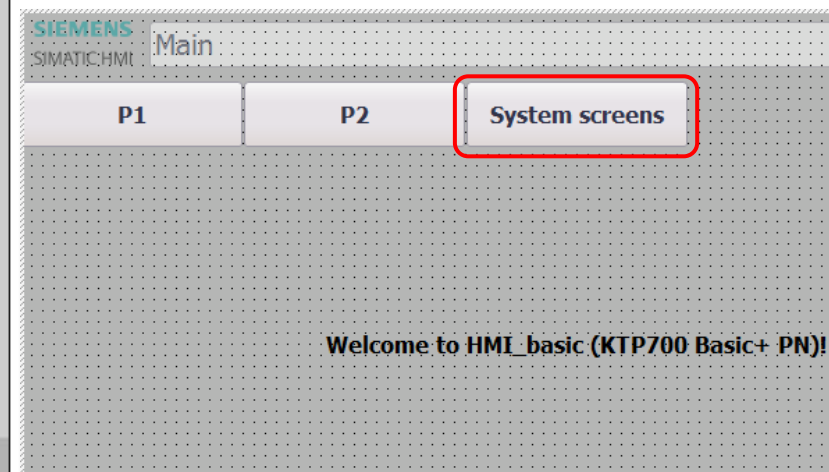
<< Back

Next >>

Finish

Cancel

☒		SIMATIC PLC	ดู Diagnostics ได้
☒		Project information	ดูข้อมูลรายละเอียดของ Project
☒		User administration	หน้าต่างของ User Administration
☒		System information	ข้อมูลรายละเอียดของระบบ
☒		Different jobs	- โหมดการทำงาน - การเปลี่ยนภาษา - หยุดการทำงาน



HMI Wizard Button (6)

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○ Log on คือ ปุ่มเข้ารหัส

○ Language คือ ปุ่มเปลี่ยนภาษา

○ Start screen คือ หน้าเริ่มต้นของจอ

○ Exit คือ ออกจากหน้าจอ

เลือกปุ่มที่จะใช้งานบนหน้าจอ
โดยลากไปวาง หรือ Double Click

ลักษณะรูปแบบของปุ่ม

HMI Wizard Button (6)

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PLC connections ✓

Screen layout ✓

Alarms ✓

Screens ✓

System screens ✓

Buttons ○

PLC connections ✓

Screen layout ✓

Alarms ✓

Screens ✓

System screens ✓

Buttons ○

○ Bottom (ด้านล่างของจอ)

เลือก ตำแหน่งของปุ่ม

Button area

☐ Left

☒ Bottom

☐ Right

Reset all

Finish

○ Left (ด้านซ้ายของจอ)

○ Right (ด้านขวาของจอ)

Siemens - C:\Users\labuser\Documents\Automation\Basic workshop\Basic workshop

Project Edit View Insert Online Options Tools Window Help

Basic workshop > HMI_basic [KTP700 Basic PN] > Screens > Main

Visualization

Devices

- Basic workshop
 - Add new device
 - Devices & networks
 - PLC_1 [CPU 1214C AQ/DQ/Rly]
 - HMI_basic [KTP700 Basic PN]
 - Device configuration
 - Online & diagnostics
 - Runtime settings
 - Screens
 - Add new screen
 - Different jobs
 - Main
 - P1
 - P1.1
 - P1.2
 - P2
 - P2.1
 - P2.2
 - Project information
 - SIMATIC PLC system diagnostics
 - System information
 - System screens
 - User administration
 - Screen management
 - HMI tags
 - Connections
 - HMI alarms
 - Recipes
 - Historical data
 - Scheduled tasks
 - Text and graphic lists
 - User administration
 - Common data
 - Documentation settings
 - Languages & resources
 - Online access
 - Card Reader/USB memory

Annotations:

- หน้าจอที่สร้างขึ้นตาม Flowchart (Screen created according to Flowchart)
- ปุ่มที่ลิงค์ไปยังหน้าอื่นๆ (Buttons that link to other screens)
- หน้าจอของ System Screen (System Screen screen)

Properties

General

Pattern

Name: Main

Background color: 181, 182, 181

Grid color: 0, 0, 0

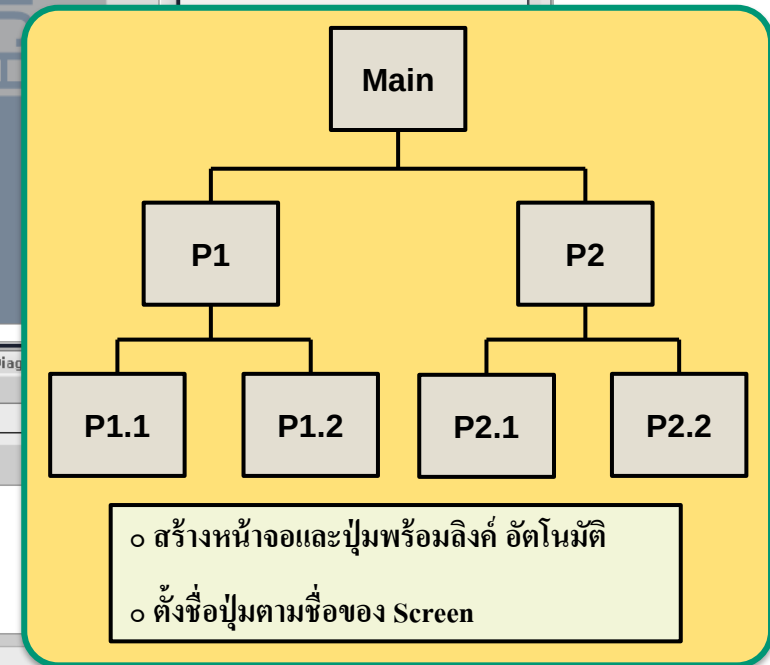
Number: 1

Template: Template_1

Tooltip

Portal view Overview PLC_1 Main

Wizard: successfully configured KTP700...



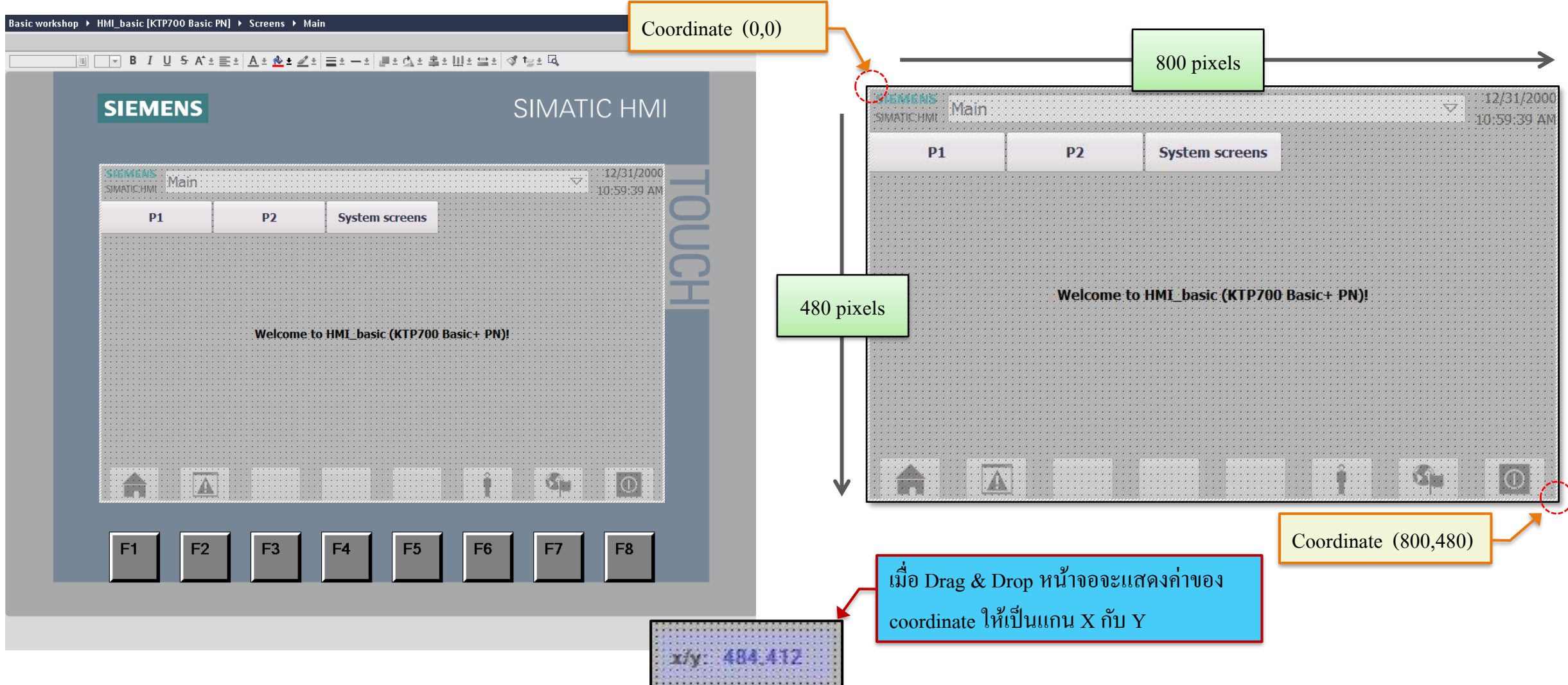
HMI Wizard Coordinate & Ruler

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Ex. KT700 Comfort : Resolution (W*H in pixels) = 800 * 480 pixels



HMI Wizard Grid

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The screenshot shows the Siemens HMI Wizard software interface. The main workspace displays a screen layout with buttons labeled 'P1', 'P2', and 'System screens'. A red box highlights the 'Layout' mode in the 'Options' panel on the right. A red arrow points from the 'Layout' mode to the 'Grid' settings dialog. The 'Grid' dialog has a 'Layout mode' section with three radio buttons: 'Snap to lines' (selected), 'Snap to grid', and 'None'. Below this is a 'Grid' section with a checked 'Show grid' checkbox. A red box highlights the 'Show grid' checkbox and the 'X' and 'Y' grid size input fields, which are both set to 10. A red arrow points from the 'Show grid' checkbox to the 'เลือกเพื่อแสดง Grid' (Select to show Grid) text. Another red arrow points from the 'X' and 'Y' input fields to the 'ในกรณีไม่เลือกแสดง Grid' (In case not selected to show Grid) text. The 'Project tree' on the left shows the project structure, including 'Main' and 'Screens'.

ในกรณีไม่เลือกแสดง Grid

เลือกเพื่อแสดง Grid

Layout

Grid

Layout mode

☒ Snap to lines

☐ Snap to grid

☐ None

Grid

☒ Show grid

X: 10

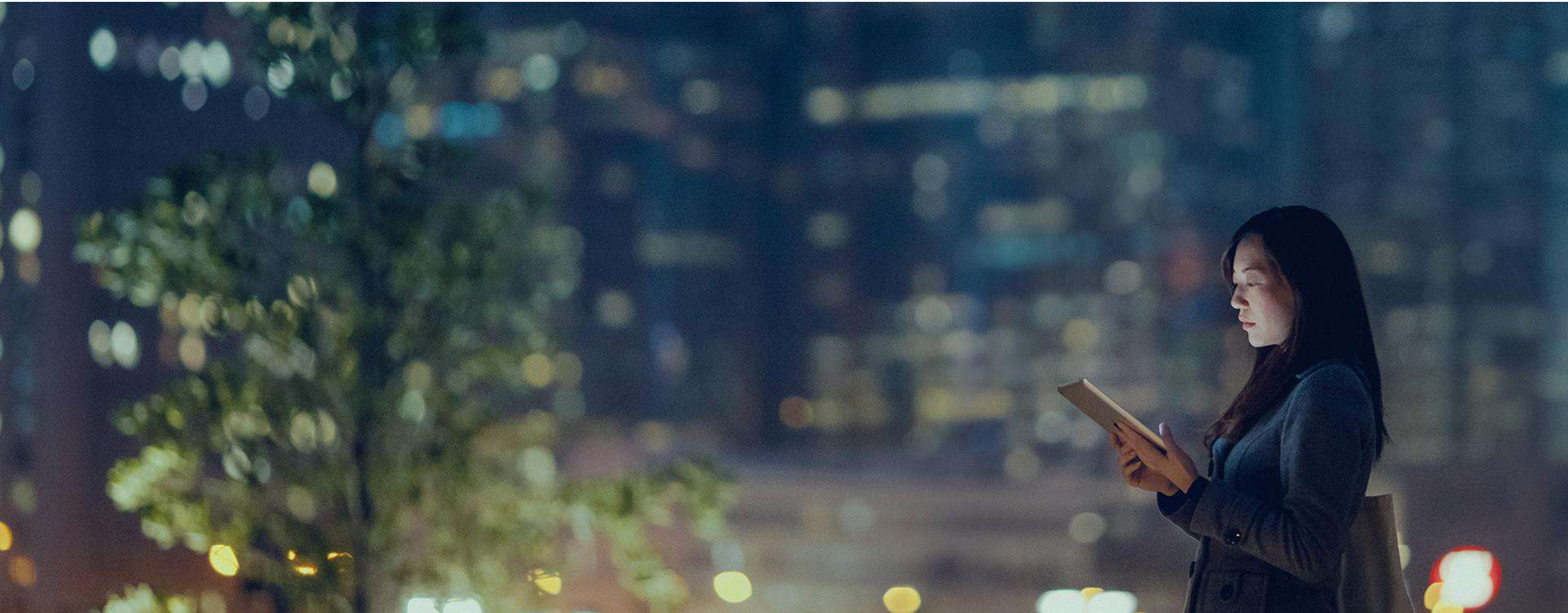
Y: 10

02 - Graphic User Interface

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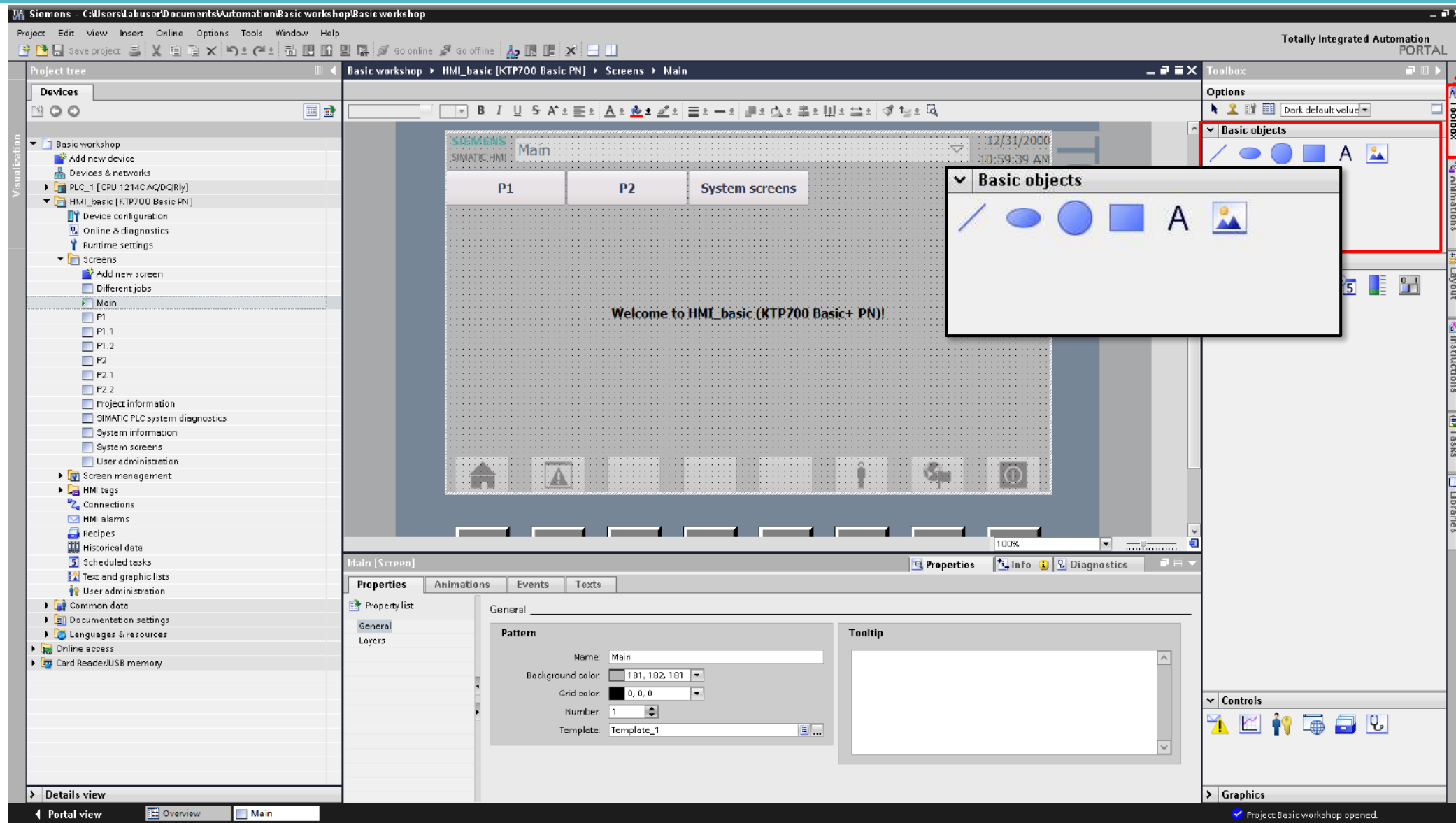


HMI Basic Object

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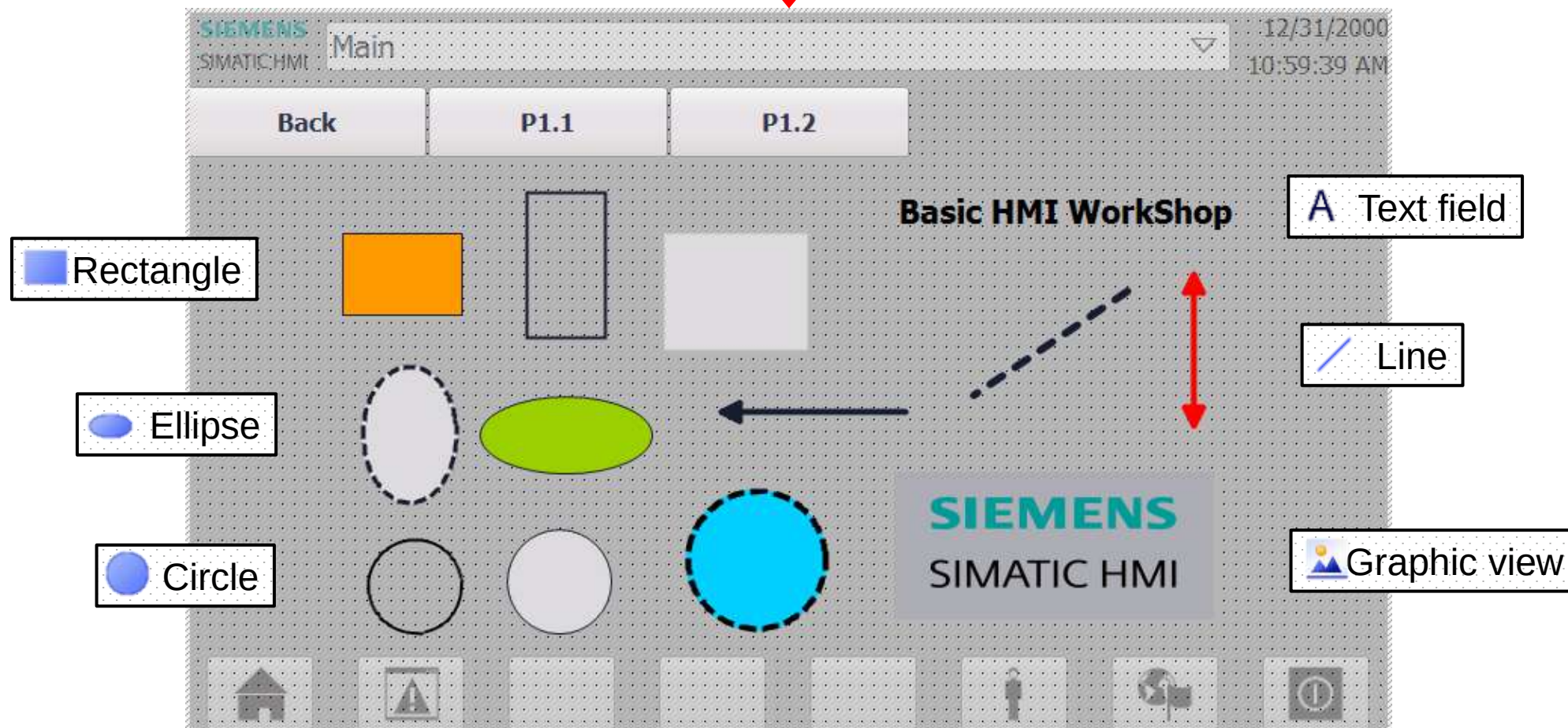
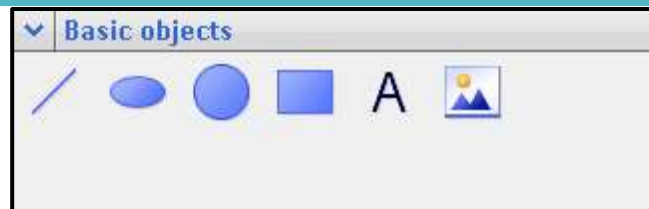


Basic Object

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Basic Object Line (Appearance)

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The screenshot displays the 'Basic objects' palette on the left, where the 'Line' icon is highlighted with a green arrow and a yellow circle labeled '1'. The main window shows the 'Line_1 [Line]' object with the 'Properties' tab selected, indicated by a yellow circle labeled '2'. The 'Properties' tab is further divided into 'Property list', 'Appearance', 'Layout', 'Style/Designs', and 'Miscellaneous', with 'Appearance' selected and marked by a yellow circle labeled '3'. The 'Appearance' section contains two main areas: 'Line' and 'Line ends'. The 'Line' section, marked with a yellow circle labeled '4', includes settings for 'Width' (set to 4), 'Style' (set to 'Solid'), 'Color' (set to 255, 0, 0), 'Background color' (set to 255, 255, 255), and 'Fill style' (set to 'Transparent'). A yellow circle labeled '5' points to the 'Style' dropdown. The 'Line ends' section, marked with a yellow circle labeled '6', includes 'Start' and 'End' dropdowns (both set to 'Default') and a 'Line end shape' dropdown (set to 'Round'). Below the 'Line' section, there are two preview boxes showing a solid line and a dashed line, with labels 'o Solid (เส้นทึบ)' and 'o Dash (เส้นประ)' respectively. Below the 'Line ends' section, there are two preview boxes showing a line with a round end and a line with a flush end, with labels 'o Round (ปลายมน)' and 'o Flush (ปลายตัด)' respectively. Red dotted lines connect the 'Style' dropdown to the line style preview boxes and the 'Line end shape' dropdown to the line end shape preview boxes. Thai text labels are present: 'เลือกขนาดของเส้น' (Select line width) points to the 'Width' field; 'เลือกหัวลูกศรหรือเส้นธรรมดา' (Select arrow or normal line) points to the 'Line ends' section.

Basic objects

Line_1 [Line]

Properties

Property list

Appearance

Layout

Style/Designs

Miscellaneous

Line

Width: 4

Style: Solid

Color: 255, 0, 0

Background color: 255, 255, 255

Fill style: Transparent

Line ends

Start: Default

End: Default

Line end shape: Round

Round

Flush

o Solid (เส้นทึบ)

o Dash (เส้นประ)

o Round (ปลายมน)

o Flush (ปลายตัด)

เลือกขนาดของเส้น

เลือกหัวลูกศรหรือเส้นธรรมดา

Basic Object Line (Layout)

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Partner

Factory
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ตำแหน่งของเส้นที่จะแสดงบนจอ

ขนาดของเส้น

ตำแหน่งของจุดเริ่มต้นเส้น

ตำแหน่งปลายเส้น

เส้น ขาว 75 และสูง 90 pixels

Properties

Animations

Events

Texts

Property list

Appearance

Layout

Style/Designs

Miscellaneous

Layout

Position & size

X: 500

Y: 200

Width: 75

Height: 90

Start point

X: 500

Y: 200

End point

X: 575

Y: 290

SIEMENS SIMATIC HMI

Main

P1

P2

System screens

12/31/2000 10:59:39 AM

X = 500

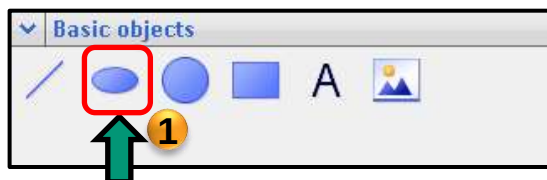
Y = 200

Basic Object Ellipse (Appearance)

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Factory
Automation

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Properties dialog box for Ellipse_1 [Ellipse]. The dialog is divided into tabs: Properties, Animations, Events, and Texts. The Properties tab is active, showing the Appearance section. The Appearance section includes a Property list on the left and a main area for configuring the ellipse's appearance. The main area is divided into Background and Border sections. The Background section includes a Color picker (set to 156, 207, 0) and a Fill pattern dropdown (set to Solid). The Border section includes a Width input (set to 4), a Style dropdown (set to Solid), and a Color picker (set to 0, 0, 255). Annotations in Thai point to various elements: 1. Ellipse icon in the toolbar. 2. Properties tab. 3. Appearance section. 4. Property list. 5. Color picker. 6. Width input. 7. Style dropdown. 8. Color picker. 9. Fill pattern dropdown. 10. Solid (สีทึบ) and Transparent (โปร่งแสง) options. 11. Solid (เส้นทึบ) and Dash (เส้นประ) options.

Annotations in Thai:

- เลือกสีของวงรี (Select ellipse color)
- ขนาดเส้นขอบวงรี (Ellipse border size)
- ลักษณะเส้นของวงรี (Ellipse line style)
- Transparent (โปร่งแสง)
- Solid (สีทึบ)
- Solid (เส้นทึบ)
- Dash (เส้นประ)

Basic Object Ellipse (Layout)

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The image shows the Siemens SIMATIC Manager interface for creating an ellipse object. The main window is titled "Ellipse_1 [Ellipse]". The left sidebar shows the "Properties" tab with sub-tabs: "Property list", "Appearance", "Layout", "Style/Designs", and "Miscellaneous". The "Layout" sub-tab is selected, showing the "Position & size" section with the following values:

- X: 500
- Y: 200
- Width: 60
- Height: 90

The "Radius" section shows the following values:

- Horizontal: 30
- Vertical: 45

Annotations in Thai explain the fields:

- ตำแหน่งของเส้นที่จะแสดงบนจอ (Position of the line to be displayed on the screen) - points to the X and Y coordinates.
- ขนาดของวงรี (Ellipse size) - points to the Width and Height fields.
- รัศมีของวงรีในแนวนอน (Horizontal) และแนวตั้ง (Vertical) (Horizontal and vertical radius of the ellipse) - points to the Horizontal and Vertical radius fields.
- ขนาดของวงรีและรัศมี มีความแปรผันต่อกัน (Ellipse size and radius are interdependent) - points to the Radius section.
- วงรี กว้าง 60 และสูง 90 pixels (Ellipse 60 wide and 90 high pixels) - points to the Width and Height fields.

The bottom left shows a preview of the ellipse on a grid, with the X and Y coordinates displayed. The top right shows the "Properties" and "Info" tabs.

Basic Object Circle (Appearance)

Approved
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Factory
Automation

SIEMENS

Basic objects

1

2

3

4

5

6

Properties

Info

Diagnostics

Circle_1 [Circle]

Properties

Animations

Events

Texts

Property list

Appearance

Layout

Style/Designs

Miscellaneous

เลือกสีของวงกลม

ขนาดเส้นขอบวงกลม

Background

Color: 0, 207, 255

Fill pattern: Solid

Transparent

Solid

Border

Width: 4

Style: --- Dash

Color: 255, 0, 255

Solid

Dash

o Transparent (โปร่งแสง)

o Solid (สีทึบ)

o Solid (เส้นทึบ)

o Dash (เส้นประ)

Basic Object Circle (Layout)

ตำแหน่งของเส้นที่จะแสดงบนจอ

ขนาดของวงกลมในรูปแบบรัศมี

7

8

9

Properties Animations Events Texts

Property list

Appearance

Layout

Style/Designs

Miscellaneous

Layout

Position & size

X: 500

Y: 200

100

100

Radius

Radius: 50

SIEMENS SIMATIC HMI Main: 12/31/2000 10:59:39 AM

P1 P2 System screens

Y = 200

X = 500

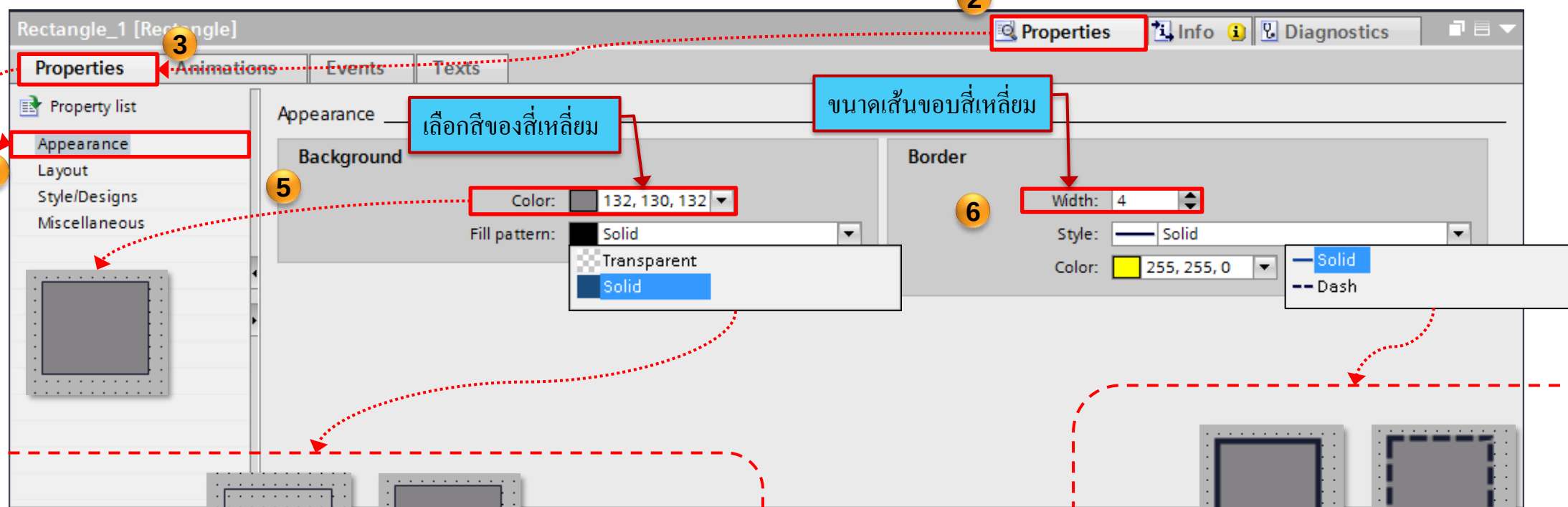
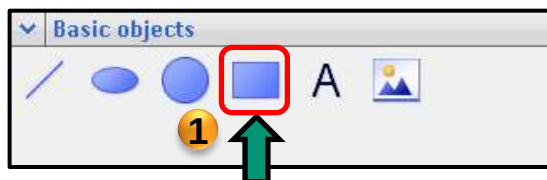
วงกลม รัศมี 50 pixels

Basic Object Rectangle (Appearance)

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o Transparent (โปร่งแสง)

o Solid (สีทึบ)

o Solid (เส้นทึบ)

o Dash (เส้นประ)

Basic Object Rectangle (Layout)

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มุมโค้งเป็นรัศมีแกน X=60% และ Y=40%

ตำแหน่งของเส้นที่จะแสดงบนจอ

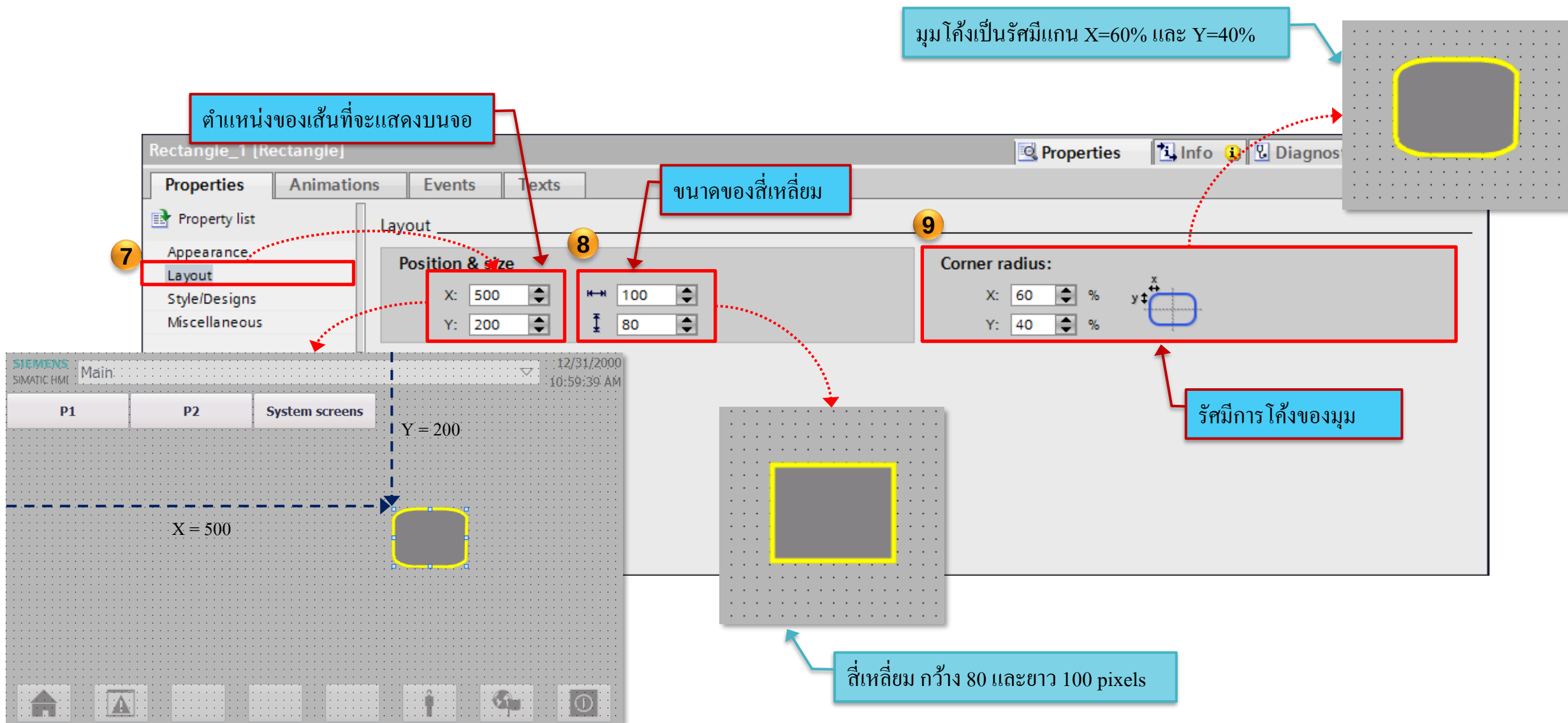
ขนาดของสี่เหลี่ยม

Corner radius:

X: 60 %
Y: 40 %

รัศมีการโค้งของมุม

สี่เหลี่ยม กว้าง 80 และยาว 100 pixels



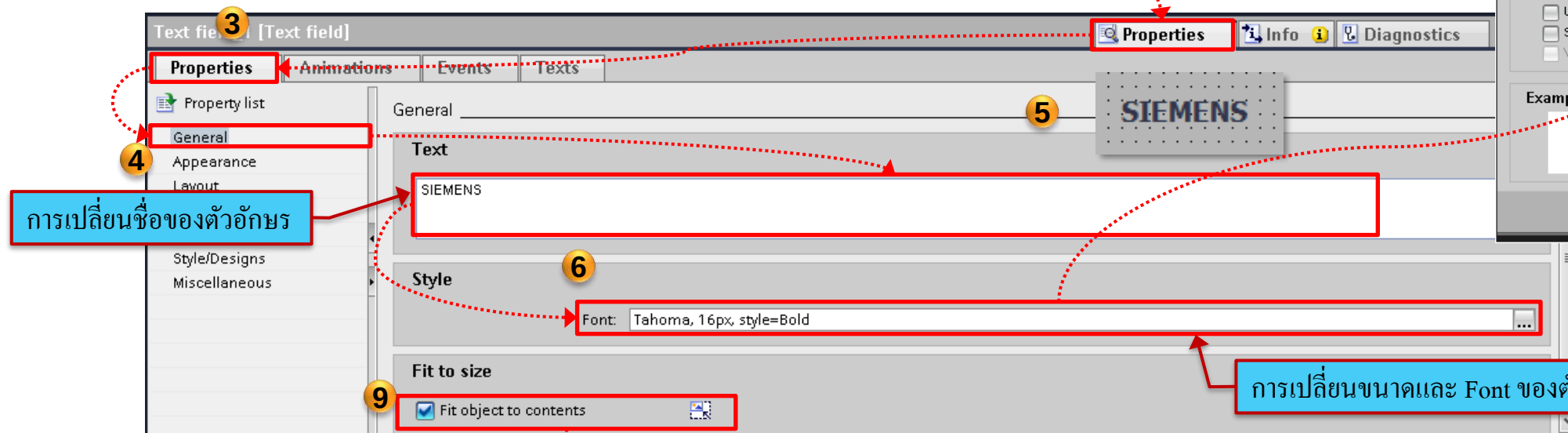
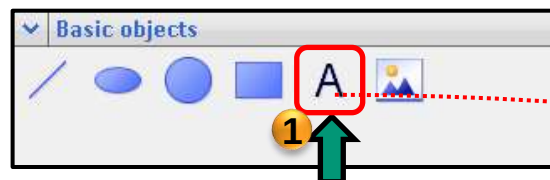
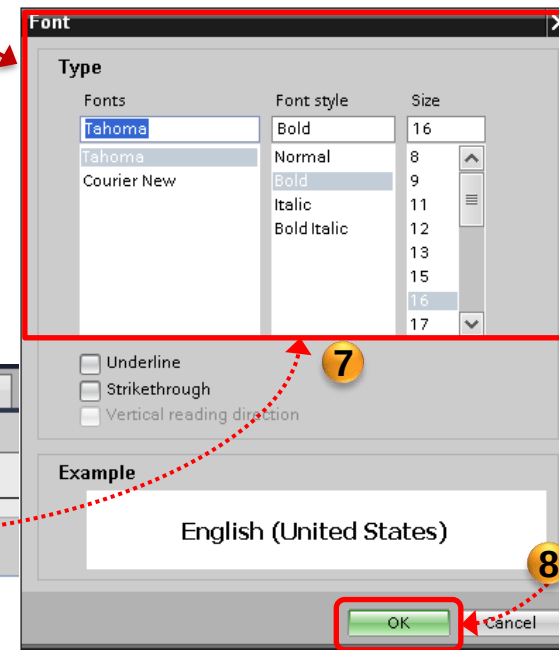
Basic Object Text

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เลือก Font และ ขนาด



การเปลี่ยนขนาดและ Font ของตัวหนังสือ

การเปลี่ยนชื่อของตัวอักษร



The image is a composite of two screenshots from the Siemens Logo Maker software, illustrating the steps to create a custom Siemens logo.

Top Screenshot: Appearance and Border Settings

- 1. Properties Panel:** The 'Appearance' tab is selected in the left-hand 'Properties' panel.
- 2. Background Color:** The 'Color' dropdown is set to yellow (255, 207, 0). A red box highlights this setting, with an arrow pointing to the yellow background of the 'SIEMENS' logo preview.
- 3. Text Color:** The 'Color' dropdown is set to blue (0, 0, 255). A red box highlights this setting, with an arrow pointing to the blue text of the 'SIEMENS' logo preview.
- 4. Border Width:** The 'Width' dropdown is set to 4. A red box highlights this setting, with an arrow pointing to the border of the 'SIEMENS' logo preview.
- 5. Border Style:** The 'Style' dropdown is set to 'Solid'. A red box highlights this setting, with an arrow pointing to the solid border of the 'SIEMENS' logo preview.
- 6. Border Color:** The 'Color' dropdown is set to red (255, 0, 0). A red box highlights this setting, with an arrow pointing to the red border of the 'SIEMENS' logo preview.

Bottom Screenshot: Layout Settings

- 7. Layout Tab:** The 'Layout' tab is selected in the left-hand 'Properties' panel.
- 8. Position & Size:** The 'X' position is set to 300 and the 'Y' position is set to 250. The width is 86 and the height is 31. A red box highlights these settings, with an arrow pointing to the 'SIEMENS' logo preview.
- 9. Fit to size:** The 'Fit object to contents' checkbox is checked.
- 10. Margins:** The 'Margins' section shows settings for the logo's placement within a container.
- 11. Preview:** A preview of the final logo is shown on the right, with the text 'SIEMENS' centered at X=300 and Y=250.

Text

Text Format

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The screenshot shows the 'Text field_1 [Text field]' properties window. The 'Properties' tab is active, and the 'Text format' section is selected in the left sidebar (marked with a red box and a yellow circle with the number 1). The 'Style' section contains a 'Font' property set to 'Tahoma, 16px, style=Bold' (marked with a red box and a yellow circle with the number 2, and a blue callout box stating 'สามารถเปลี่ยน Font และขนาดได้'). The 'Orientation' dropdown menu is open, showing options: 'Horizontal' (selected), 'Vertical, left', and 'Vertical, right' (marked with a red box and a yellow circle with the number 3). The 'Alignment' section shows 'Horizontal' and 'Vertical' settings. A blue callout box with the text 'รูปแบบการเรียงของตัวอักษร' points to the 'Orientation' dropdown. Below the screenshot, three examples of text orientation are shown within a dashed red box: 'Horizontal' (SIEMENS), 'Vertical, left' (SIEMENS), and 'Vertical, right' (SIEMENS). A red dotted arrow points from the 'Vertical, right' option in the dropdown to its corresponding example.

Text field_1 [Text field]

Properties Animations Events Texts

Property list

- General
- Appearance
- Layout
- Text format**
- Flashing
- Style/Designs
- Miscellaneous

Text format

Style

Font: Tahoma, 16px, style=Bold

Orientation: Horizontal

- Horizontal
- Vertical, left
- Vertical, right

Alignment

Horizontal: Vertical: Middle

รูปแบบการเรียงของตัวอักษร

สามารถเปลี่ยน Font และขนาดได้

Horizontal

Vertical, left

Vertical, right

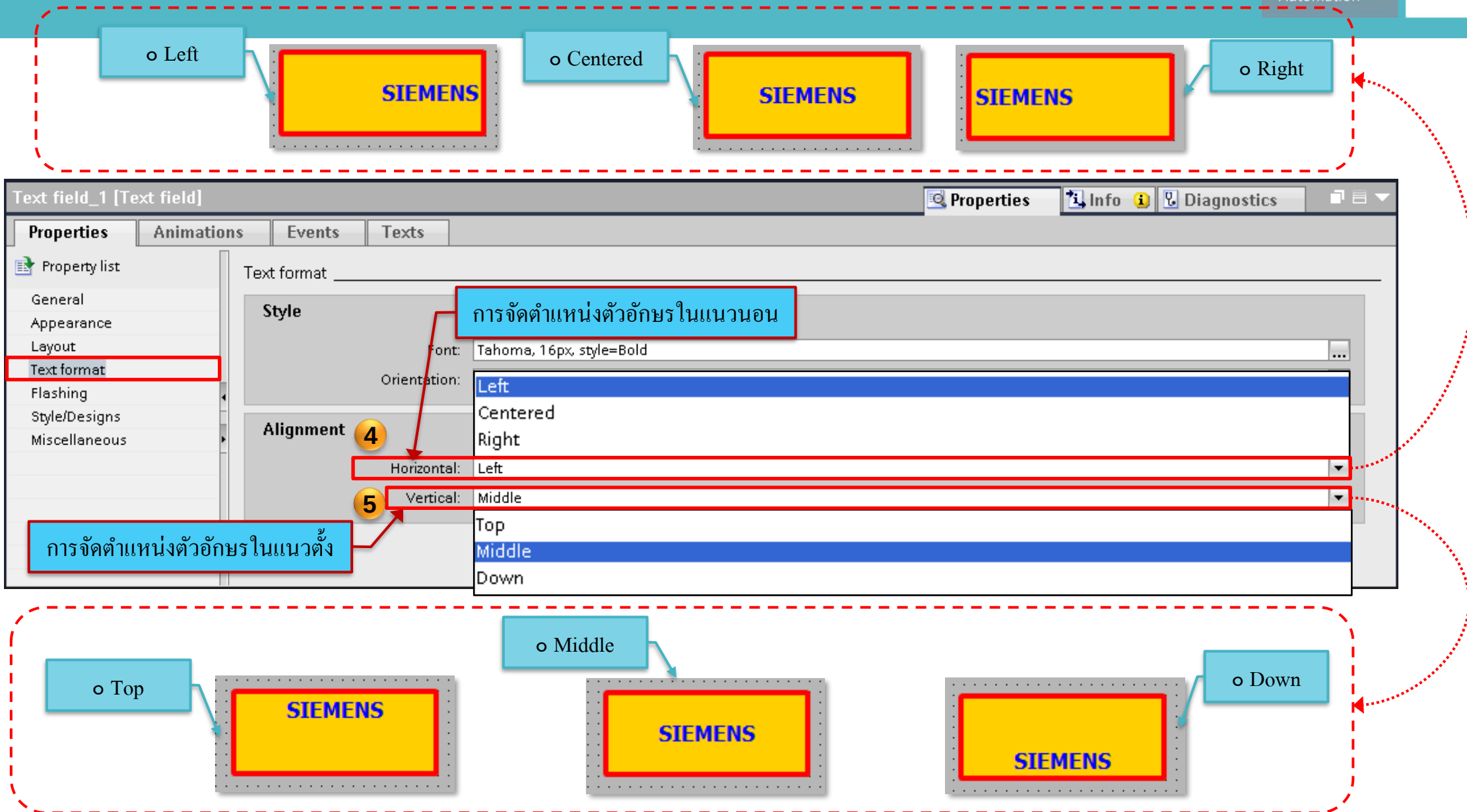
SIEMENS

SIEMENS

SIEMENS

Text

Text Format

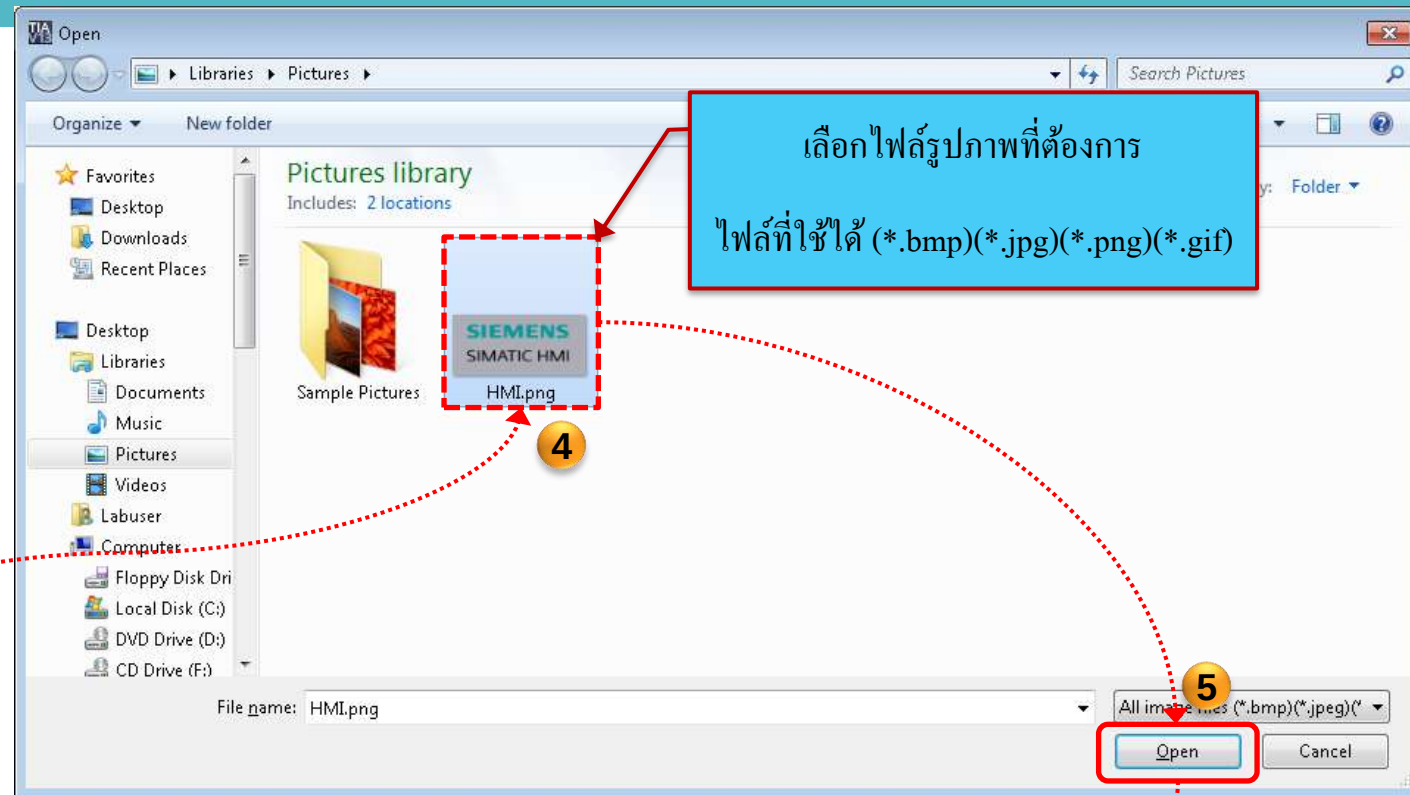
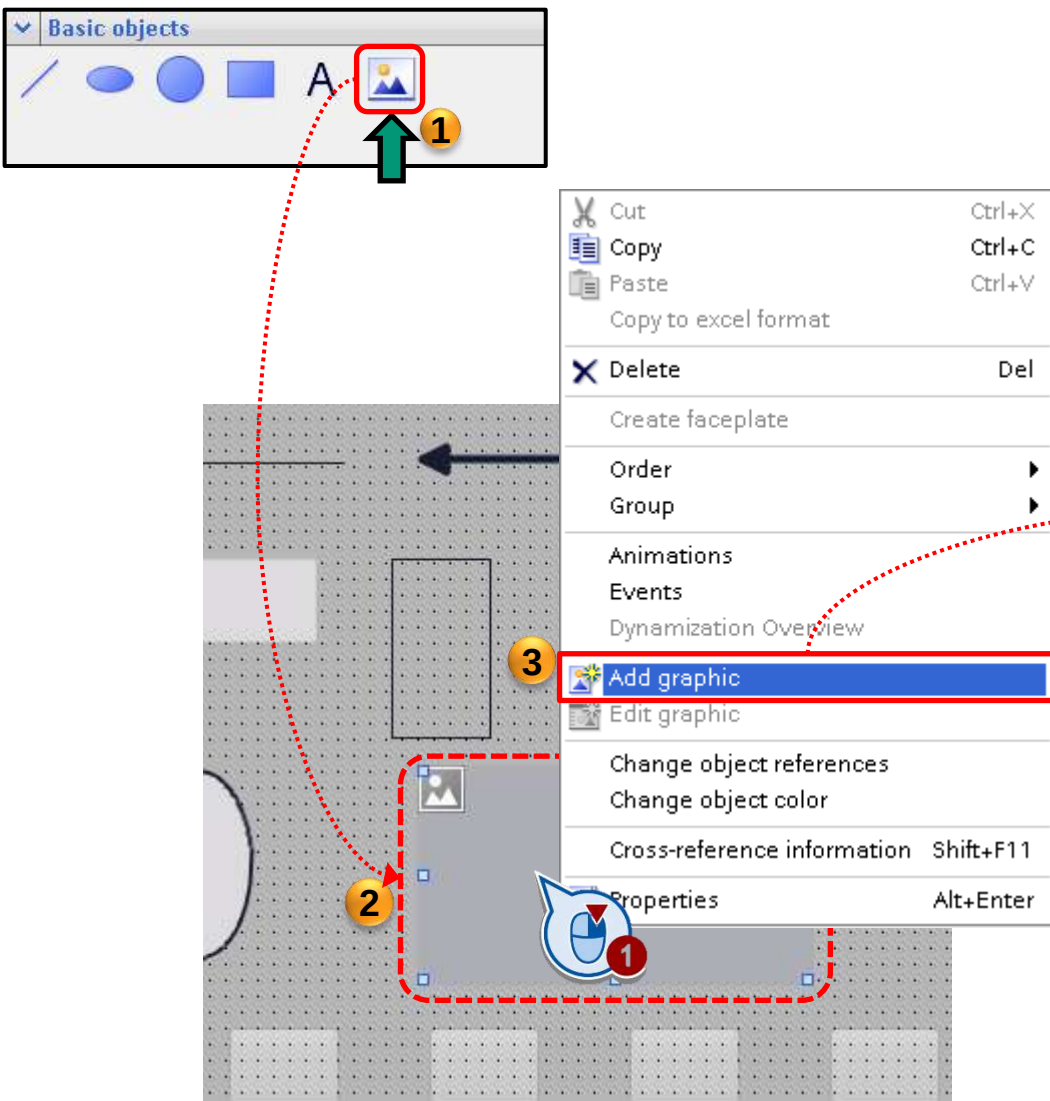


Basic Object Graphic view (Image file)

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SIEMENS

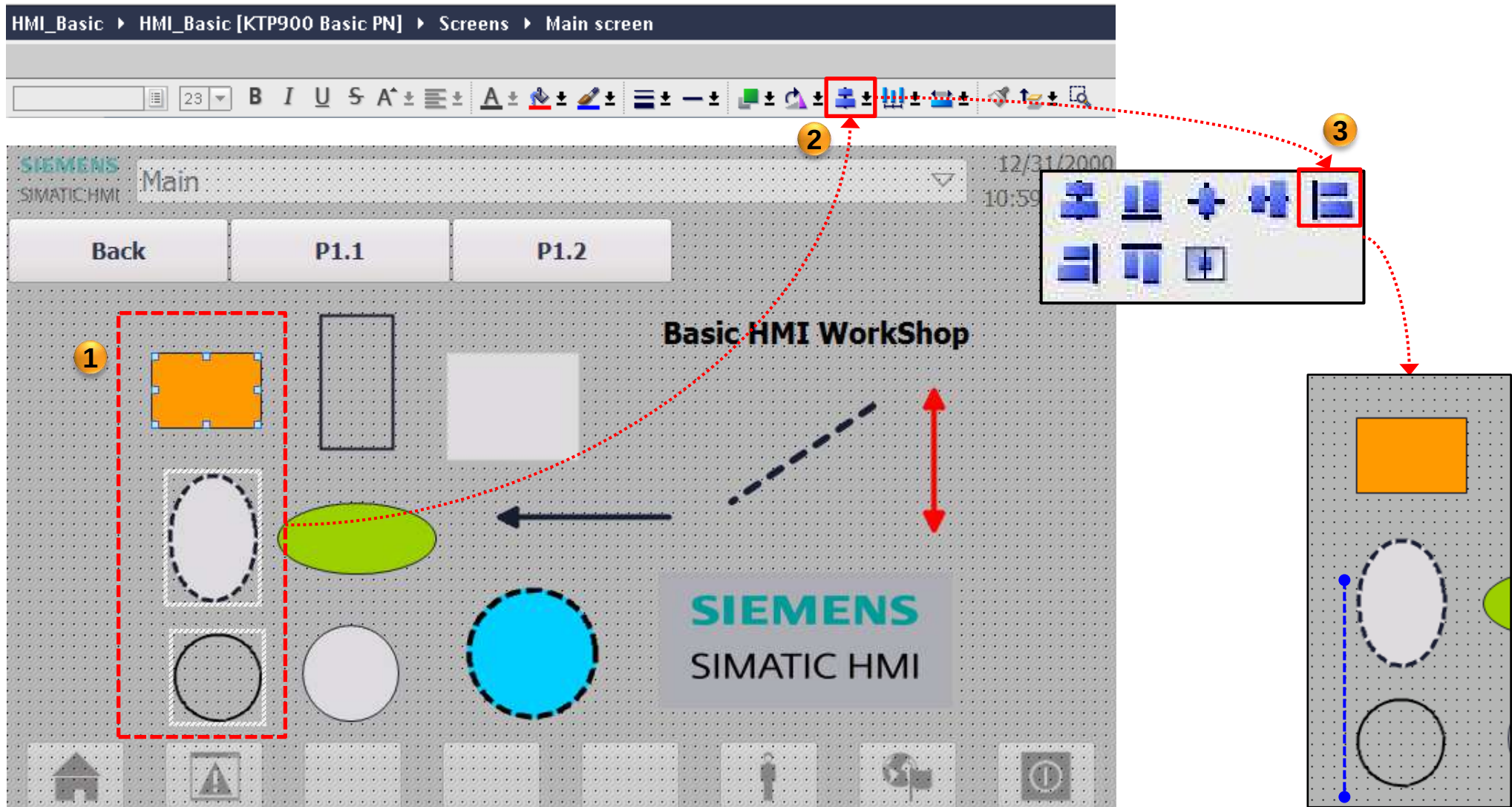


Alignment

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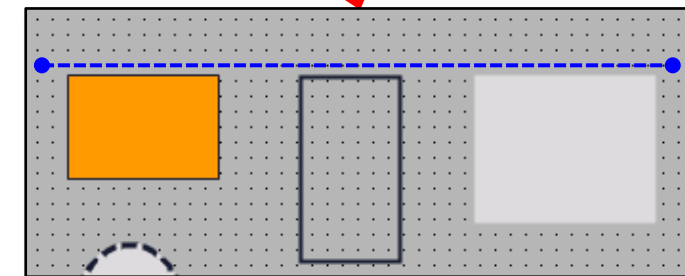
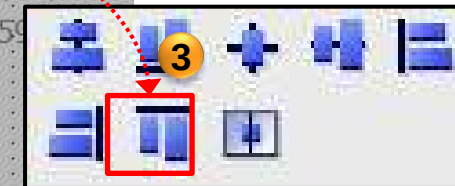
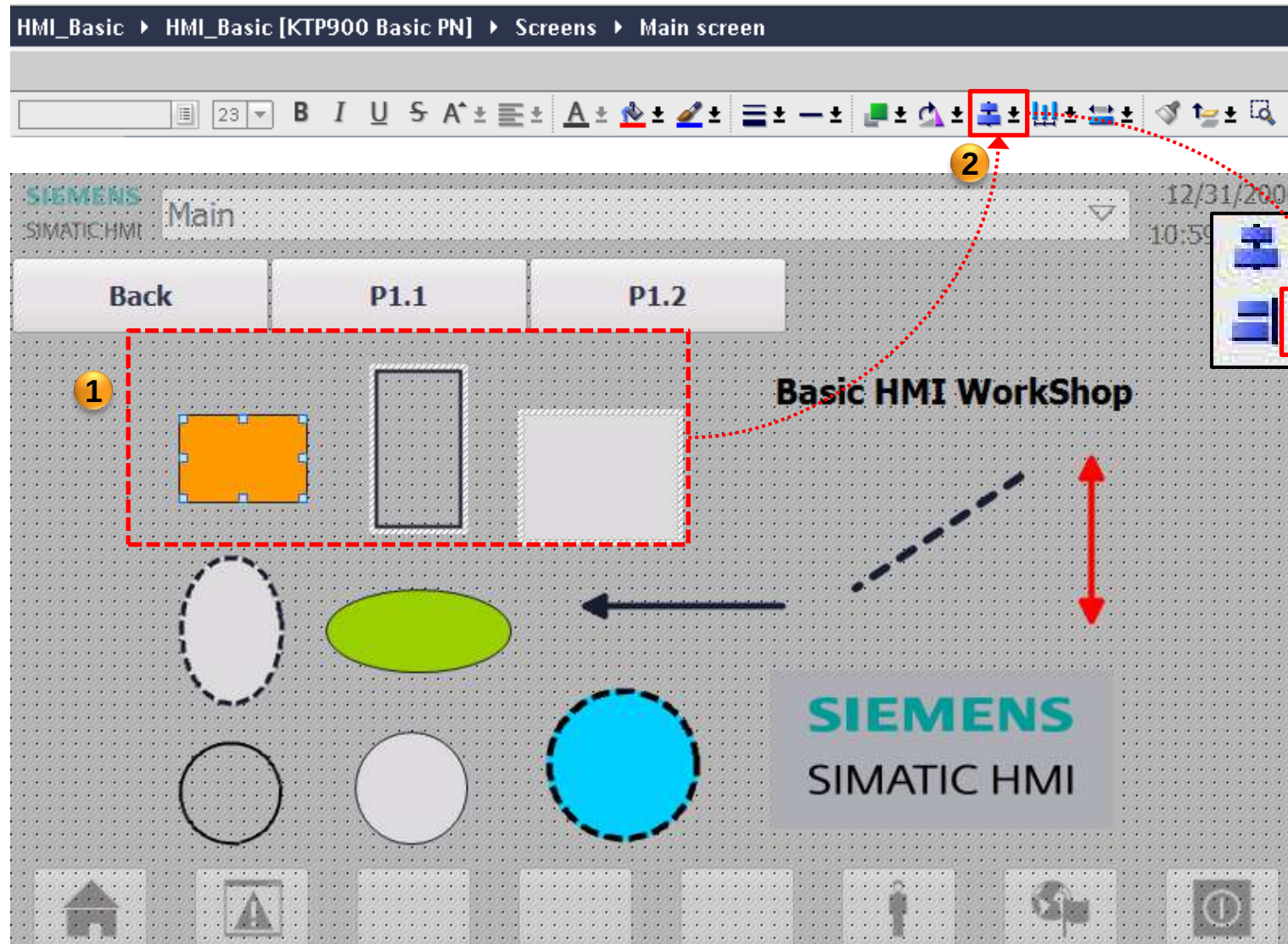


Alignment

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Automation

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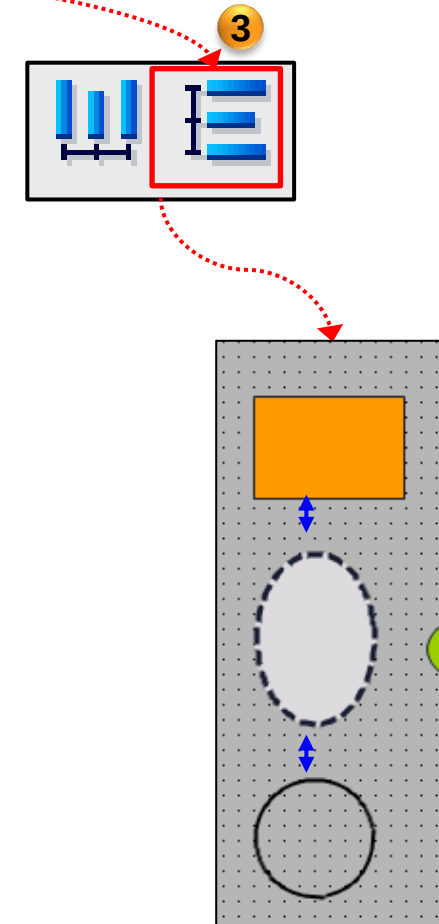
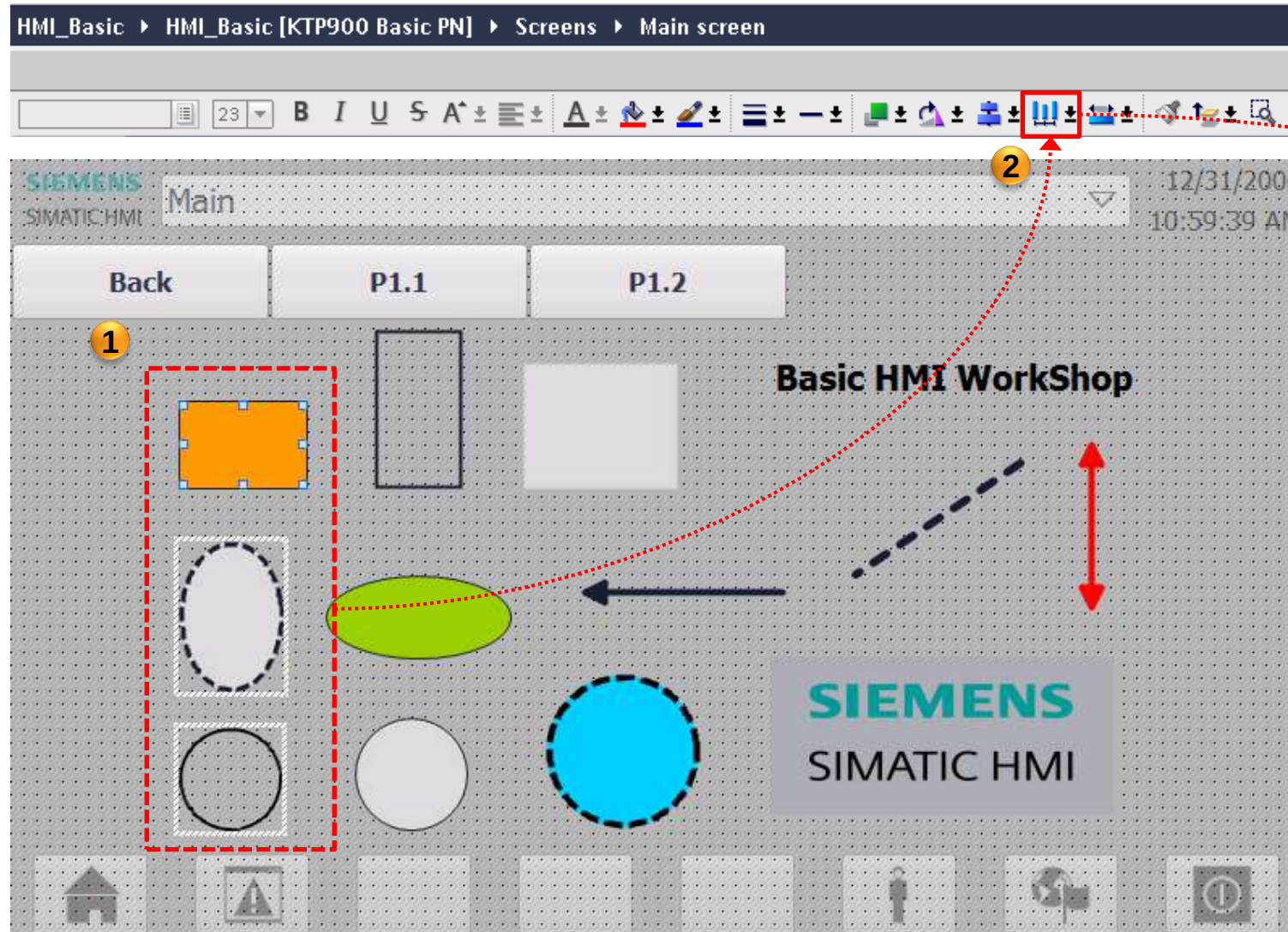


Alignment

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Factory
Automation

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Simulation

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Partner

Factory
Automation

SIEMENS

1

ในกรณีถ้า (สัญลักษณ์ Simulation)
ไม่ขึ้น ให้กดที่หน้าจอ 1 ครั้ง

Basic workshop > HMI_basic [KTP700 Basic PN] > Screens > P1

Basic HMI Workshop

SIEMENS SIMATIC HMI

Properties

General

Pattern

Name: P1

Background color: 181, 182, 181

Grid color: 0, 0, 0

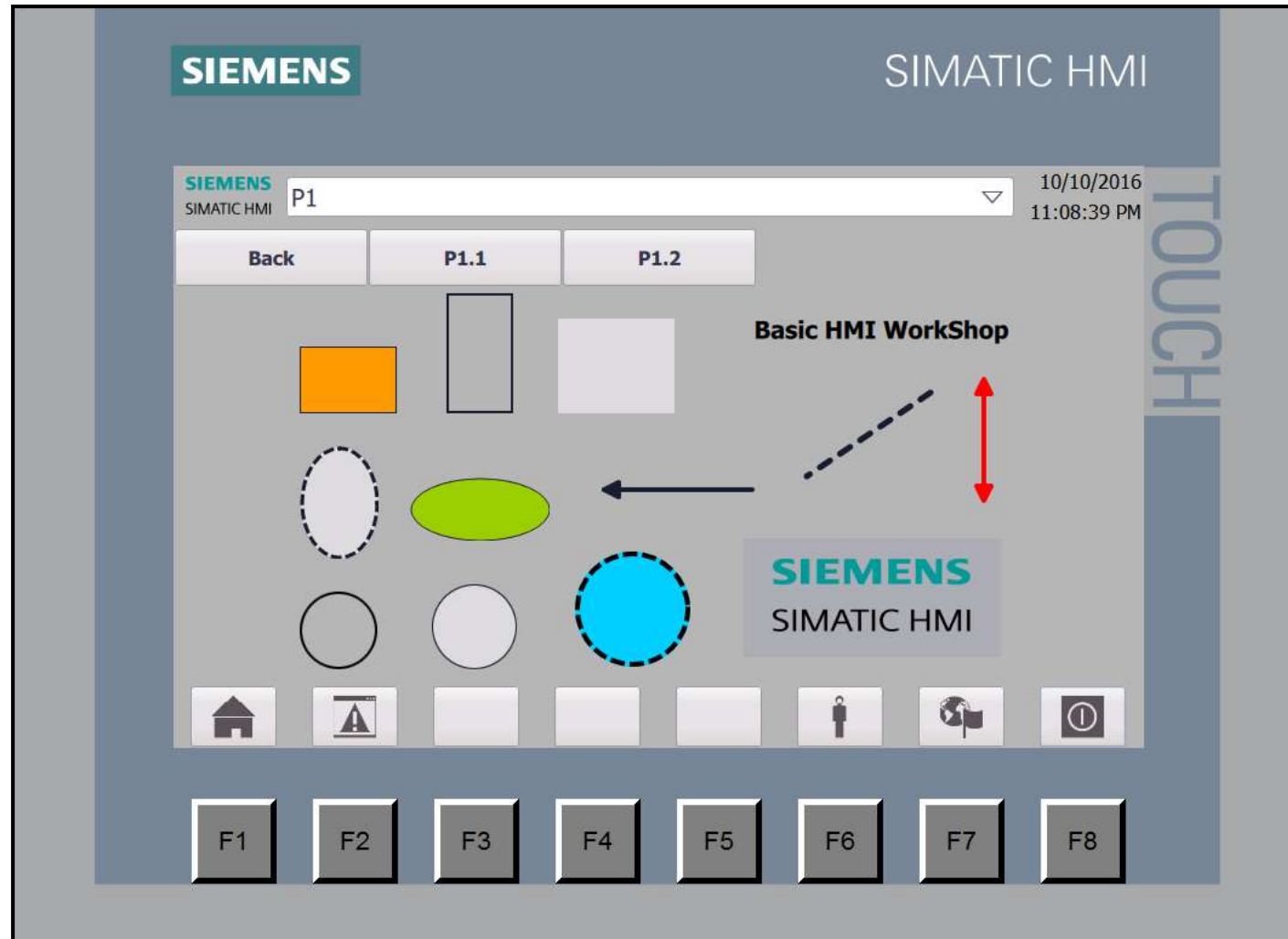
Number: 2

Template: Template_1

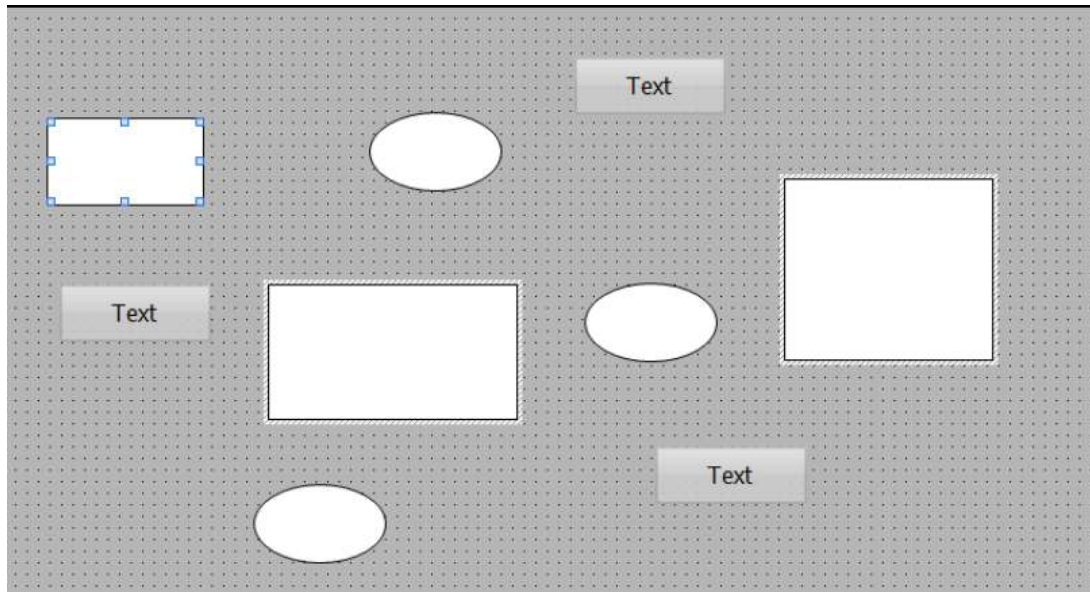
Tooltip

Portal view | Overview | Main | P1 | System screen...

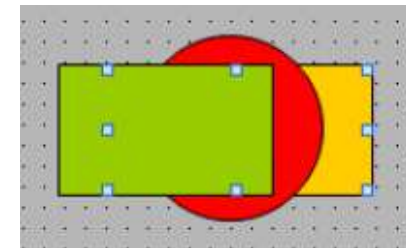
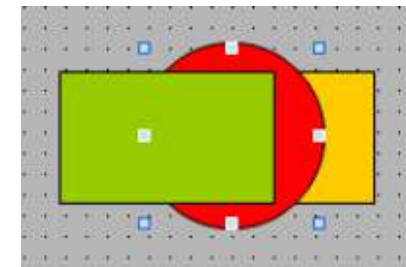
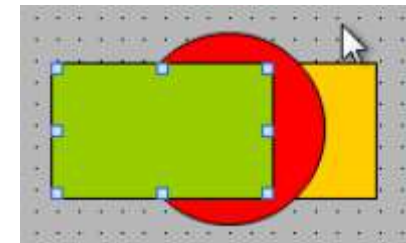
Project Basic workshop opened.



เลือกพาร์ทที่ต้องการแล้วกด กด Ctrl + Shift +A ผลที่ได้คือ
พาร์ทชนิดเดียวกันจะถูกเลือกไปด้วย



กด Alt + mouse click ไปเรื่อยๆเพื่อเลือก
พาร์ทที่ถูกทับซ้อนอยู่สลับกันไป

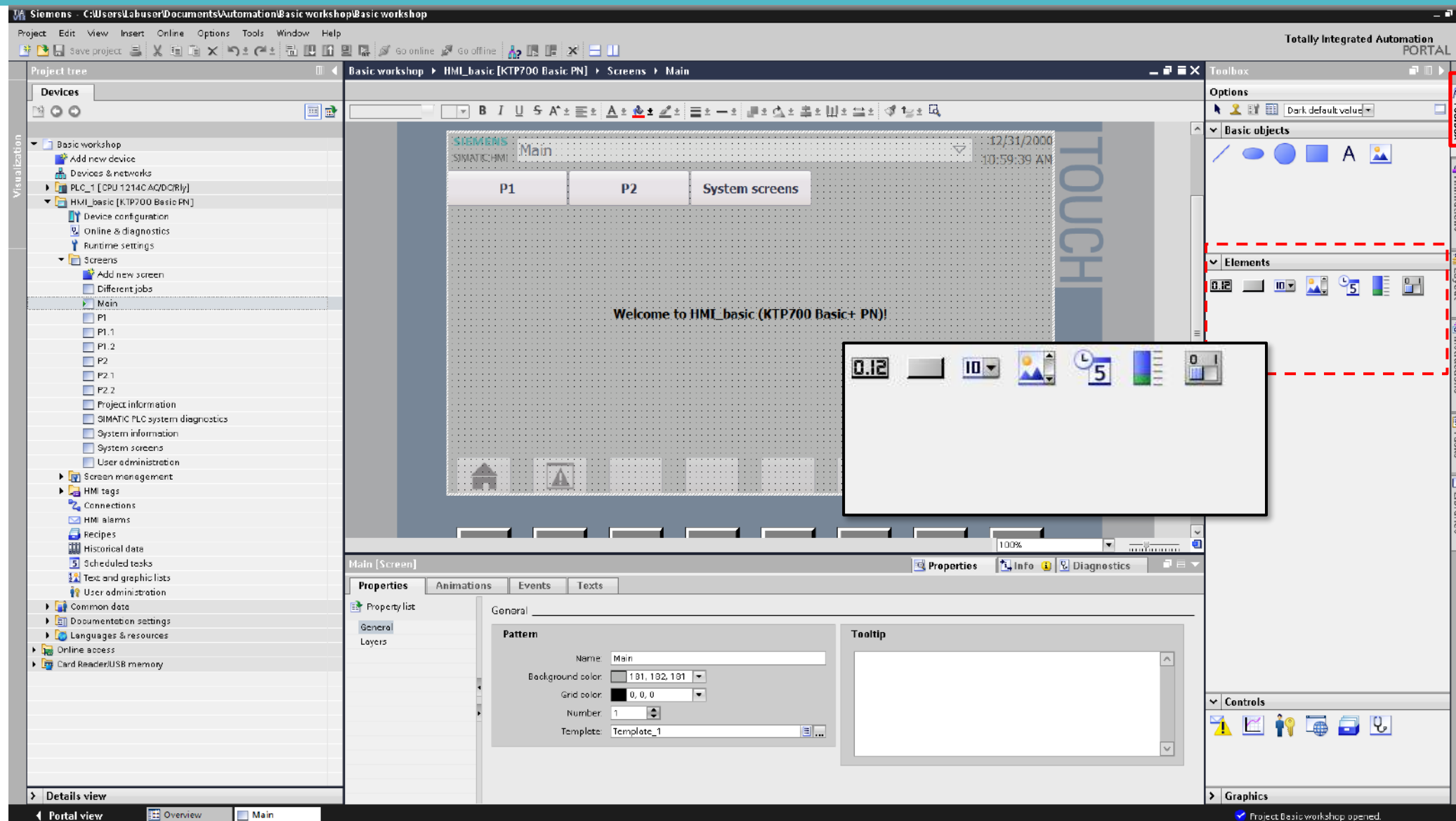


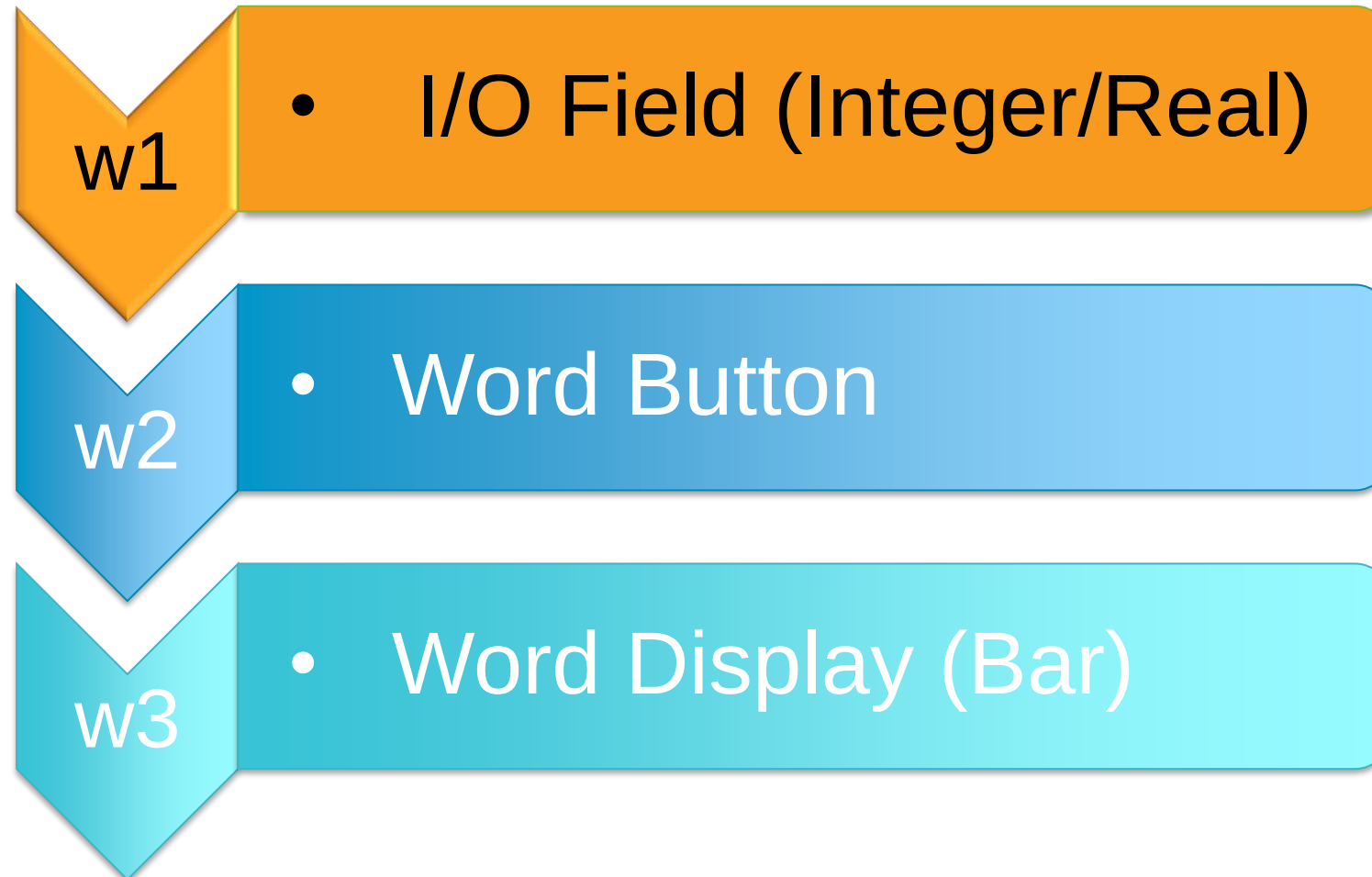
HMI Basic Element

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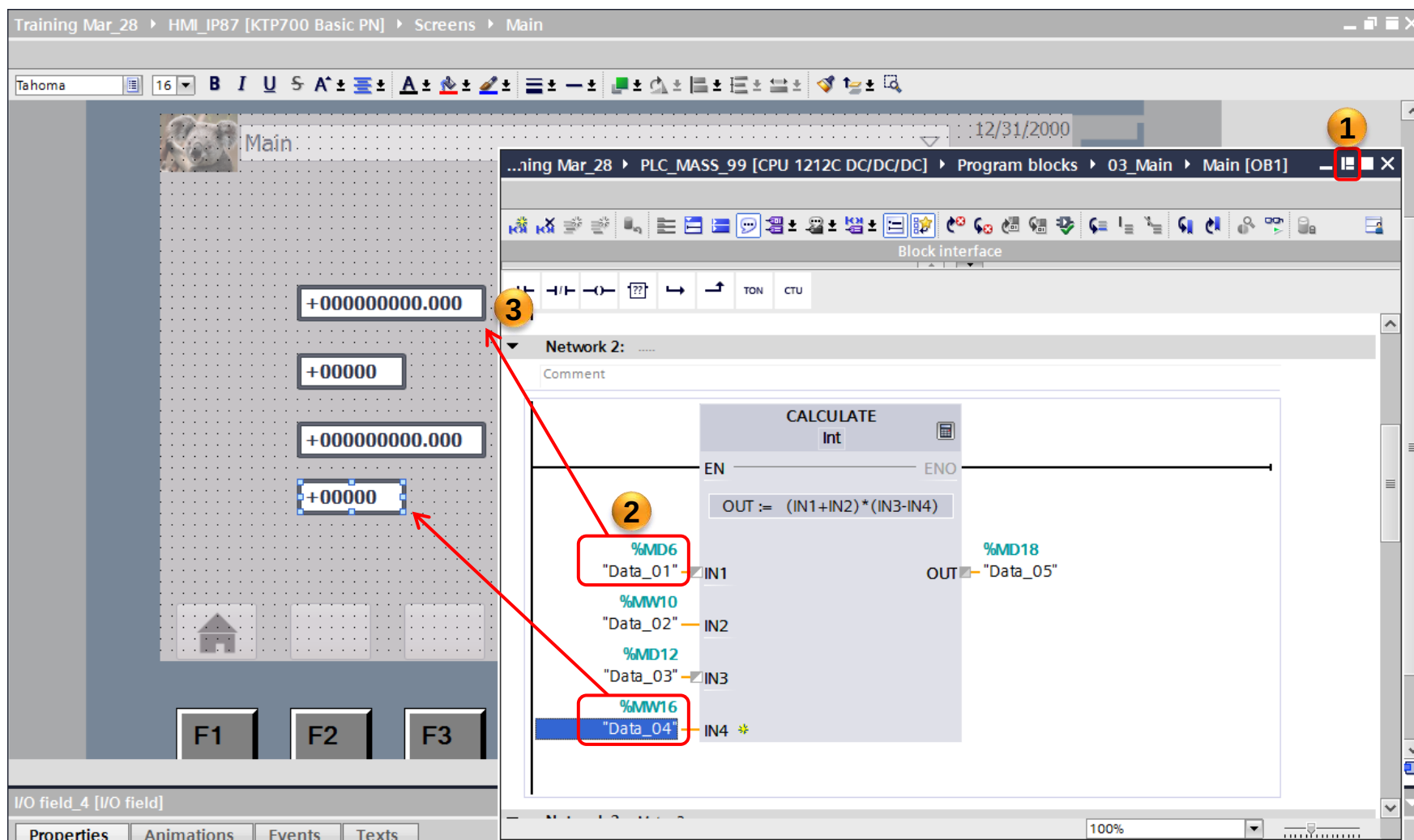
SIEMENS





- การใช้งาน I/O field คือกล่องที่ใช้แสดงตัวเลขต่างๆ ไม่ว่าจะเป็นตัวเลขแบบจำนวนเต็ม (Integer) หรือตัวเลขแบบจำนวนจริง (Real) ก็ตาม นอกจากสามารถแสดงค่าตัวเลขได้แล้วยังสามารถใส่ค่าตัวเลขจากหน้าจอเพื่อไปเปลี่ยนค่าใน PLC ได้ด้วย
- ซึ่งการนำ I/O filed เพื่อ link address จาก PLC มาวางไว้ที่หน้าจอ สามารถทำได้หลายวิธี แต่จะกล่าวถึงวิธีที่ใช้งานได้จริงแบบไม่ยากนัก ดังตัวอย่างหน้าต่อไป

- วิธีที่ 1 : เปิดหน้าจอ PLC ให้เป็นแบบ Float แล้วลาก PLC tag มายังหน้าจอตรงๆ



- วิธีที่ 2: ลาก tag จากหน้า PLC ลงมายังหน้า HMI แล้วลากกลับขึ้นไปทีหน้า HMI

Training Mar_28 ▶ PLC_MASS_99 [CPU 1212C DC/DC/DC] ▶ Program blocks ▶ 03_Main ▶ Main [OB1]

Network 2: ...

Comment

CALCULATE
Int

EN — ENO

OUT := (IN1+IN2)*(IN3-IN4)

%MD6
"Data_01" — IN1

%MW10
"Data_02" — IN2

1 %MD12
"Data_03" — IN3

%MW16
"Data_04" — IN4

%MD18
OUT — "Data_05"

Network 3: Motor 2

Control Motor 2

%DB1

Main [OB1]

General

General

Information

Time stamps

Compilation

Protection

Attributes

2

Consistent n

Event

Language: LAD

Number: 1

Overview Main (OB1) Main

ลาก tag PLC มาไว้ที่ ta
หน้าจอ แล้วค้างไว้ก่อน



Training Mar_28 > HMI_IP87 [KTP700 Basic PN] > Screens > Main

Tahoma 16 B I U A+ A- [color] [font] [align] [bullet] [list] [undo] [redo]

Main

+000000000.000

4

ลากกลับ

F1 F2 F3 F4 F5

I/O field_1 [I/O field]

Properties Animations Events Texts

Property list

- General
- Appearance
- Characteristics
- Layout
- Text format
- Limits
- Styles/Designs
- Miscellaneous
- Security

Appearance

Background

Color: 255, 255, 255

Fill pattern: Solid

Corner radius: 3

Text

3

Overview Main (OB1) Main

Word Status I/O filed

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Automation

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- วิธีที่ 3: วาง I/O filed จาก Toolbox ที่หน้าจอ แล้วเลือก PLC tag ที่ต้องการเอาเอง

Training Mar_28 ▸ HMI_IP87 [KTP700 Basic PN] ▸ Screens ▸ Main

Tahoma 16 B I U S A ±

12/31/2000 10:59:39 AM

2

0000000

Drag & Drop

1

0.12

4

PLC tags

5

Name	Data type	Address	Comment
CMD_Stop_M4	Bool	%I0.7	
Count_M1	Int	%MW4	
Data_01	Real	%MD5	
Data_02	Int	%MW10	
Data_03	Real	%MD12	
Data_04	Int	%MW16	
Data_05	Real	%MD18	
DiagStatusUpdate	Bool	%M1.1	

6

3

Tag: ...

PLC tag: ...

Address: ...

Type

Mode: Input/output

Display format: Decimal

Decimal places: 0

Field length: 10

Leading zeros: ☐

Format pattern: 9999999

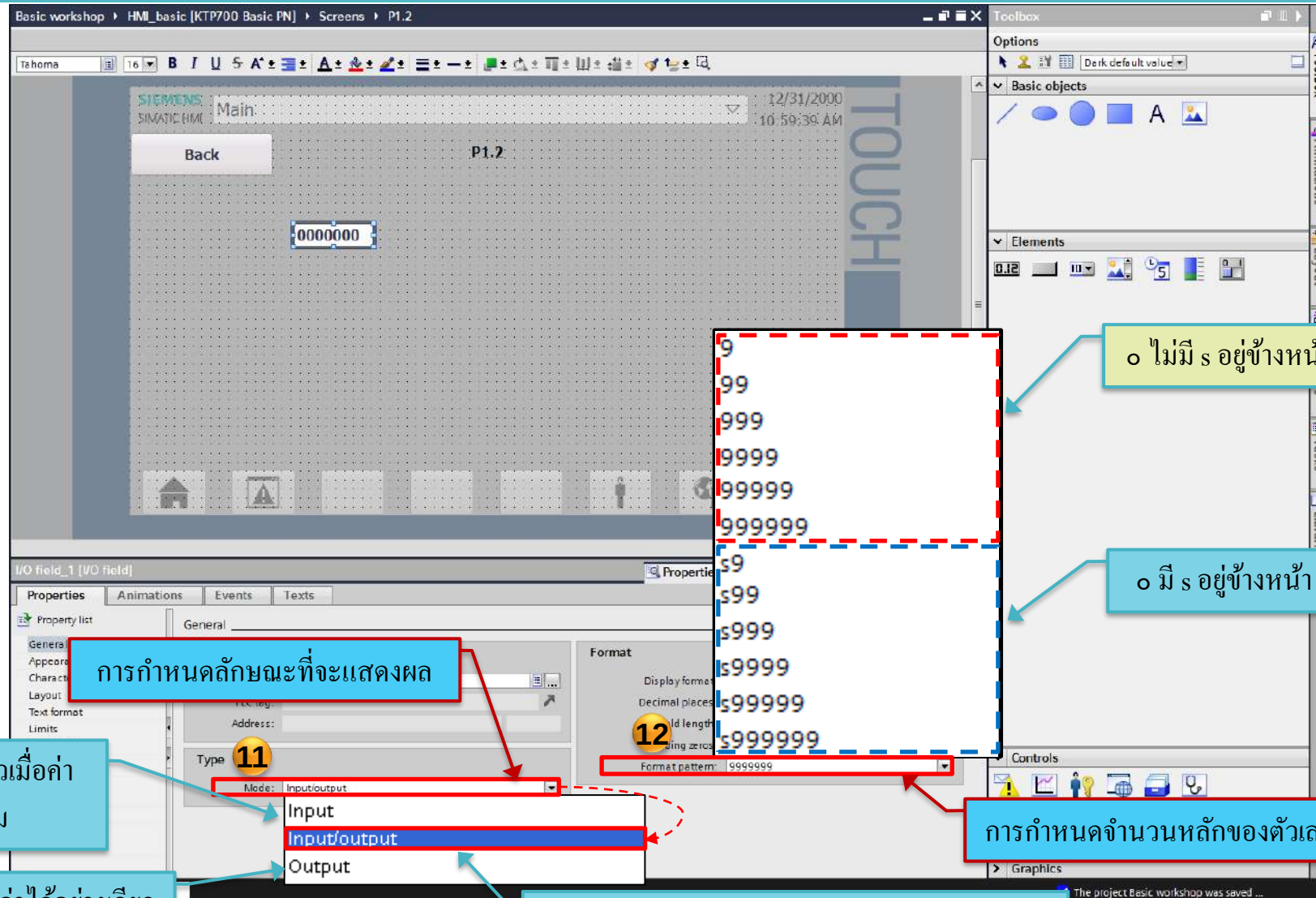
I/O field

Select Type & Format (Integer)

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o ไม่มี s อยู่ข้างหน้า คือ ค่าจะไม่แสดงเครื่องหมาย + -

o มี s อยู่ข้างหน้า คือ ค่าจะแสดงเครื่องหมาย + -

o Input คือ ใส่ค่าได้อย่างเดียวเมื่อค่าเปลี่ยนจะไม่เปลี่ยนตาม

o Output คือ แสดงค่าได้อย่างเดียว

การกำหนดจำนวนหลักของตัวเลขที่จะแสดง

o Input/Output คือ สามารถใส่ค่าและแสดงผลได้



- o มี s อยู่ข้างหน้า คือ ค่าจะแสดงเครื่องหมาย + -

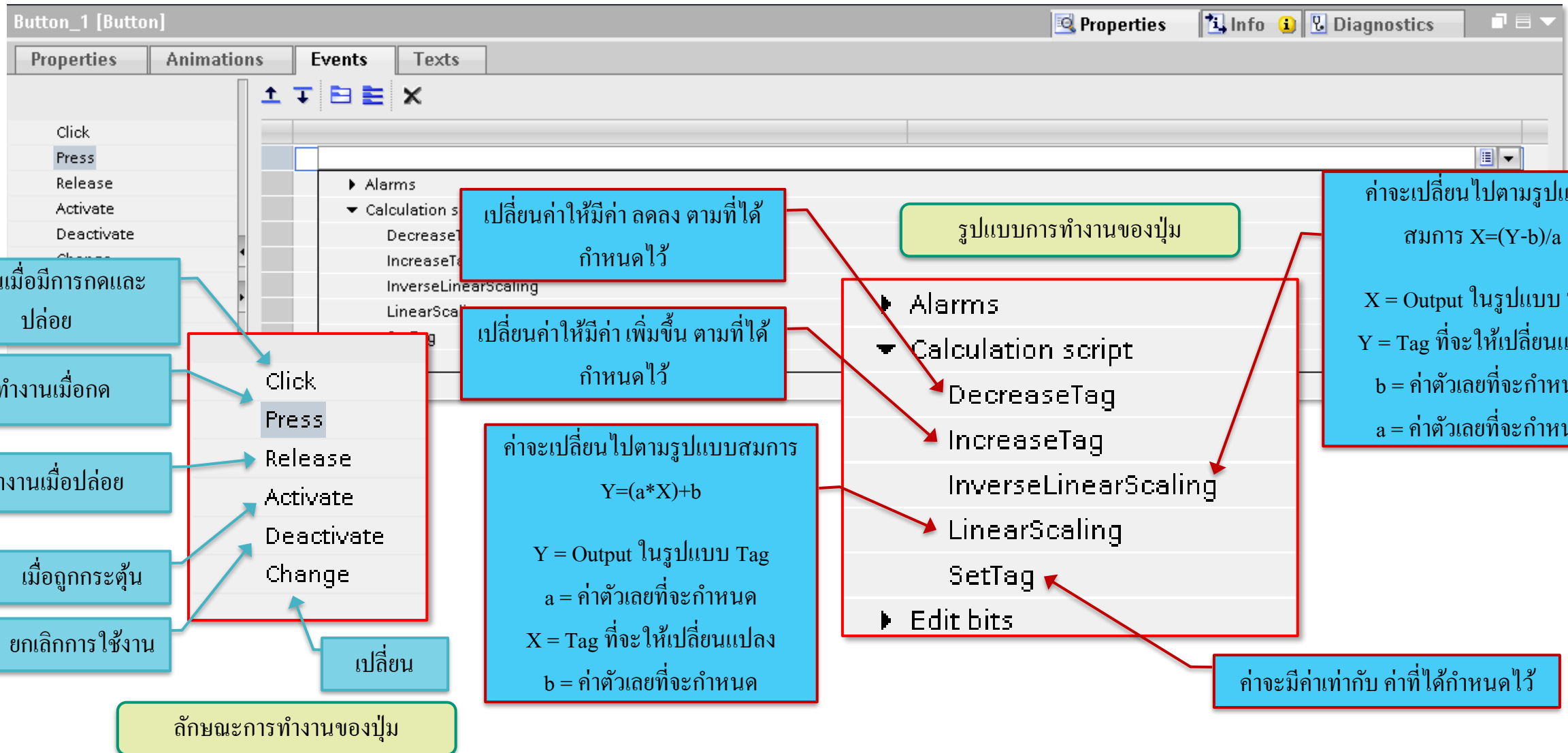
การกำหนดจำนวนหลักของตัวเลขที่จะแสดง

Button Word Events

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Word Command Button

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The image shows the Siemens SIMATIC Manager interface for configuring a HMI screen. The main screen, titled 'P1.2', displays a grid of buttons for controlling a motor speed. The buttons are labeled '+100', '-100', and '500'. A 'Back' button is also visible. The configuration for each button is shown in the 'Properties' and 'Events' windows.

Button_1 [Button] Configuration:

- Click event: **IncreaseTag**
- Tag (Input/output): **Speed_Motor**
- Value: **100**
- Function: **<Add function>**
- Annotation: **เมื่อกดปุ่ม Speed_Motor จะมีค่าเพิ่มขึ้น 100**

Button_2 [Button] Configuration:

- Click event: **DecreaseTag**
- Tag (Input/output): **Speed_Motor**
- Value: **100**
- Function: **<Add function>**
- Annotation: **เมื่อกดปุ่ม Speed_Motor จะมีค่าลดลง 100**

Button_3 [Button] Configuration:

- Click event: **SetTag**
- Tag (Output): **Speed_Motor**
- Value: **500**
- Function: **<Add function>**
- Annotation: **เมื่อกดปุ่ม Speed_Motor จะมีค่าเท่ากับ 500**

Main Screen (P1.2) Details:

- Buttons: +100, -100, 500
- Grid: 181, 182, 181
- Grid color: 0, 0, 0
- Number: 4
- Template: Template_1

Word Status Bar

Approved Partner

Factory Automation

SIEMENS

Basic workshop > HMI_basic [KTP700 Basic PN] > Screens > P1.2

Toolbox

Options

Basic objects

Elements

Drag & Drop

Properties

Bar_1 [Bar]

Properties

General

Process

Maximum scale value: 1000

Minimum scale value: 0

เลือก Tag ที่จะให้แสดงผล

Process tag:

PLC tags

Tag table_1 [4]

Speed_Motor

Int

%IWO

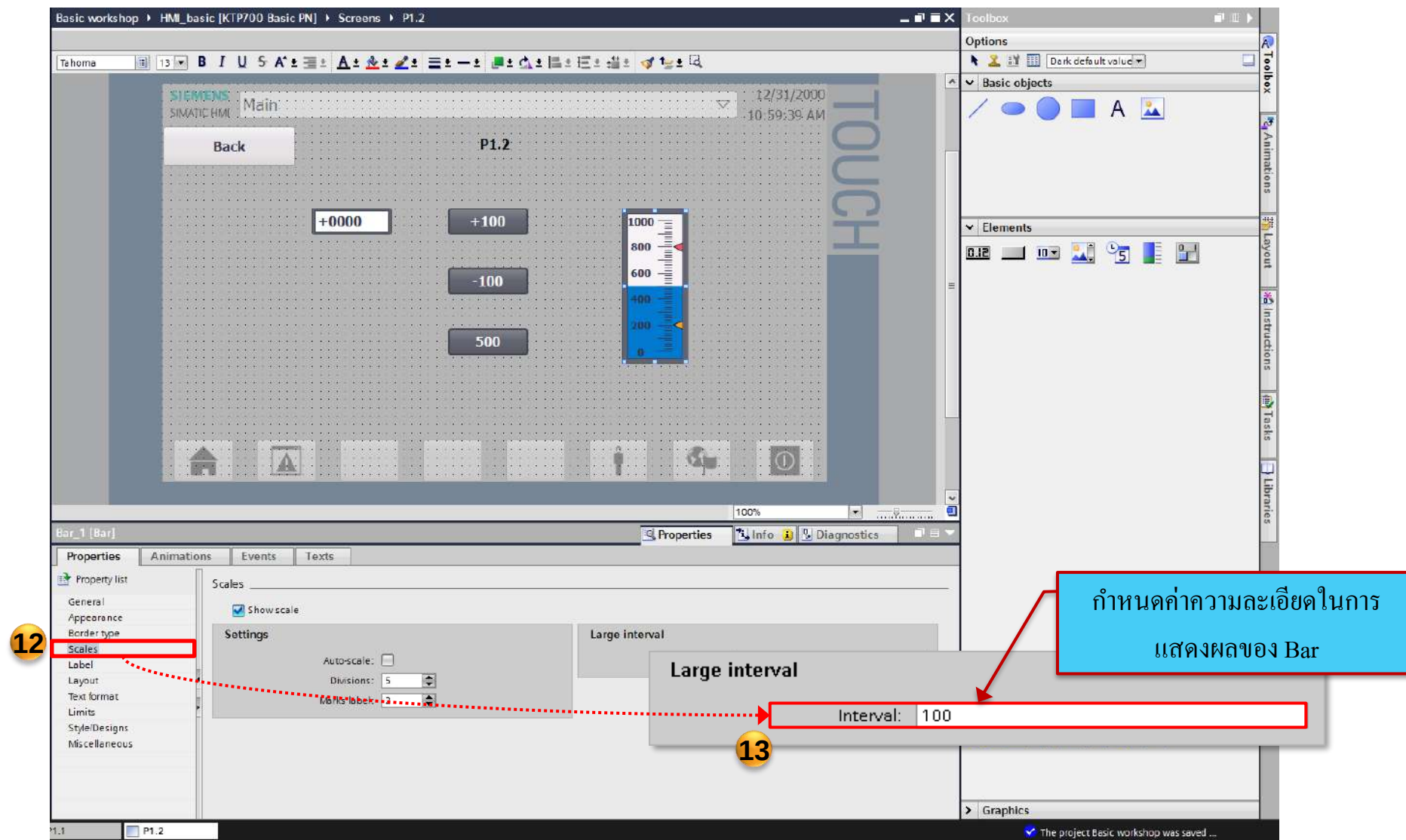
Add new

OK

การกำหนดค่า Scale สูงสุด

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Factory
Automation



Simulation

Approved
Partner

Factory
Automation

SIEMENS

1

ในกรณีถ้า (สัญลักษณ์ Simulation) ไม่
ขึ้น ให้กดที่หน้าจอ 1 ครั้ง

SIEMENS SIMATIC HMI

Main

Back

P1.2

+0000 +100 -100 500

1000 800 600 400 200 0

Properties Animations Events Texts

General

Pattern

Name: P1.2

Background color: 181, 182, 181

Grid color: 0, 0, 0

Number: 4

Template: template_1

Tooltip

Controls

Graphics

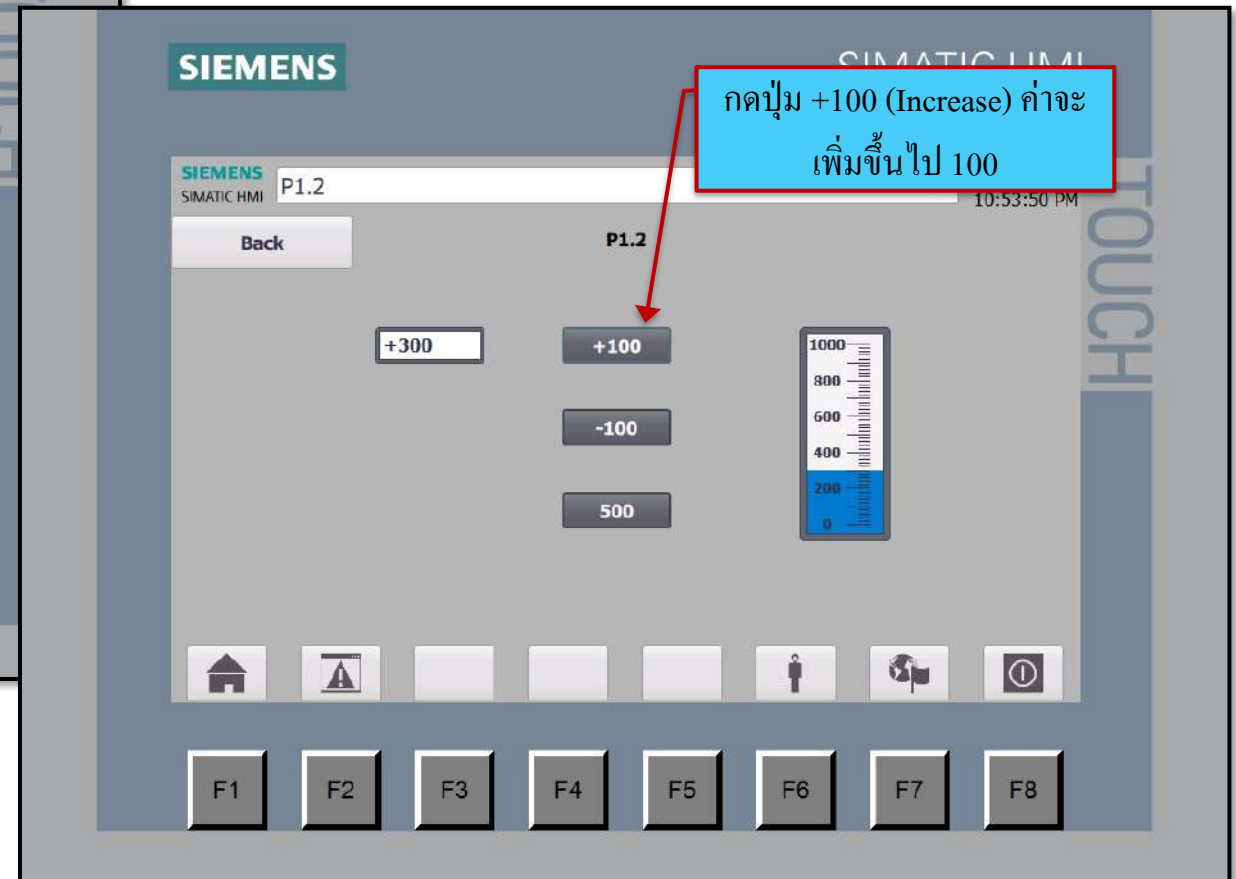
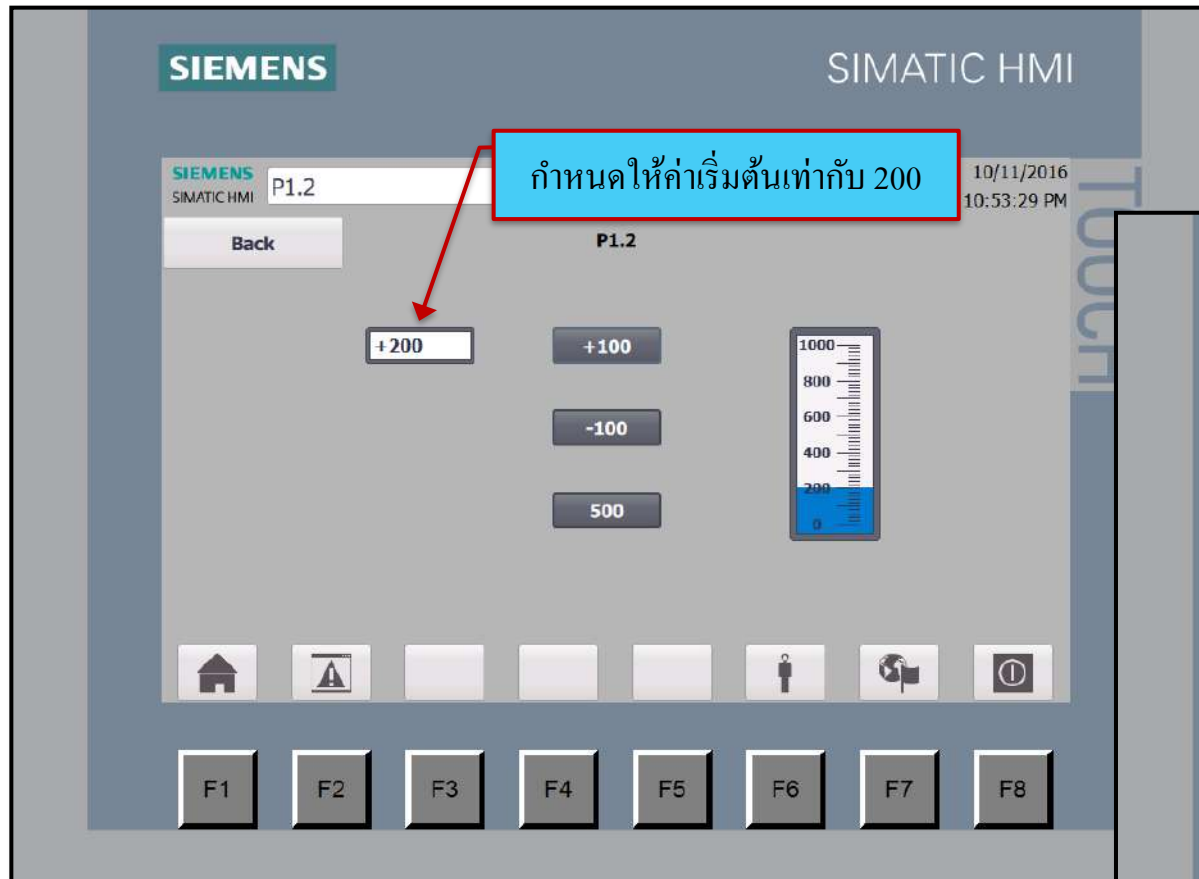
Project: Basics workshop opened.

Simulation

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Automation

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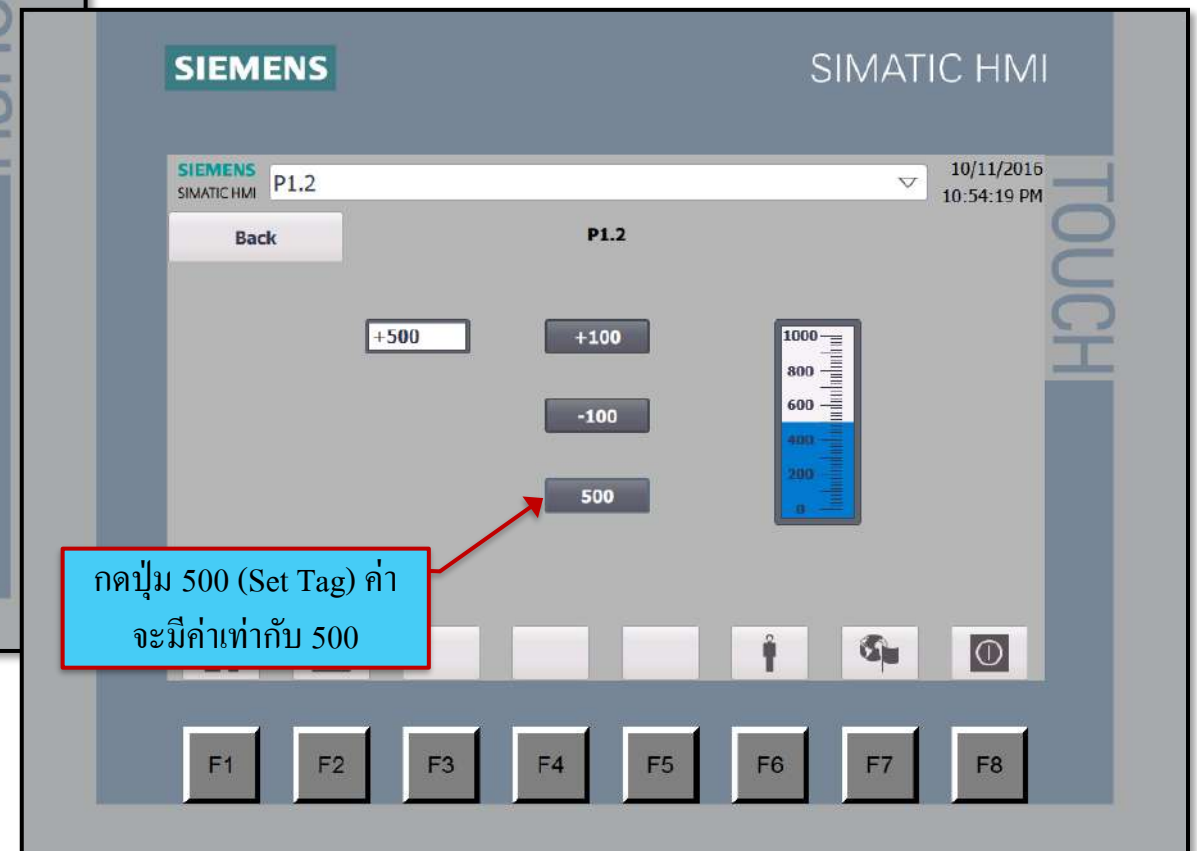
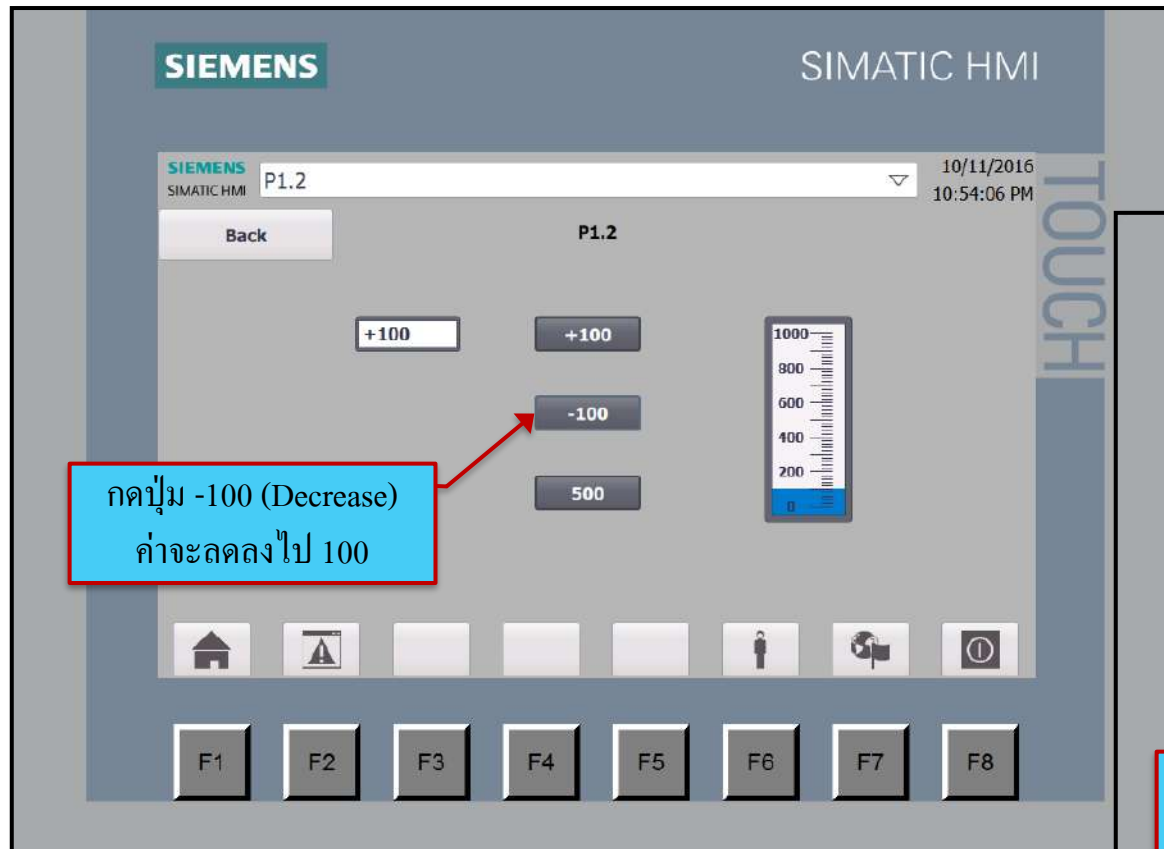


Word Status Bar

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Automation

SIEMENS

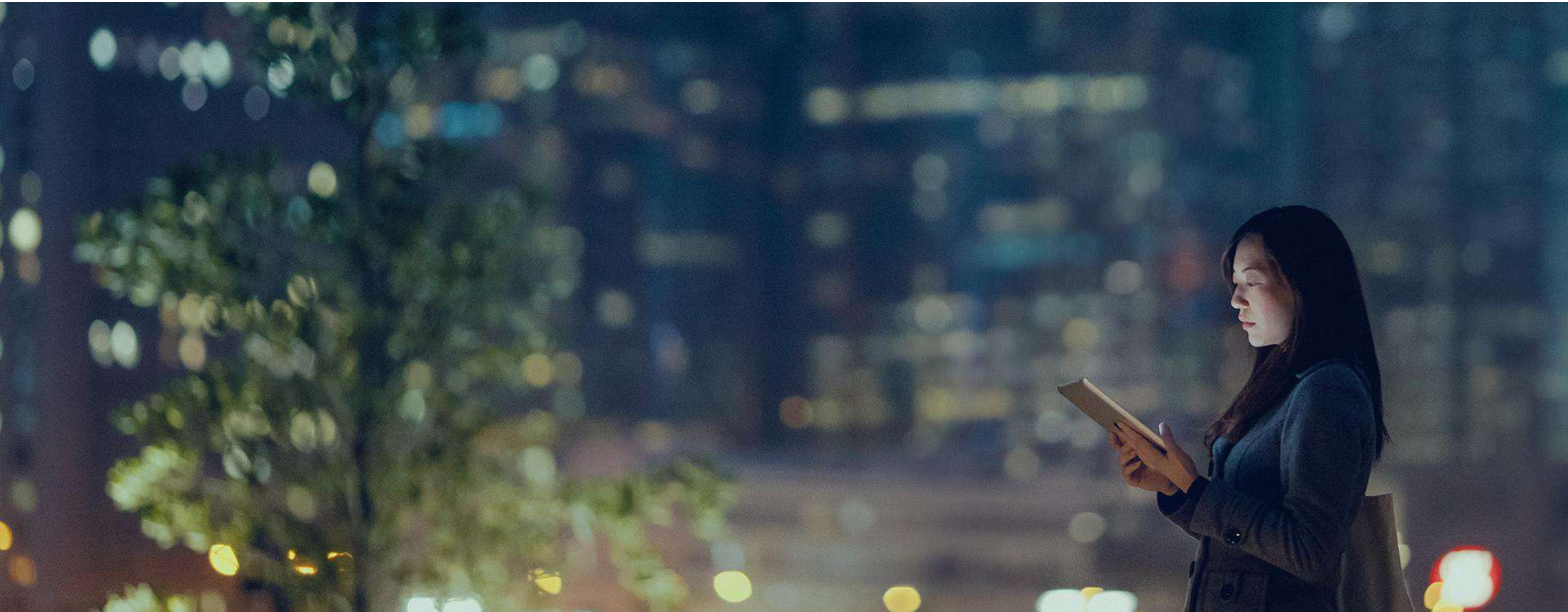


Apply with Movement Animation

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SIEMENS



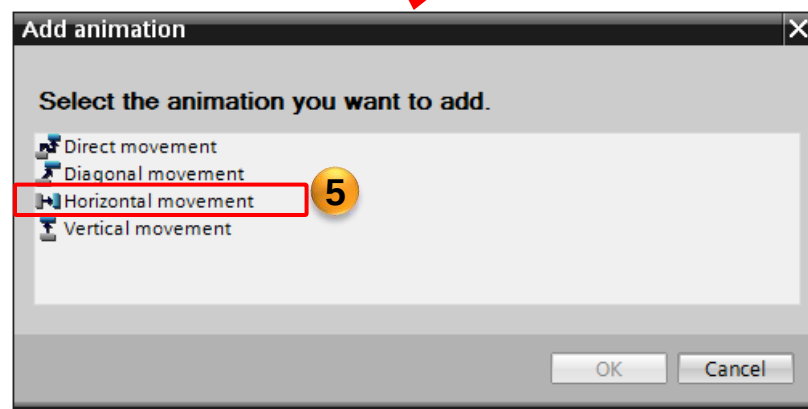
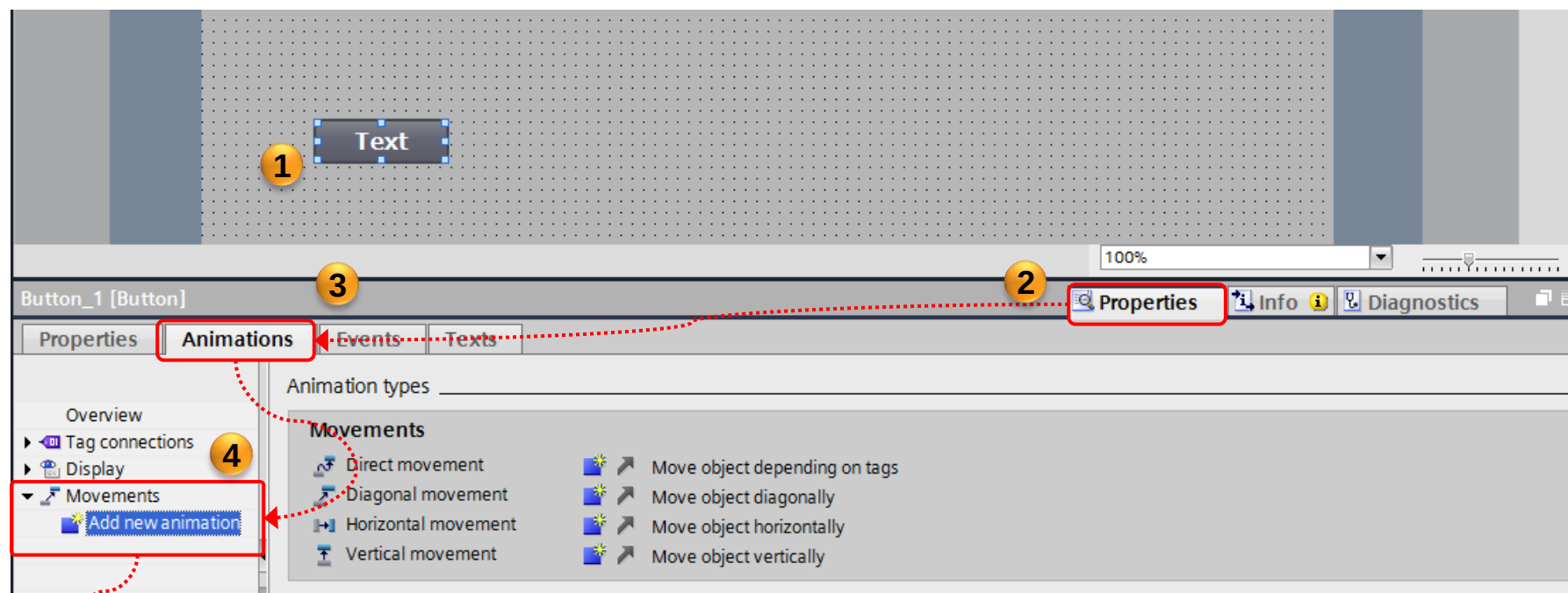
- เราสามารถใช้งาน Animation เพื่อเพิ่มความน่าสนใจ และความสมจริงในการทำงานของเครื่องจักรได้ ซึ่ง Animation ที่ใช้กันบ่อยๆก็คือการเคลื่อนที่ (Movement) นั่นเอง
- ความง่ายของการใช้งาน Animation แบบ Movement ก็คือ เราไม่จำเป็นต้องคำนวณการเคลื่อนจากตำแหน่งเริ่มต้นไปยังตำแหน่งสิ้นสุดของตัว object เลย การใช้งานนั้นเราเพียงแค่ลาก object ที่ต้องการไปยังตำแหน่งปลายทางเท่านั้น ตัวโปรแกรมจะคำนวณระยะทางเทียบกับ process tag ให้อัตโนมัติ

Animation Movement

Approved
Partner

Factory
Automation

SIEMENS



The screenshot displays the Siemens animation software interface. At the top, a canvas shows a text object labeled 'Text' with a dashed blue line indicating a horizontal movement path to a target position labeled 'xt' with coordinates 'x/y: 587, 374'. A yellow circle with the number '6' is placed near the target position. A red arrow points from a blue text box to this circle.

Below the canvas, the 'Button_1 [Button]' properties panel is visible. It has tabs for 'Properties', 'Animations', 'Events', and 'Texts'. The 'Animations' tab is selected, showing 'Horizontal movement'. A yellow circle with the number '7' is placed next to the 'Tag' section. A red arrow points from a blue text box to this circle.

The 'Tag' section includes a 'Name' field set to 'Speed_Motor', a 'Range from' field set to '0', and a 'To' field set to '500'. A yellow circle with the number '8' is placed next to the 'Start position' section. A red arrow points from a blue text box to this circle.

The 'Start position' section shows 'X: 79' and 'Y: 374'. The 'Target position' section shows 'X: 587' and 'Y: 374'.

Four blue text boxes with red borders provide Thai annotations:

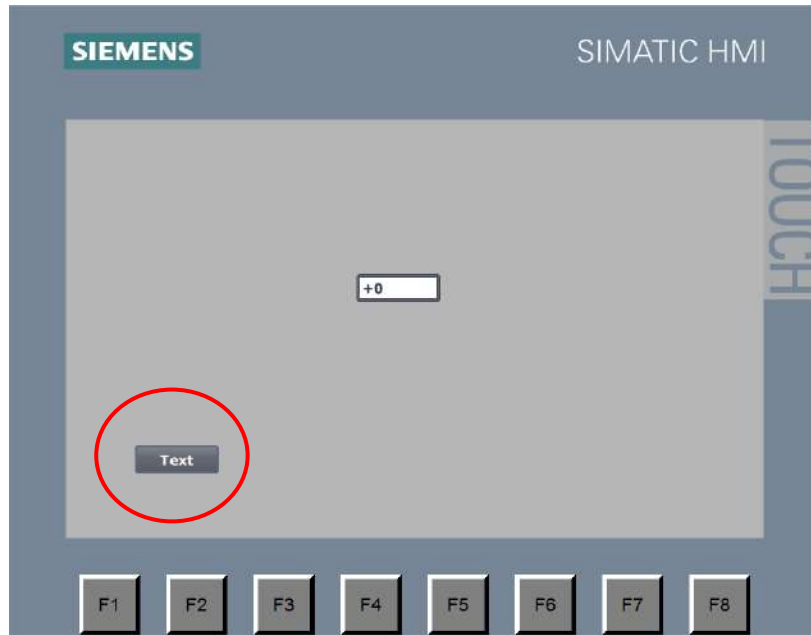
- Top right: สามารถลากตำแหน่งปลายทางได้เลย (You can drag the target position directly).
- Middle right: ระบุ tag ที่ต้องการสั่งงานตำแหน่ง (Specify the tag you want to control the position).
- Bottom right: ระบุค่าของ tag ที่ใช้งานเพื่อให้เคลื่อนที่จากต้นทางไปยังปลายทาง (Specify the value of the tag used to move from the start to the end).
- Bottom left: (Points to the start position fields).

Simulation

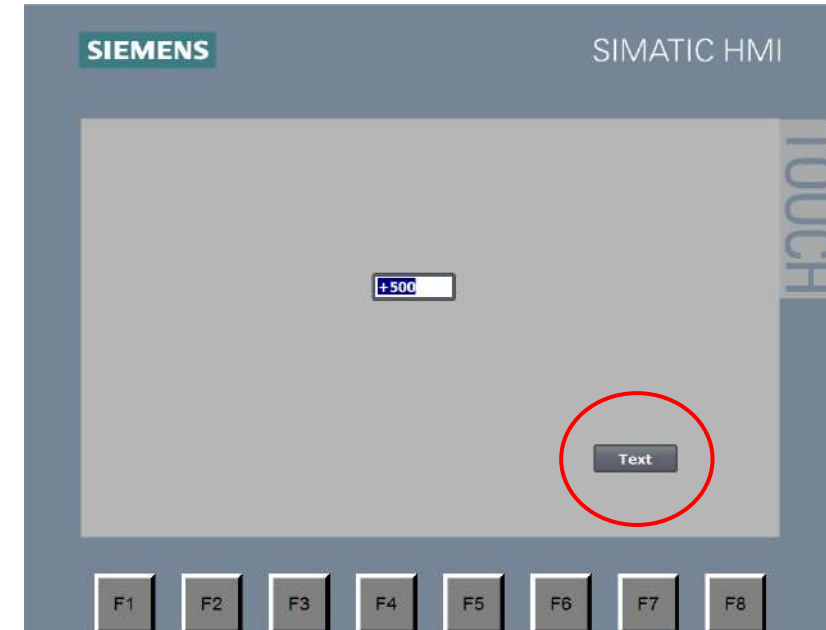
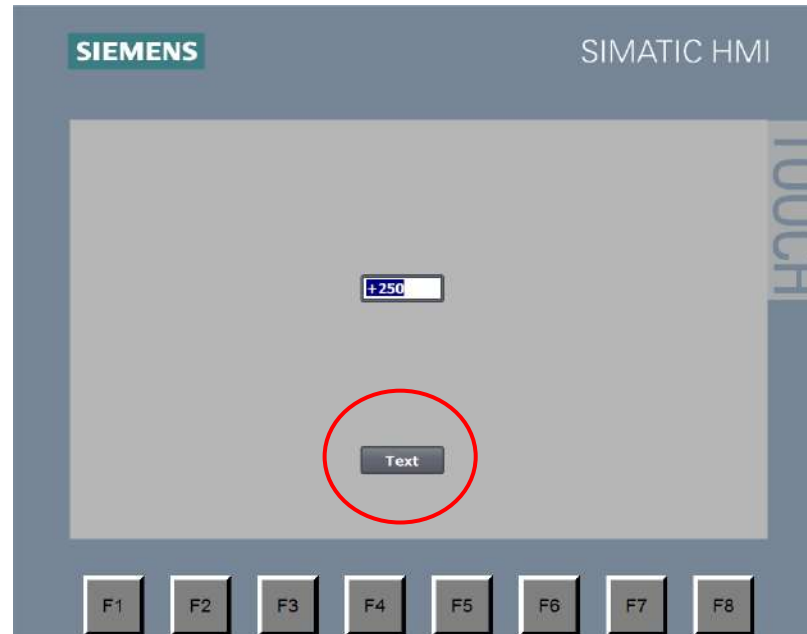
Approved
Partner

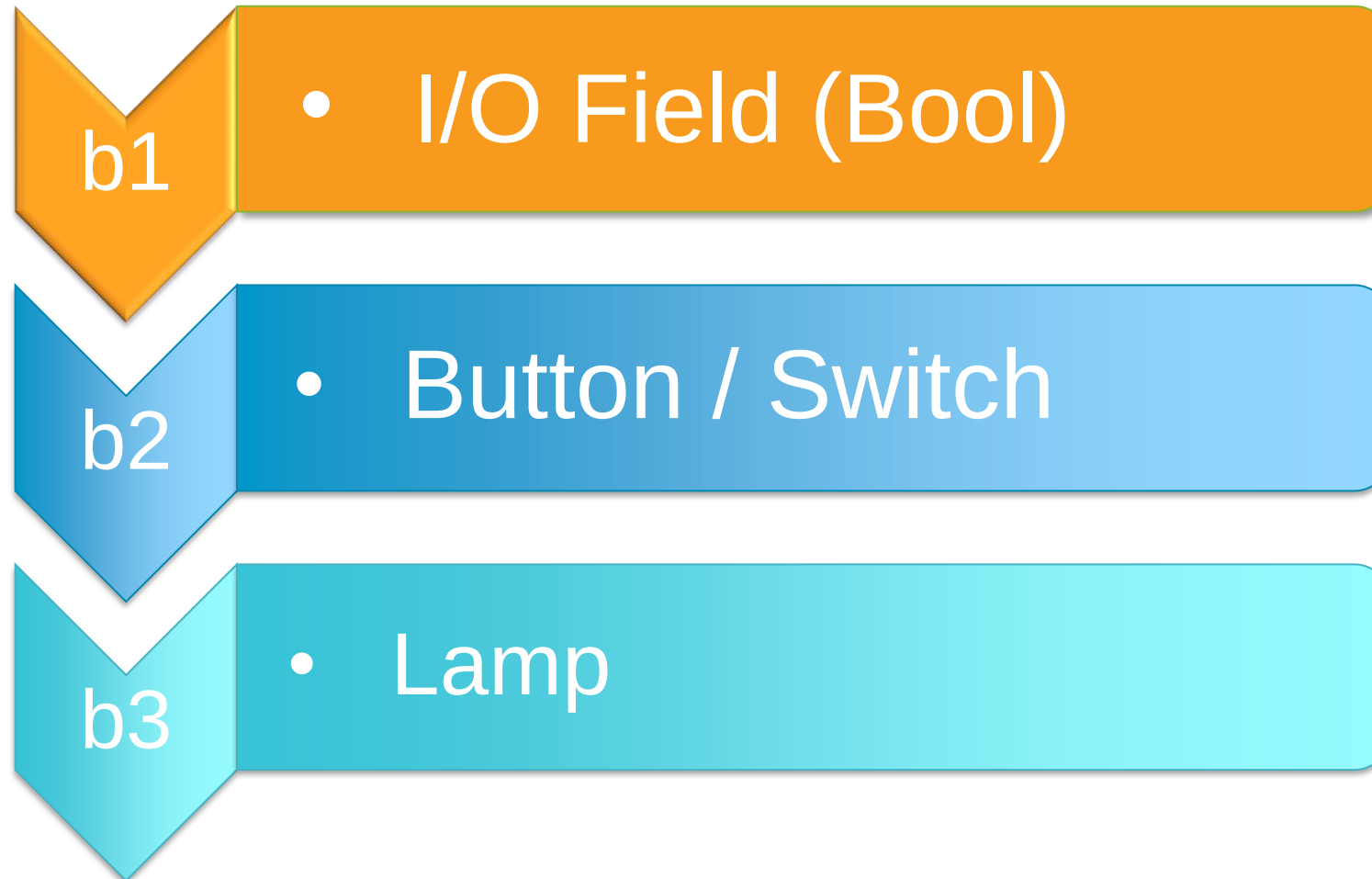
Factory
Automation

SIEMENS



ตำแหน่งของอุปกรณ์จะเปลี่ยนไปตามค่าที่เราผูก tag เอาไว้



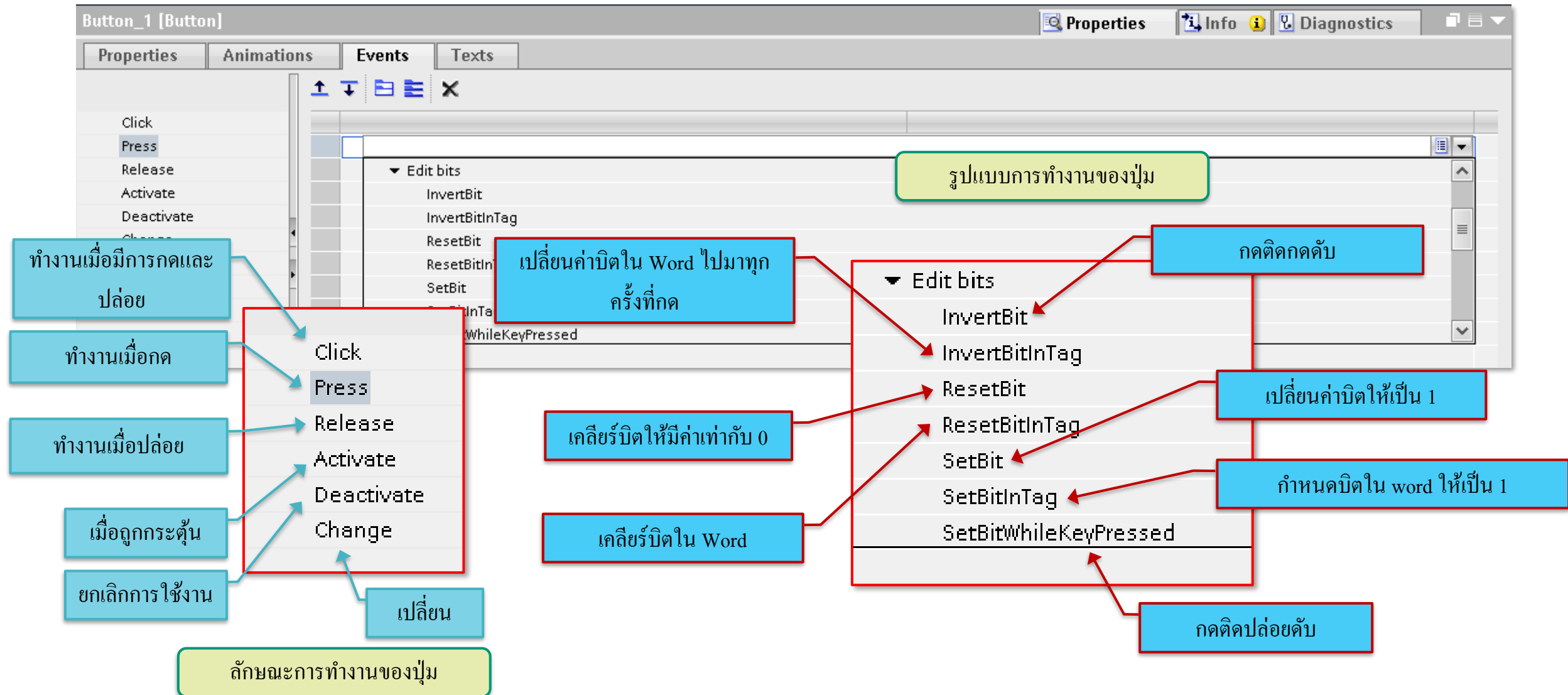


Button Bit Events

Approved
Partner

Factory
Automation

SIEMENS



Invert bit Setting Events

Approved
Partner

Factory
Automation

SIEMENS

The screenshot shows the Siemens SIMATIC Manager interface. The main window displays a screen design with a 'Back' button and a 'Start/Stop' button. The 'Properties' window is open for 'Button_1 [Button]', and the 'Events' tab is selected. The 'Press' event is highlighted in the list. The 'Edit bits' list is open, showing various actions, with 'InvertBit' selected. Red callouts with numbers 7 through 10 indicate the steps: 7 points to the 'Events' tab, 8 points to the 'Press' event, 9 points to the 'Edit bits' list, and 10 points to the 'InvertBit' action. A blue box with Thai text 'เลือกกลุ่มแบบการทำงานของปุ่ม' (Select button operation group) points to the 'Basic objects' section of the toolbox. Another blue box with Thai text 'เลือกลักษณะการทำงานของปุ่ม' (Select button operation characteristic) points to the 'Press' event.

7

8

9

10

เลือกกลุ่มแบบการทำงานของปุ่ม

เลือกลักษณะการทำงานของปุ่ม

Events

Press

Edit bits

- InvertBit
- InvertBitInTag
- ResetBit
- ResetBitInTag
- SetBit
- SetBitInTag
- SetBitWhileKeyPressed

Invert bit Setting Events

Approved
Partner

Factory
Automation

SIEMENS

The screenshot shows the Siemens SIMATIC Manager interface. The main window displays a SIMATIC HMI screen with a 'Start/Stop' button. The 'Events' tab is selected for the button, and the 'InvertBit' event is configured. The configuration window shows a list of PLC tags with 'Start' and 'Stop' selected. A Thai text box points to the 'Tag (Input/output)' field.

เลือก Tag ที่จะใช้งานปุ่ม

Name	Data type	Address
Run_Motor	Bool	%Q0.2
Speed_Motor	Int	%IW0
Start	Bool	%I0.0
Stop	Bool	%I0.1

Set bit Create button

Approved
Partner

Factory
Automation

SIEMENS

The screenshot shows the Siemens SIMATIC Manager interface for creating a button. The main workspace displays a button labeled 'Text' with a red box around it and a yellow circle '2'. A red arrow labeled 'Drag & Drop' points from the 'Text' element in the 'Elements' panel (yellow circle '1') to the button. A red box around the 'Properties' tab (yellow circle '3') leads to a detailed 'Properties' window. In this window, the 'General' tab is selected (yellow circle '4'), and the 'Text' mode is chosen. The 'Text when button is "not pressed"' field is highlighted with a red box and a yellow circle '6', containing the text 'Start'. A red arrow points from this field to a blue box with the text 'ตั้งชื่อปุ่มตามต้องการ' (Name the button as required). A blue box with the text 'จะให้เป็นปุ่มแบบ กด Start อย่างเดียว' (Make it a button that only presses Start) points to the 'Text' mode selection. A red box around the 'Properties' list (yellow circle '5') is also shown. A red box around the 'Toolbox' icon (yellow circle '1') is shown in the top right corner.

จะให้เป็นปุ่มแบบ กด Start อย่างเดียว

Drag & Drop

Properties

Label

Text

Text when button is "not pressed"

Start

Text when button is "pressed"

ตั้งชื่อปุ่มตามต้องการ

Set bit Setting Events

Approved
Partner

Factory
Automation

SIEMENS

Basic workshop ▶ HMI_basic [KTP700 Basic PN] ▶ Screens ▶ P1.1

Tahoma 16 B I U A+ A- [Icons]

SIEMENS SIMATIC HMI Main 12/31/2000 10:50:39 AM

Back P1.1

0

Start/Stop Start

TOUCH

Toolbox

Options Dark default value

Basic objects [Icons]

Elements [Icons]

Button_2 [Button]

Properties Animations **Events** Texts

8 Click Press Release Activate Deactivate Change

7

9

10

▼ Edit bits

- InvertBit
- InvertBitInTag
- ResetBit
- ResetBitInTag
- SetBit**
- SetBitInTag
- SetBitWhileKeyPressed

เลือกประเภทการทำงานของปุ่ม

เลือกลักษณะการทำงานของปุ่ม

Set bit Setting Events

Approved
Partner

Factory
Automation

SIEMENS

Basic workshop > HMI_basic [KTP700 Basic PN] > Screens > P1.1

Toolbox

Options

Basic objects

Elements

Button_2 [Button]

Properties Animations Events Texts

Click Press Release Activate Deactivate Change

SetBit

Tag (Input/output)

เลือก Tag ที่จะใช้งานปุ่ม

11

12

13

14

15

Name	Data type	Address
Run_Motor	Bool	%Q0.2
Speed_Motor	Int	%IW0
Start	Bool	%I0.0
Stop	Bool	%I0.1

Reset bit

Create button

Approved
Partner

Factory
Automation

SIEMENS

Basic workshop > HMI_basic [KTP700 Basic PN] > Screens > P1.1

Tahoma 16 B I U A+ A- [font icons]

SIEMENS SIMATIC HMI Main 12/31/2000 10:56:39 AM

Back P1.1

0 1

Start/Stop Start 2

CTRL + Drag & Drop

Stop 3

100%

4 Properties

5 Properties

6

Label

☒ Text

☐ Text list

Text when button is "not pressed"

Stop

Text when button is "pressed"

Text when button is "pressed"

Hotkey

None

จะให้เป็นปุ่ม Stop อย่างเดียว

เปลี่ยนชื่อจาก Start เป็น Stop

Reset bit Setting Events

Approved
Partner

Factory
Automation

SIEMENS

จะทำปุ่มเป็นลักษณะ Push switch เหมือนปุ่ม Start คือกดแล้ว
ไม่มีการค้างค่าเอาไว้ จึงต้องมีการเคลียค่าบิตเมื่อกดปล่อย

เปลี่ยน Tag จาก Start ให้เป็น Stop

Name	Data type	Address
Run_Motor	Bool	%Q0.2
Speed_Motor	Int	%IW0
Start	Bool	%I0.0
Stop	Bool	%I0.1

- Click
- Release
- Tag (Input/output)
- Tag table_1 (4)
- Stop
- OK

SetBitWhileKeyPressed

Create button

Approved
Partner

Factory
Automation

SIEMENS

Basic workshop > HMI_basic [KTP700 Basic PN] > Screens > P1.1

Tahoma

SIEMENS SIMATIC HMI Main

Back P1.1

0

Start/Stop Start Stop

Text

Drag & Drop

1

2

3

4

5

6

Properties

General

Property list

General

Appearance

Fill pattern

Design

Layout

Text format

Style/Designs

Miscellaneous

Security

Mode

☒ Text

☐ Graphic

☐ Graphics or text

☐ Graphics and text

☐ Invisible

Hotkey

None

Label

☒ Text

☐ Text list

Text when button is "not pressed"

Start

Text when button is "pressed"

Text when button is "pressed"

ตั้งชื่อปุ่มตามต้องการ

จะเป็นปุ่มแบบเมื่อกดปุ่มไฟจะติดและ
ปล่อยไฟก็จะดับ

Toolbox

Options

Dark default value

Basic objects

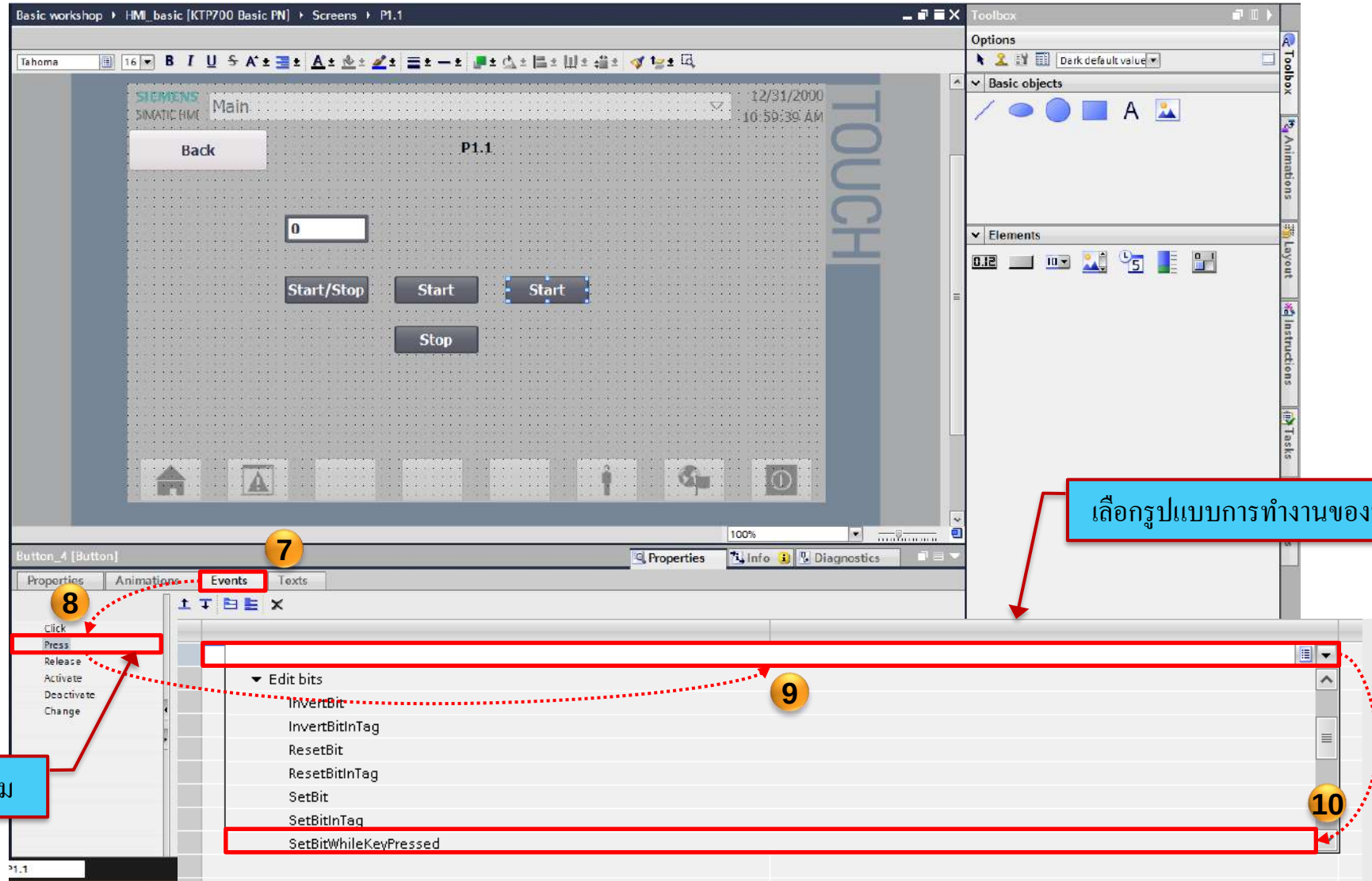
Elements

SetBitWhileKeyPressed Setting Events

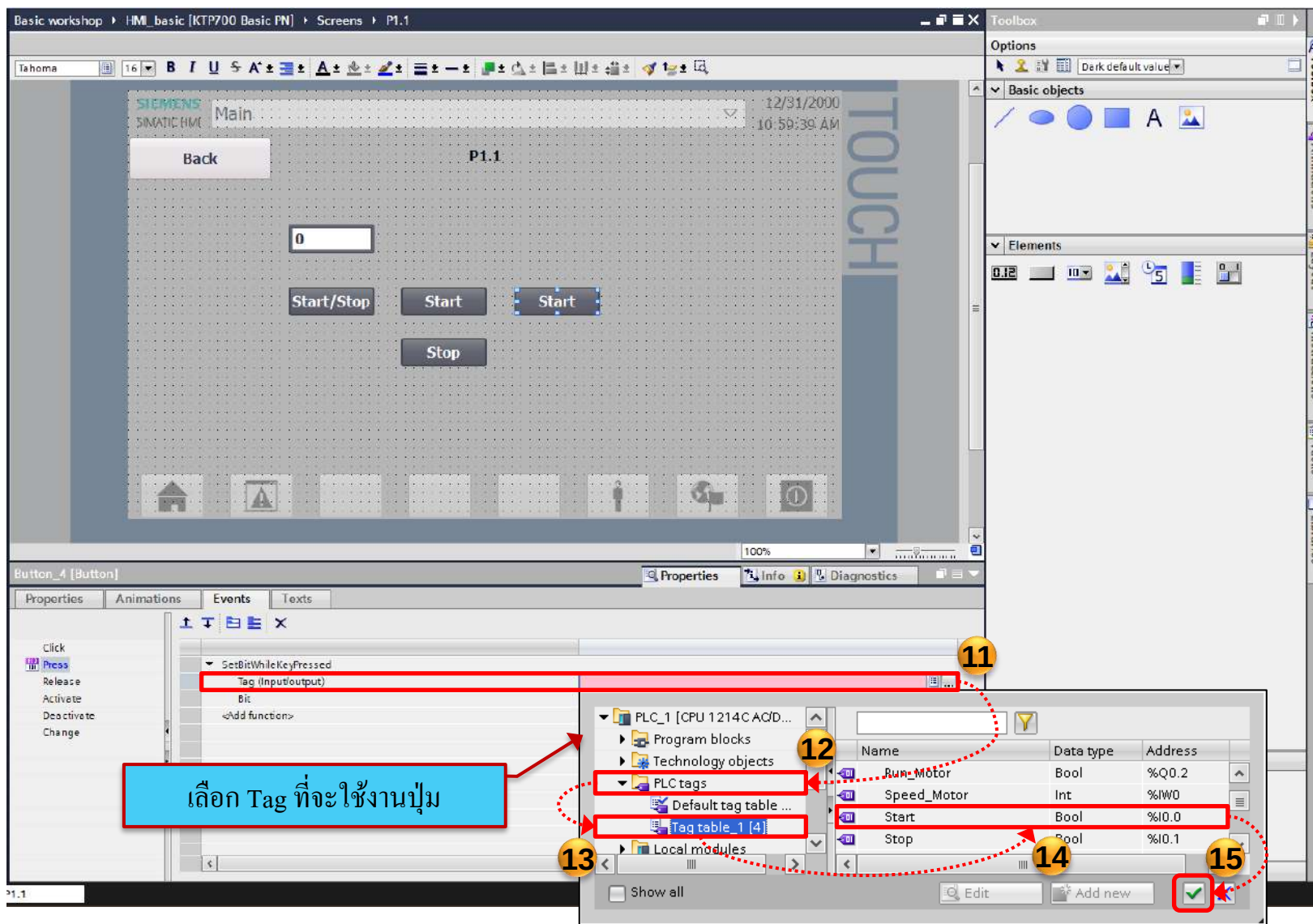
Approved
Partner

Factory
Automation

SIEMENS



SetBitWhileKeyPressed Setting Events



Switch Select tag

Approved
Partner

Factory
Automation

SIEMENS

Drag & Drop

1

2

3

4

5

6

7

8

9

10

Properties

General

Tag:

Format: Switch

PLC tag:

Address:

Value for "ON":

PLC_1 [CPU 1214C AC/D...]

Program blocks

Technology objects

PLC tags

Default tag table ...

Tag table_1 (4)

Local modules

Show all

Edit

Add new

Run_Motor Bool %Q0.2

Speed_Motor Int %IW0

Start Bool %I0.0

Stop Bool %I0.1

เลือก Tag ที่จะให้แสดงผล

Lamp Pilot Light

Approved
Partner

Factory
Automation

SIEMENS

Basic workshop > HMI_basic [KTP700 Basic PN] > Screens > P1.1

12/31/2000 10:59:39 AM

Back P1.1

0

Start/Stop Start Start OFF Stop

TOUCH

100%

Properties

General

Property list

General

Appearance

Layout

Limits

Miscellaneous

Security

Tag:

Value for "ON": 1

PLC_1 [CPU 1214C AG/D...]

Run blocks

Technology objects

PLC tags

Default tag table ...

Tag table_1 [4]

Local modules

Show all

Edit Add new

Libraries

Options

Library view

Project library

Global libraries

Buttons-and-Switches

Master copies

DiagnosticsButtons

FunctionButtons

LeverSwitches

PilotLights

PlotLight_Round_G

PlotLight_Round_GN

PlotLight_Round_N_Mono

PlotLight_Round_R

PlotLight_Round_RN

PlotLight_Square_G

PlotLight_Square_GN

PlotLight_Square_N_Mono

PlotLight_Square_R

PlotLight_Square_RN

PushButtonSwitches

RotarySwitches

SlideSwitches

SystemButtons

ToggleSwitches

Long Functions

Monitoring-and-control-objects

Documentation templates

WinAC_MP

Libraries

to (Global libraries)

Library Buttons-and-Switches was open...

เลือก Tag ที่จะให้แสดงผล

Name	Data type	Address	Co...
Run_Motor	Bool	%Q0.2	
Speed_Motor		%MW0	
Start	Bool	%I0.0	
Stop	Bool	%I0.1	

Simulation

Approved
Partner

Factory
Automation

SIEMENS

The screenshot shows the Siemens SIMATIC Manager Basic Workshop interface. The main window displays a HMI screen titled 'P1.1' with a 'TOUCH' label. The screen contains a 'Back' button, a numeric input field showing '0', and several control buttons: 'Start/Stop', 'Start', 'Start', 'OFF', and 'Stop'. A red dashed rectangle highlights the central control area. The left sidebar shows the project tree with 'HMI_basic [KTP700 Basic PN]' expanded, and 'Screens' containing 'Main', 'P1', 'P1.1', 'P1.2', 'P2', 'P2.1', and 'P2.2'. The bottom status bar shows 'Project Basic workshop opened.'.

1

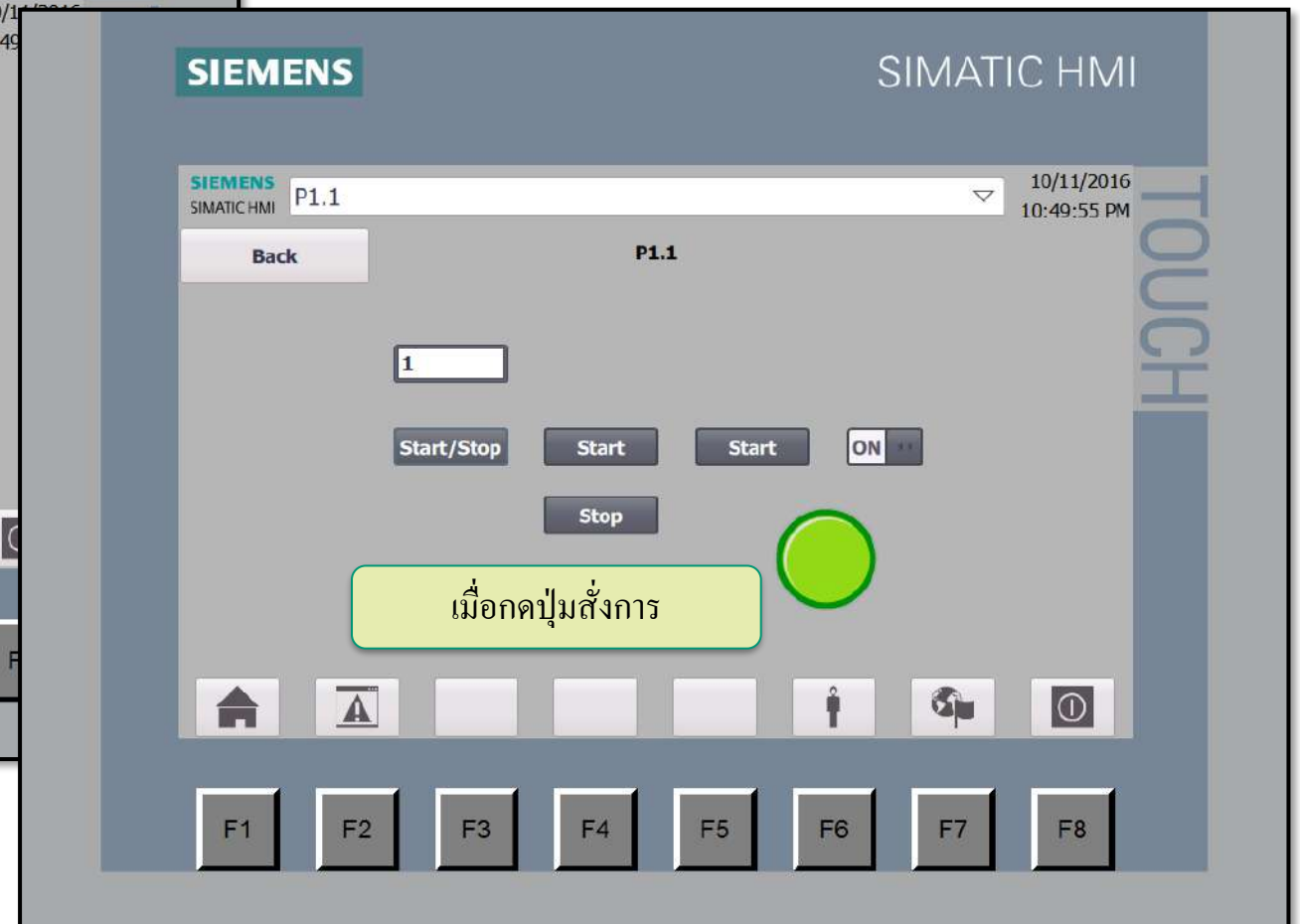
ในกรณีถ้า  (สัญลักษณ์ Simulation) ไม่
ขึ้น ให้กดที่หน้าจอ 1 ครั้ง

Simulation

Approved
Partner

Factory
Automation

SIEMENS



Exercise

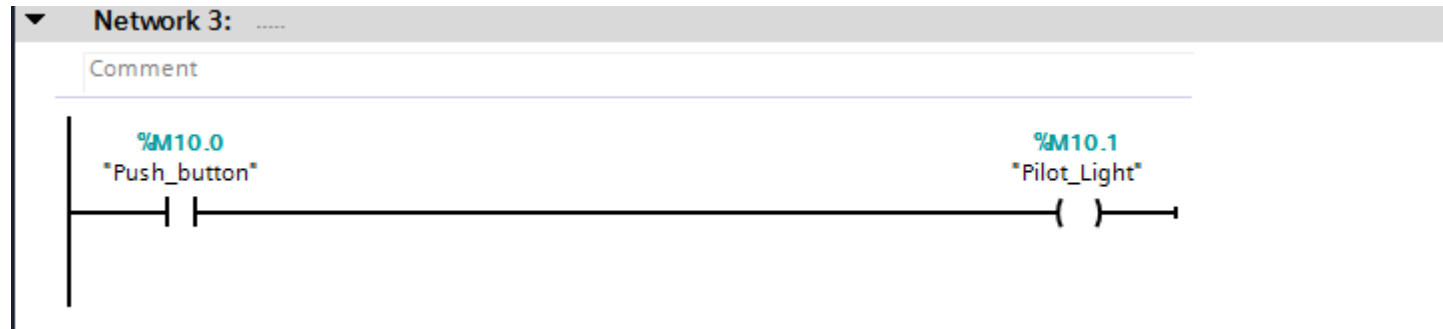
Button / Pilot Light

Approved
Partner

Factory
Automation

SIEMENS

1. สร้างโปรแกรมใน PLC แบบง่ายๆใน network ใหม่ที่มีแค่ contact NO และ Output ขึ้นมา



2. สร้าง Button ที่หน้าจอ HMI 4 ตัว โดยให้ Button ทั้ง 4 ตัว link ไปหา tag ที่เป็น contact NO ของ PLC เหมือนกัน
ทุกตัว แต่เราจะกำหนด Events ให้ 4 ตัวมีหน้าที่แตกต่างกัน 4 แบบดังนี้

- SetBit
- ResetBit
- SetbitWhilekeyPressed
- InvertBit

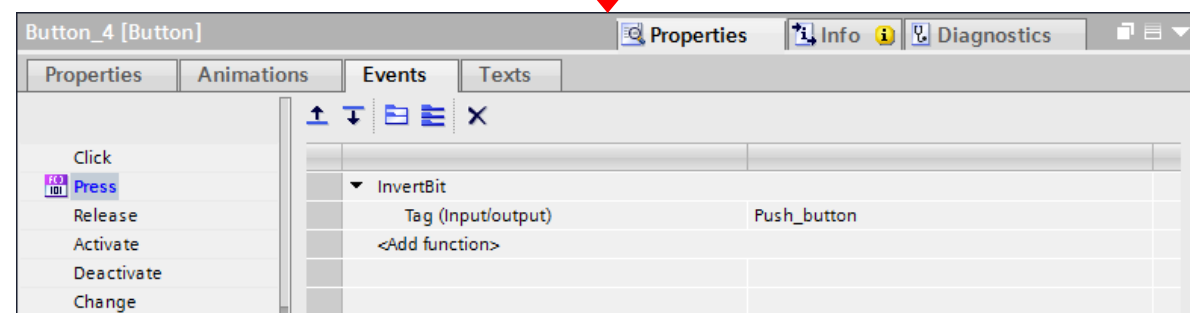
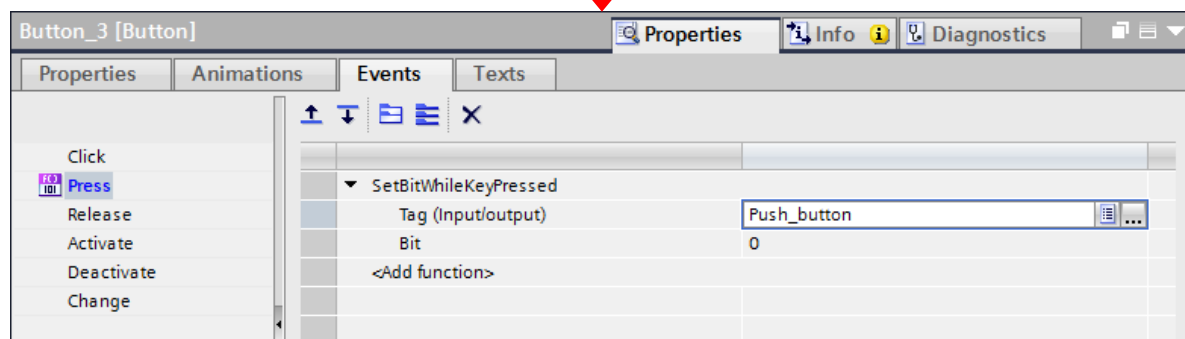
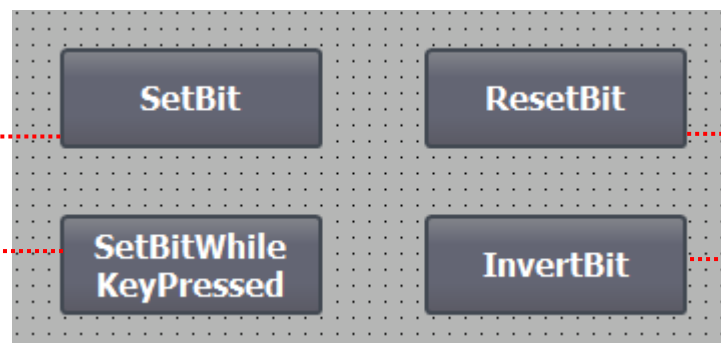
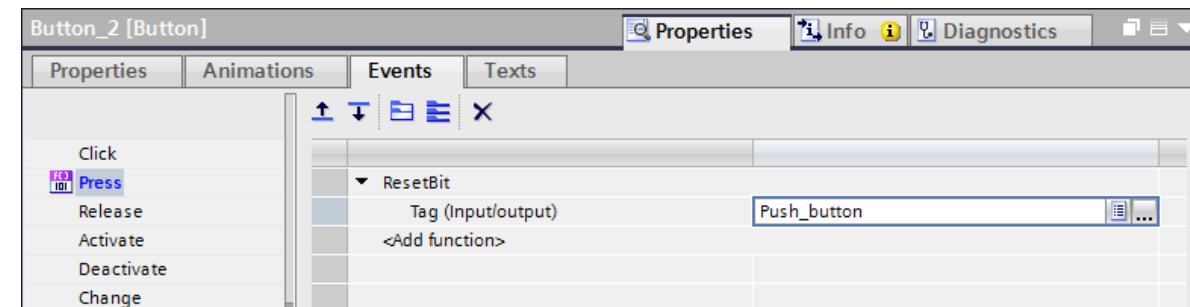
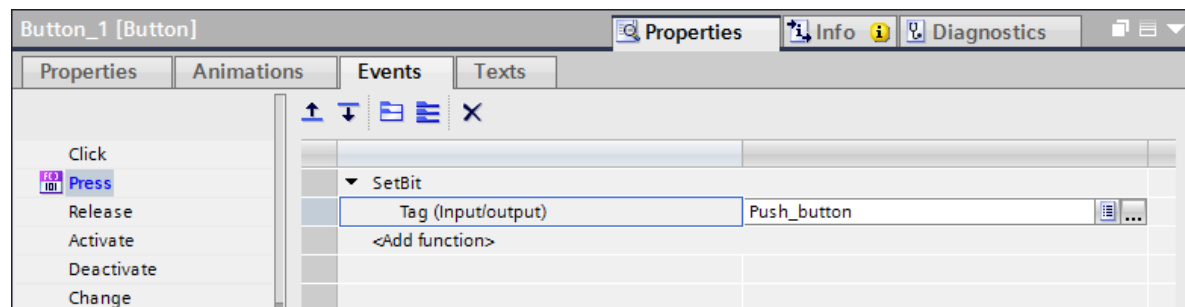
Exercise

Button / Pilot Light

Approved
Partner

Factory
Automation

SIEMENS



Exercise

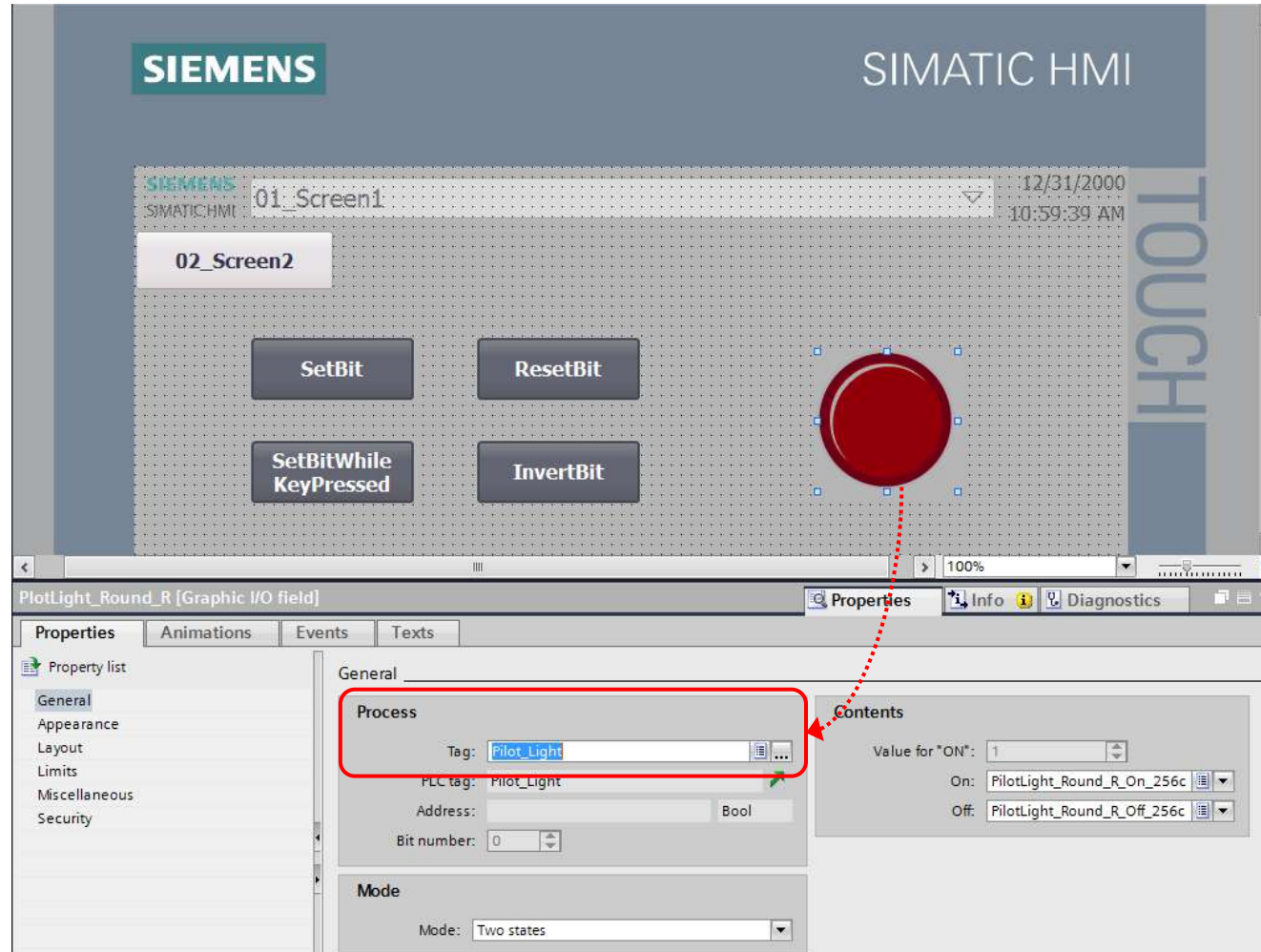
Button / Pilot Light

Approved
Partner

Factory
Automation

SIEMENS

3. วาง Pilot light ที่หน้าจอ โดยให้ link กับ Output contact ของ PLC



Exercise

Button / Pilot Light

Approved
Partner

Factory
Automation

SIEMENS

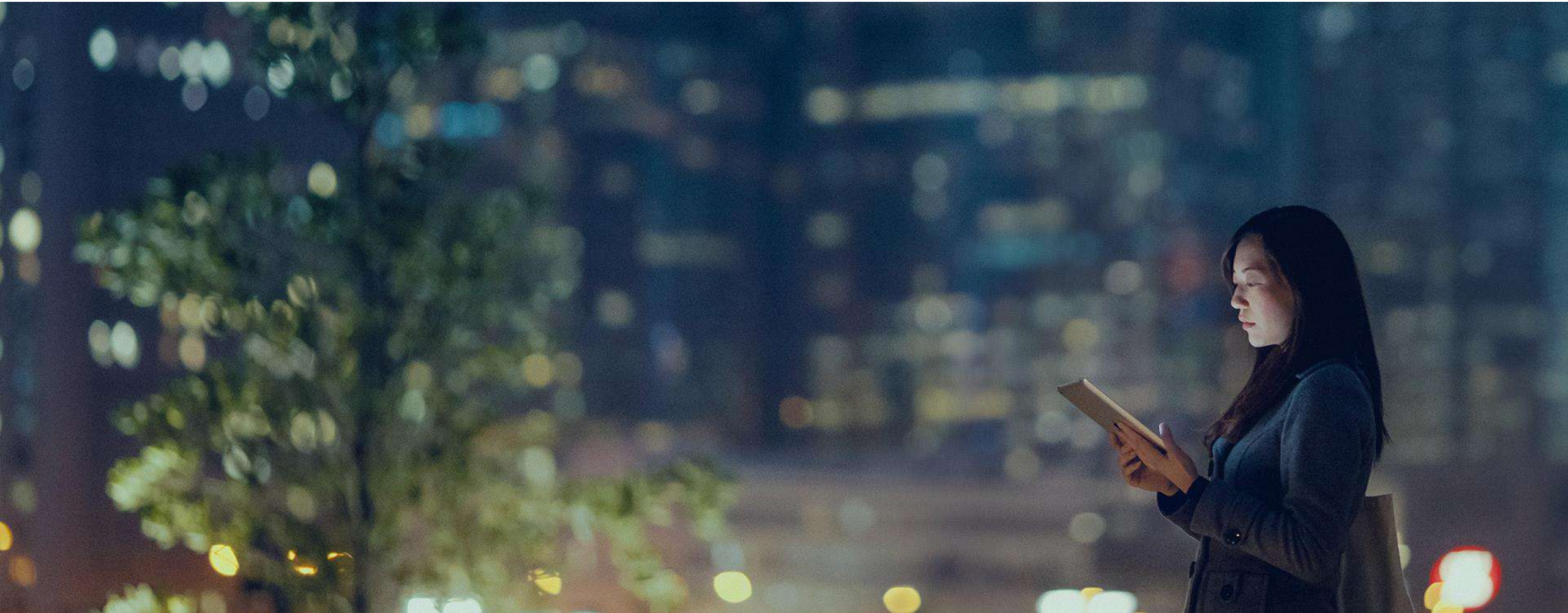
4. ทำการ download โปรแกรมลงทั้ง PLC และ HMI
5. สังเกตการเปลี่ยนแปลงว่า Button แต่ละแบบมีรูปแบบการทำงานอย่างไรบ้าง

03 - Alarm

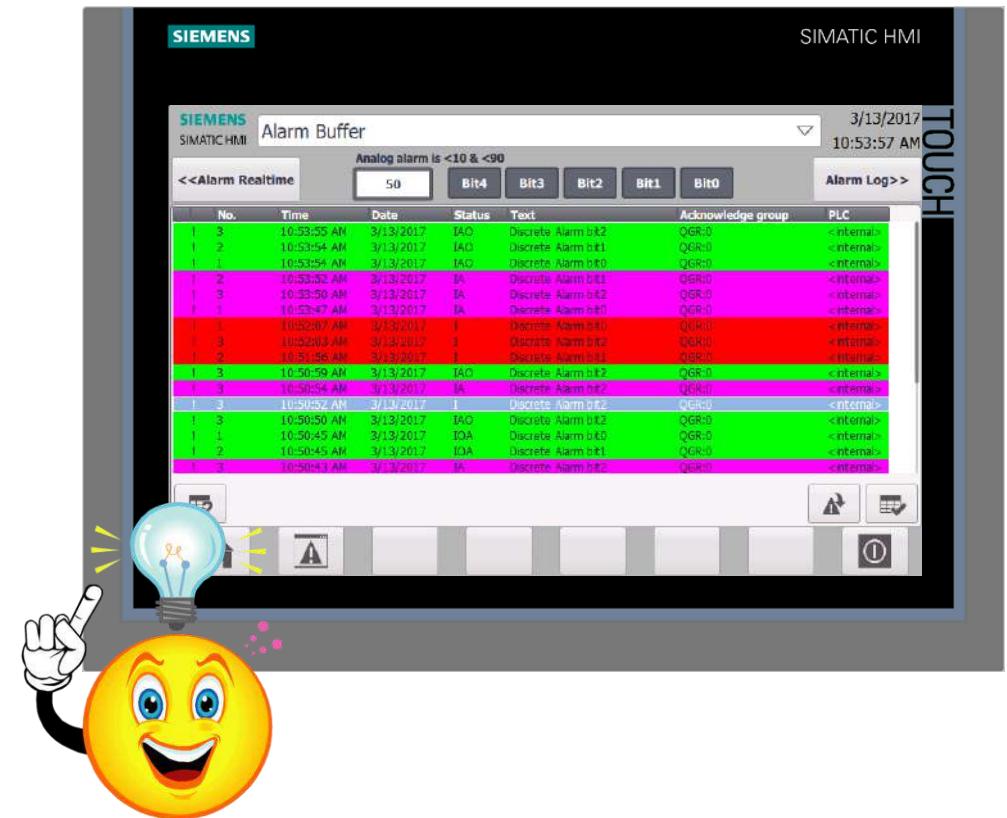
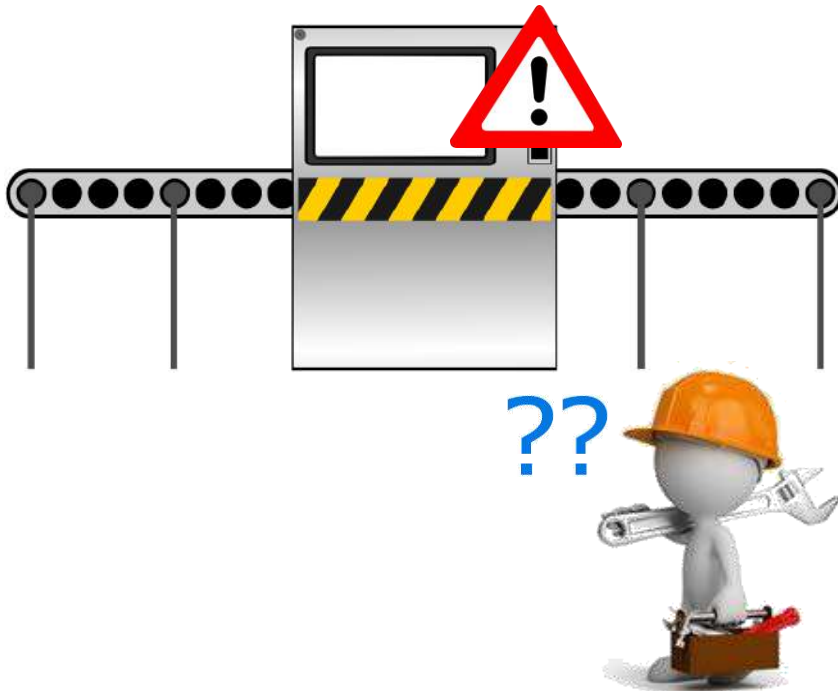
Approved
Partner

Factory
Automation

SIEMENS



- Alarm เป็น feature พื้นฐานที่ใช้เพื่อบอกความผิดปกติที่เกิดขึ้นในเครื่องจักร โดยจะทำการแสดงผลเป็นข้อความที่ทำให้ผู้ใช้งานสามารถเข้าใจได้ทันทีว่ามีความผิดปกติเกิดขึ้นที่ใด



- ดังนั้นสิ่งที่ผู้ใช้งานต้องทำเป็นอันดับแรกคือ ทำการลงทะเบียนบอกให้จอ HMI ทราบก่อนว่า หาก Boolean ตัวใด on ขึ้นมาจะ
ให้แสดงข้อความ error ใดบนหน้าจอ หรือ Integer ตัวใดมีค่าออกนอกขอบเขตที่ต้องการ แล้วจะให้แสดงข้อความอะไรที่
หน้าจอเป็นต้น
- ชนิดของ Alarm แบ่งเป็น 2 แบบคือ
 1. Discrete alarm : ใช้ตรวจสอบข้อมูลแบบ Boolean
 2. Analog alarm : ใช้ตรวจสอบข้อมูลแบบ Word / Integer

Alarm Registration Analog Alarm

Approved
Partner

Factory
Automation

SIEMENS

- ทำการลงทะเบียน Analog alarm โดยเลือก PLC tag และตั้ง Limit ที่ต้องการ จากนั้นใส่ Info text ไปด้วย

The screenshot shows the Siemens HMI configuration software interface. The 'Project tree' on the left lists the project structure, with 'HMI_1 [KTP700 Basic PN]' selected. The 'HMI alarms' folder is expanded, and the 'Analog alarms' tab is active. A table of analog alarms is displayed, with the first entry 'Analog_alarm_1' highlighted. The 'Alarm text' column shows 'Flow too high', and the 'Limit' column shows '100'. The 'Info text' field is set to 'Please confirm motor speed.'.

ID	Name	Alarm text	Alarm class	Trigger tag	Limit	Limit mode
1	Analog_alarm_1	Flow too high	Errors	Flow_Sens..	100	Higher

The 'Info text' field is set to: Please confirm motor speed.

Create Alarm view

Approved
Partner

Factory
Automation

SIEMENS

The screenshot displays the Siemens WinCC Graphics Designer interface for creating an alarm view. The main workspace shows a 'Root screen' template with a 'TOUCH' label. The left sidebar contains the 'Project tree' with a hierarchy: Project2 > Test HMI [KTP700 Basic PN] > Screens > Root screen. The right sidebar features several toolboxes: 'Basic objects' (lines, shapes, text), 'Elements' (alarms, clocks, etc.), 'Controls' (a warning icon is highlighted with a red arrow and the text 'Drag & Drop'), and 'Graphics' (WinCC graphics folder, My graphics folder). The bottom panel shows the 'Properties' window for the 'Root screen' with the following settings:

- Name: Root screen
- Background color: 49, 207, 206
- Grid color: 0, 0, 0
- Number: 1
- Template: Template_1

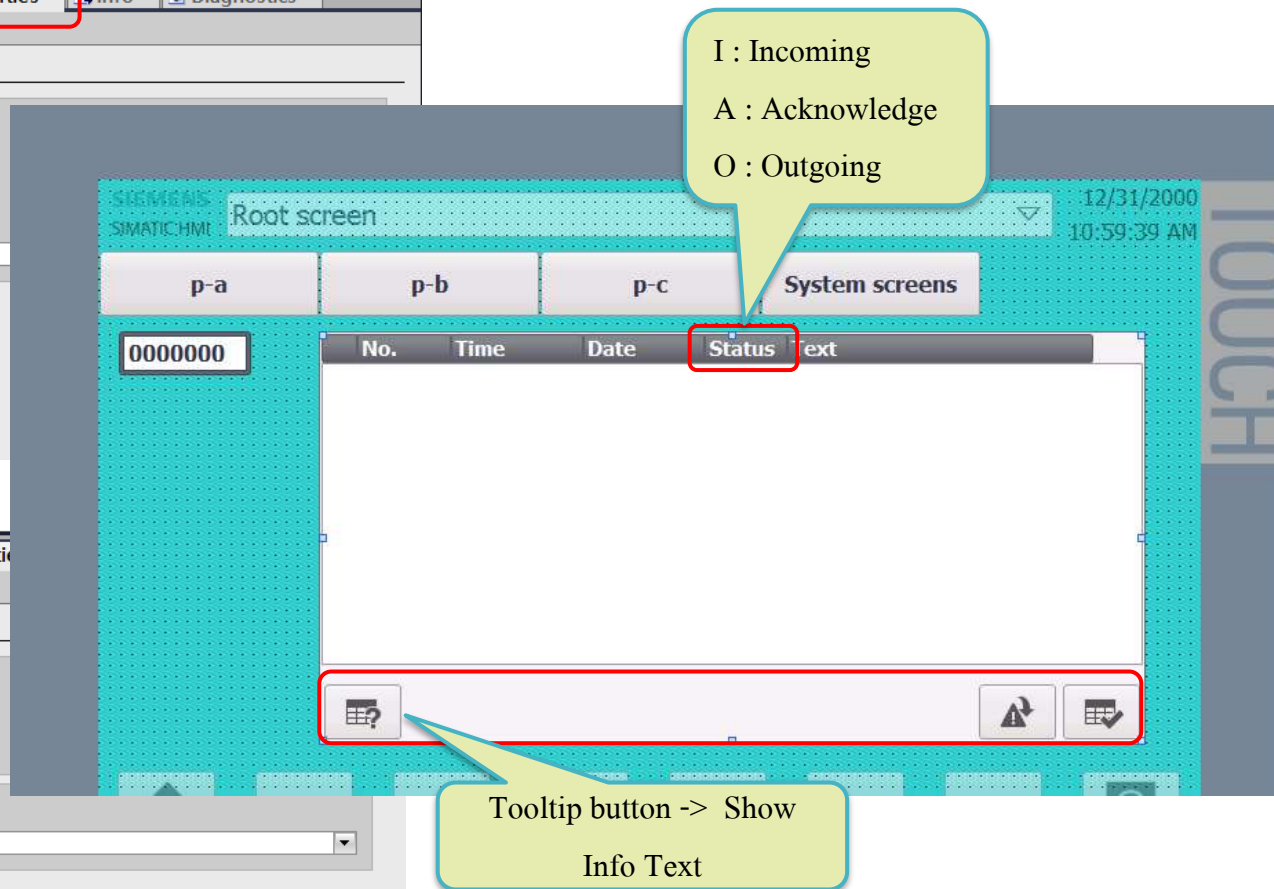
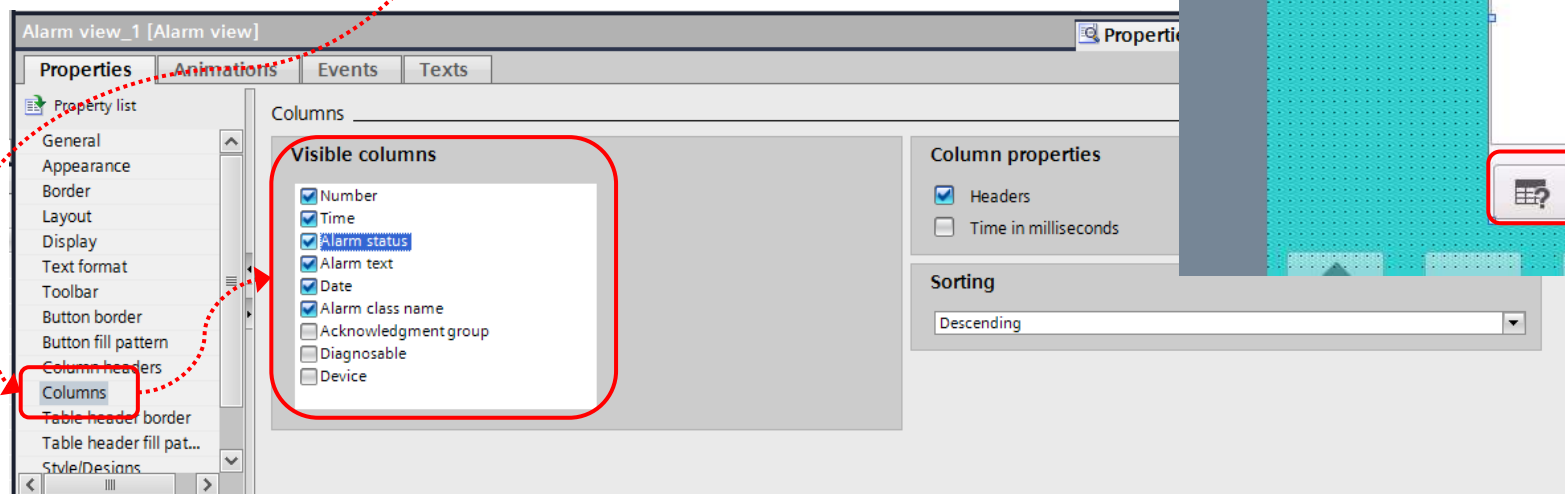
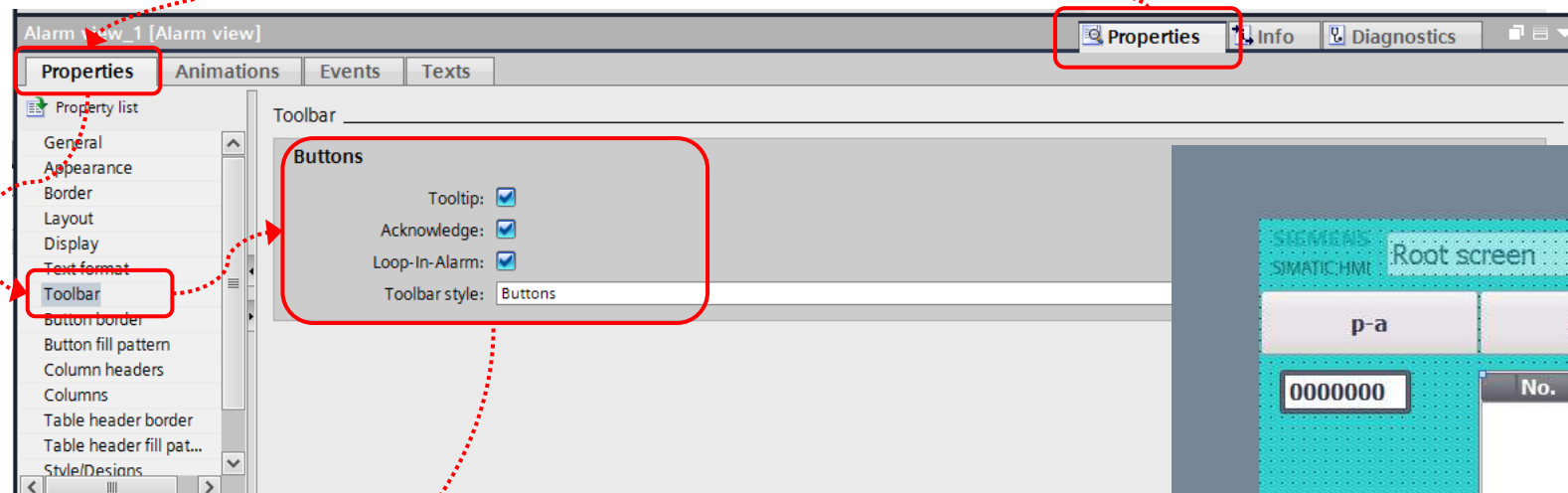
The status bar at the bottom indicates 'Wizard: successfully configured KTP700...'.

Setting Alarm view

Approved
Partner

Factory
Automation

SIEMENS

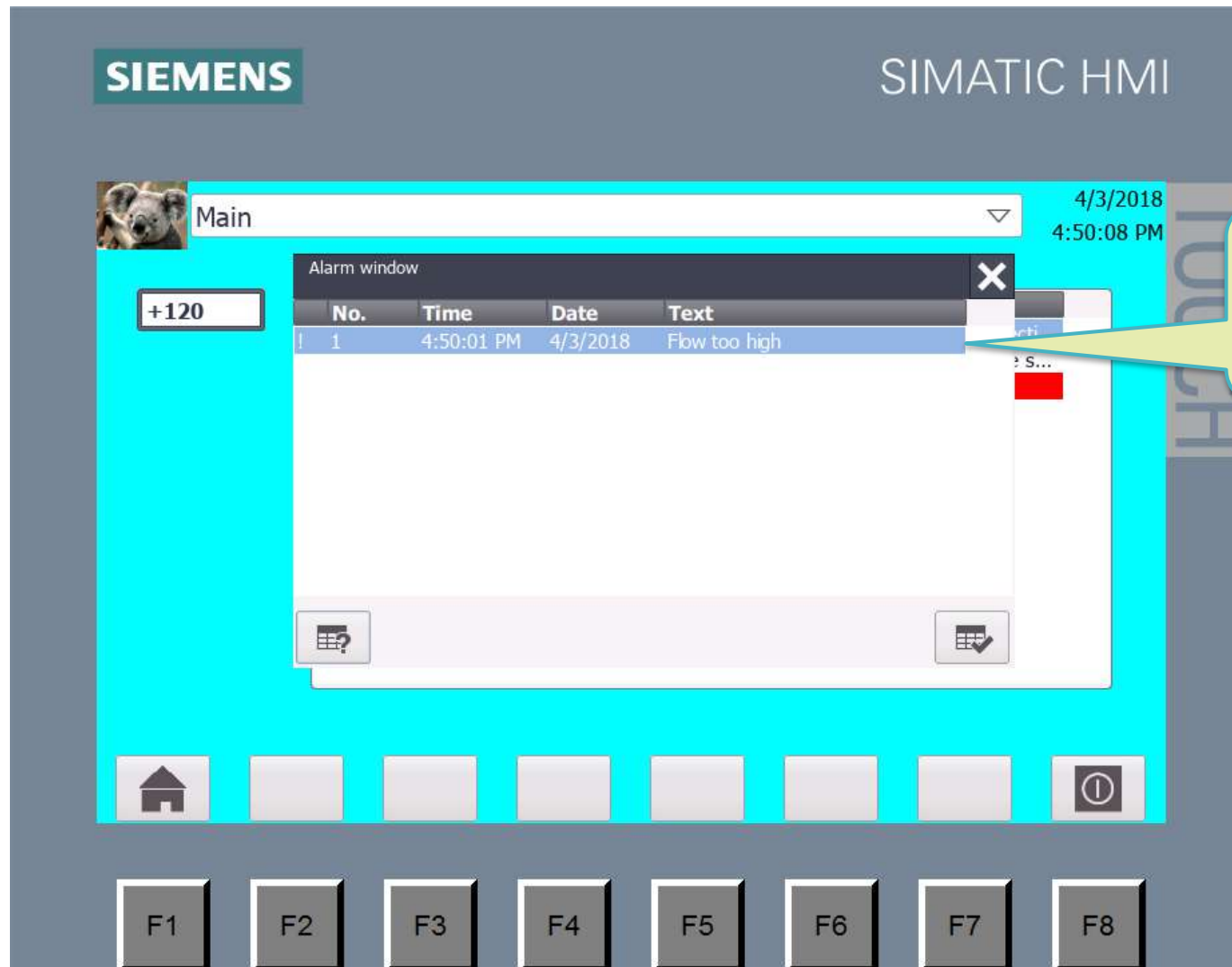


Simulation Pop-up alarm

Approved
Partner

Factory
Automation

SIEMENS



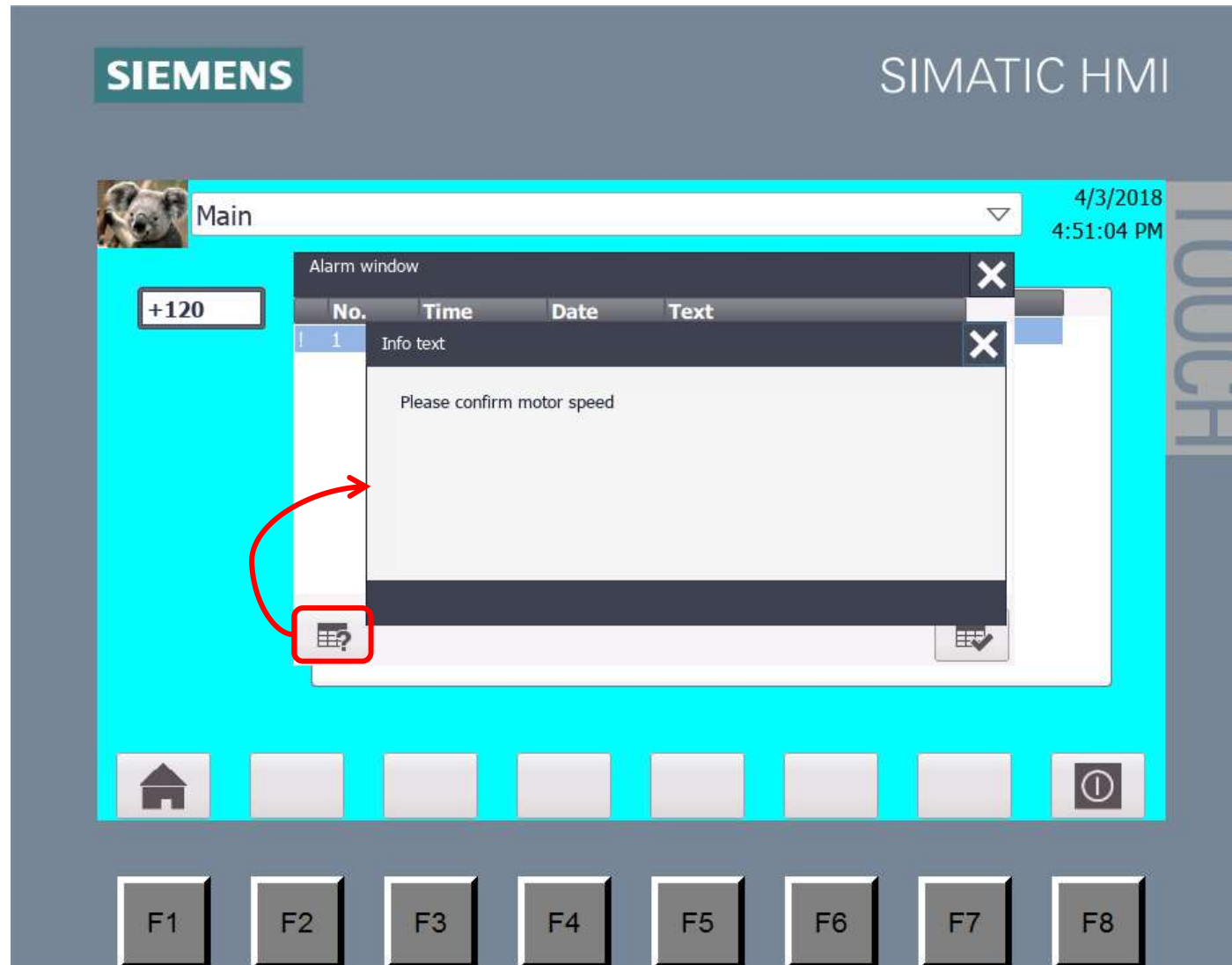
หากตอนทำ Wizard ได้ตั้งให้แสดง
Alarm เอาไว้ หน้าต่าง Alarm จะ popup
ขึ้นมาทันทีที่เกิด alarm

Simulation Info text

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Partner

Factory
Automation

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Alarm Registration

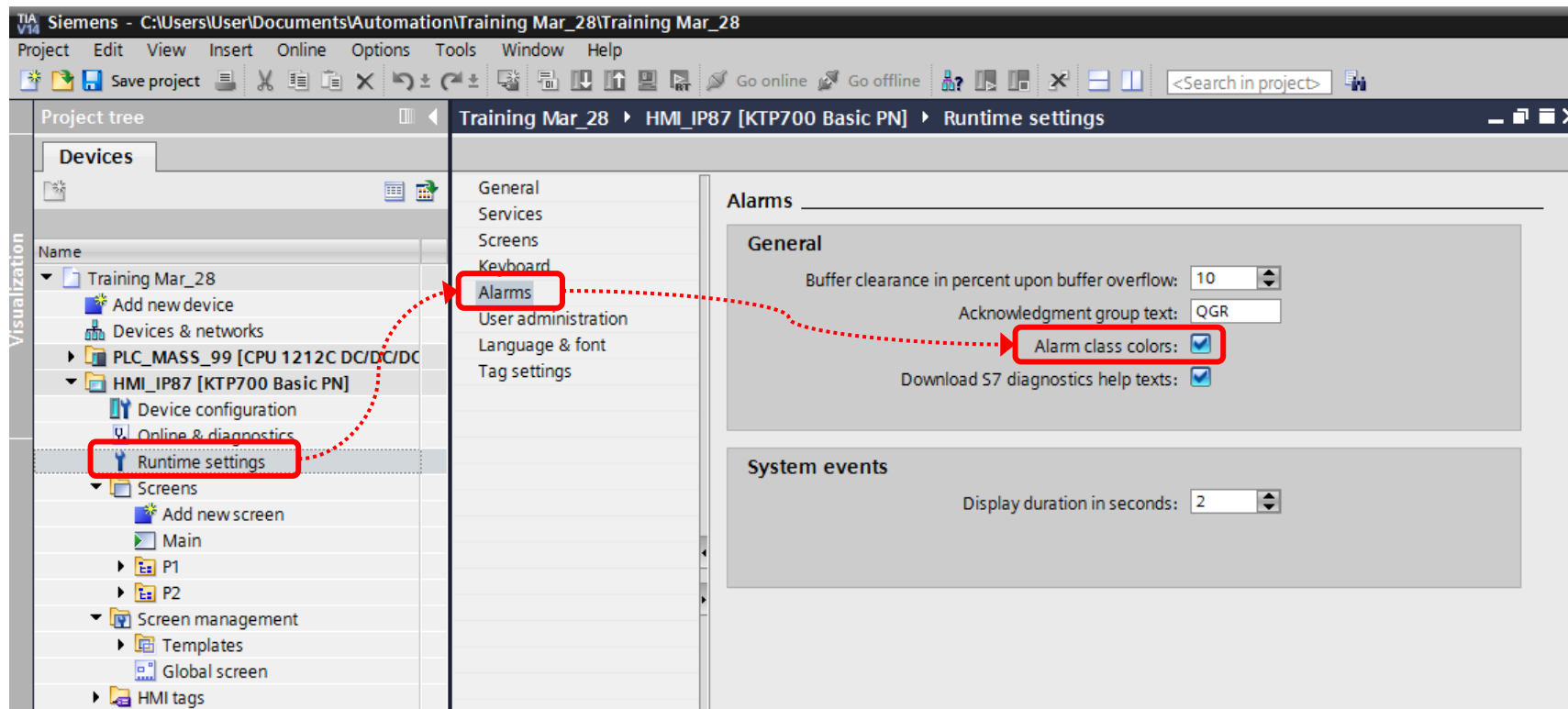
Set Alarm color

Approved
Partner

Factory
Automation

SIEMENS

- การที่จะให้ alarm แสดงสีนั้น เราต้องมีการตั้งค่าเพิ่มเติมด้วย โดยจะตั้งค่า 2 ที่คือ Runtime settings กับ Alarm classes



Alarm Registration

Set Alarm color

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Partner

Factory
Automation

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- การที่จะให้ alarm แสดงสีนั้น เราต้องมีการตั้งค่าเพิ่มเติมด้วย โดยจะตั้งค่า 2 ที่คือ Runtime settings กับ Alarm classes

The screenshot shows the Siemens SIMATIC Manager interface. On the left, the 'Project tree' displays the hierarchy: Training Mar_28 > HMI_IP87 [KTP700 Basic PN] > HMI alarms. The 'HMI alarms' folder is highlighted with a red box. On the right, the 'Alarm classes' configuration window is open. It has tabs for 'Discrete alarms', 'Analog alarms', 'System events', 'Alarm classes', and 'Alarm groups'. The 'Alarm classes' tab is selected. Below the tabs is a table with the following columns: 'Display name', 'Name', 'State machine', 'Log', 'Background color', 'Background color *In...', 'Background color *In...', and 'Background color *In...'. The 'Errors' row is highlighted with a red box, showing a red background color and a green background color *In... value. A red dotted line connects the 'HMI alarms' folder in the project tree to the 'Alarm classes' window.

Display name	Name	State machine	Log	Background color	Background color *In...	Background color *In...	Background color *In...
!	Errors	Alarm with single-mode ...	<No log>	255, 0, 0	0, 255, 0	255, 154, 255	0, 255, 0
S	System	Alarm without acknowle...	<No log>	255, 255, 255	255, 255, 255	255, 255, 255	255, 255, 255
A	Acknowledgement	Alarm with single-mode ...	<No log>	255, 0, 0	255, 0, 0	255, 255, 255	255, 255, 255
NA	No Acknowledgement	Alarm without acknowle...	<No log>	255, 0, 0	255, 0, 0	255, 255, 255	255, 255, 255
<Add new>							

Alarm Registration

Set Alarm color

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SIEMENS SIMATIC HMI

Main 4/3/2018 5:52:25 PM

+333

No.	Time	Date	Status	Text
! 1	5:52:22 PM	4/3/2018	IA	Flow too high
! 1	5:52:19 PM	4/3/2018	I	Flow too high
! 1	5:52:11 PM	4/3/2018	IO	Flow too high
! 1	5:52:06 PM	4/3/2018	I	Flow too high
! 1	5:52:03 PM	4/3/2018	IO	Flow too high
\$ 140001	5:52:02 PM	4/3/2018	I	Connection disconnected: HMI_...
\$ 140023	5:52:02 PM	4/3/2018	I	Error during time synchronizatio...
! 1	5:52:01 PM	4/3/2018	I	Flow too high
! 1	5:51:59 PM	4/3/2018	IO	Flow too high
! 1	5:51:57 PM	4/3/2018	I	Flow too high
\$ 110001	5:51:49 PM	4/3/2018	I	Change to operating mode

Navigation bar: Home, F1, F2, F3, F4, F5, F6, F7, F8, Alarm icon.

Alarm Registration

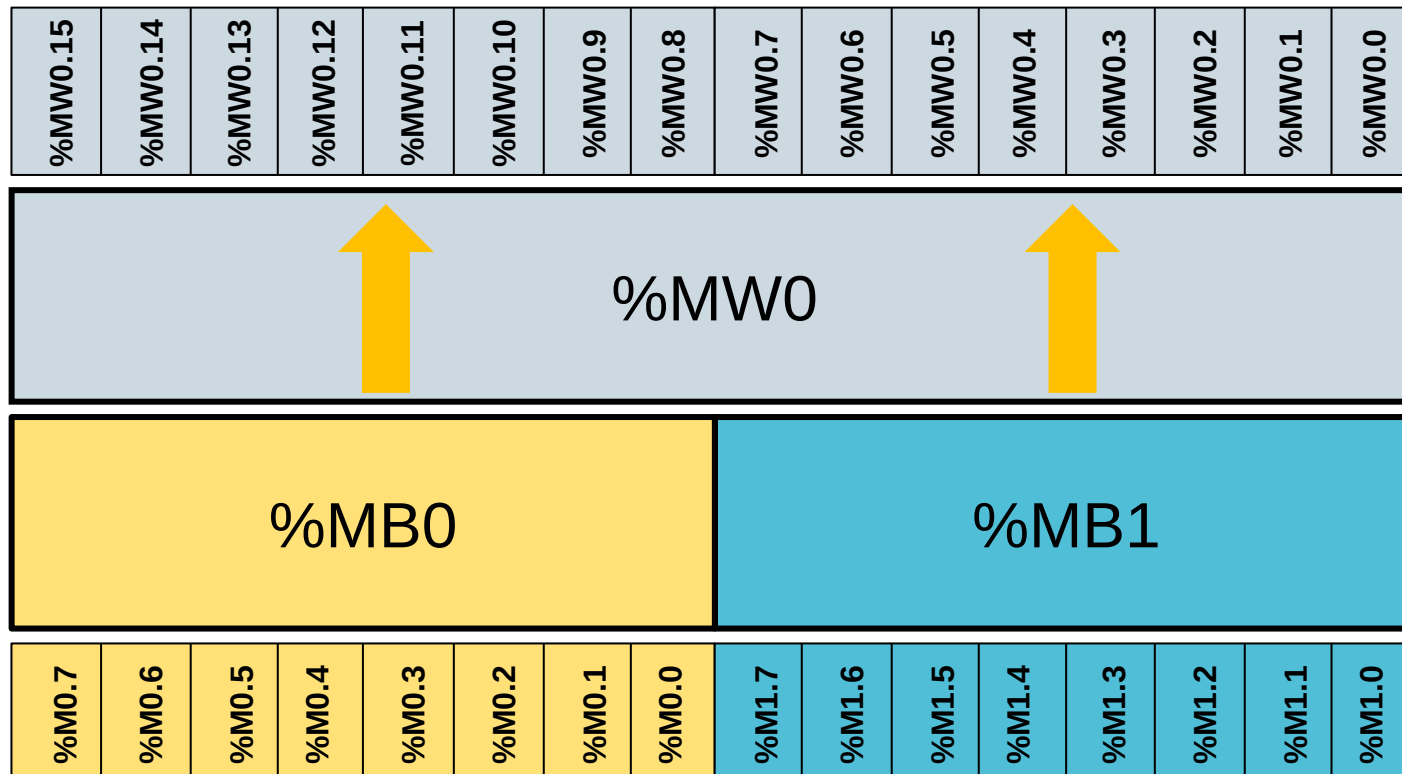
Discrete Alarm

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- แม้ว่า Discrete alarm จะดูข้อมูลแบบ Boolean แต่การตั้งค่าจะกำหนดเป็น “Bit ใน Word” แทน ดังนั้นเราจำเป็นต้องเข้าใจโครงสร้างของหน่วยความจำภายใน PLC เสียก่อน โดยการเรียง address แบบ Word, Byte และ Bit จะมีโครงสร้างดังรูป



สังเกตว่าการเรียง address ระหว่าง Word กับ Byte จะส่งผลกระทบต่อ Boolean เพราะมีการนำ Byte น้อยไปอยู่ในตำแหน่ง MSB ทำให้

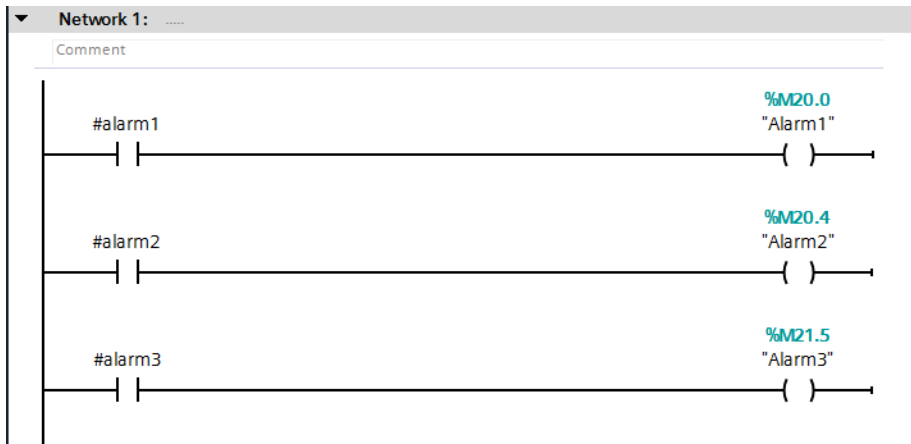
- MW0 bit0 = %M1.0
- MW0 bit7 = %M1.7
- MW0 bit8 = %M0.0
- MW0 bit15 = %M0.7

Alarm Registration Discrete Alarm

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Project tree

Project2 > PLC_1 [CPU 1214C DC/DC/DC] > PLC tags > Default tag table [32]

Devices

PLC_1 [CPU 1214C DC/DC/DC]

Device configuration

Online & diagnostics

Program blocks

Technology objects

External source files

PLC tags

Show all tags

Add new tag table

Default tag table [32]

PLC data types

Watch and force tables

Online backup

Tags

Default tag table

	Name	Data type	Address	Retain	Access...	Write...	Visibl...	Comme
1	Alarm1	Bool	%M20.0		☒	☒	☒	
2	Alarm2	Bool	%M20.4		☒	☒	☒	
3	Alarm3	Bool	%M21.5		☒	☒	☒	
4	AlarmWord1	Word	%MW20		☒	☒	☒	
5	<Add new>				☒	☒	☒	

Project tree

Project2 > HMI_1 [KTP700 Basic PN] > HMI alarms

Devices

Project2

Add new device

Devices & networks

PLC_1 [CPU 1214C DC/DC/DC]

HMI_1 [KTP700 Basic PN]

Device configuration

Online & diagnostics

Runtime settings

Screens

Screen management

HMI tags

Connections

HMI alarms

Recipes

Discrete alarms

Analog alarms

System events

Discrete alarms

	ID	Name	Alarm text	Alarm class	Trigger tag	Trigge..	Trigger address
	1	Discrete_alarm_1	Alarm message1	Errors	AlarmWord1	8	AlarmWord1.x8
	2	Discrete_alarm_2	Alarm message2	Errors	AlarmWord1	12	AlarmWord1.x12
	3	Discrete_alarm_3	Alarm message3	Errors	AlarmWord1	5	AlarmWord1.x5
		<A...					

ปัญหา

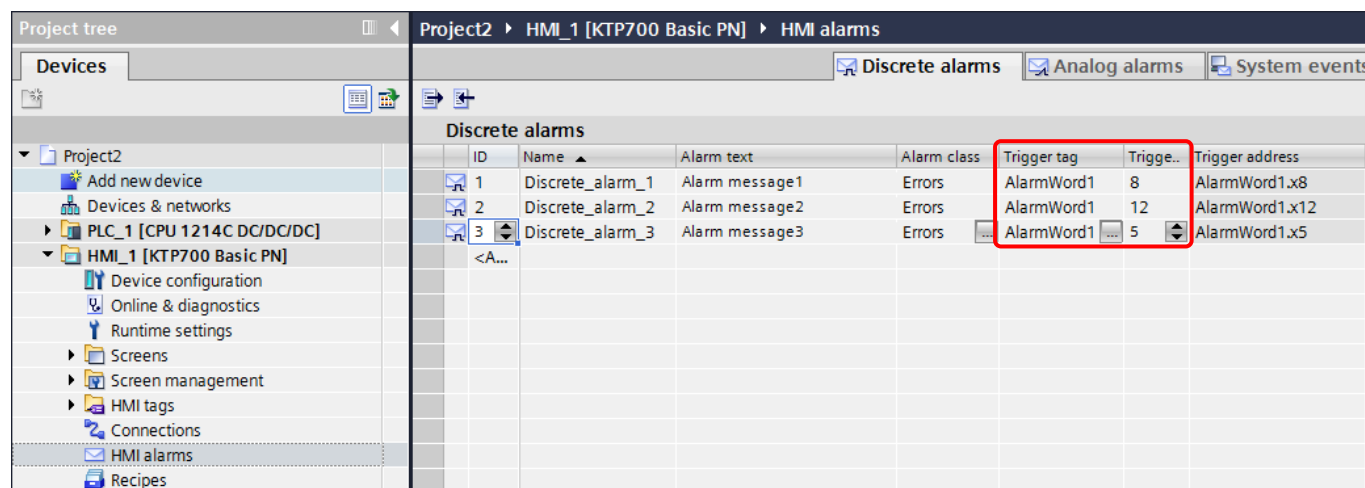
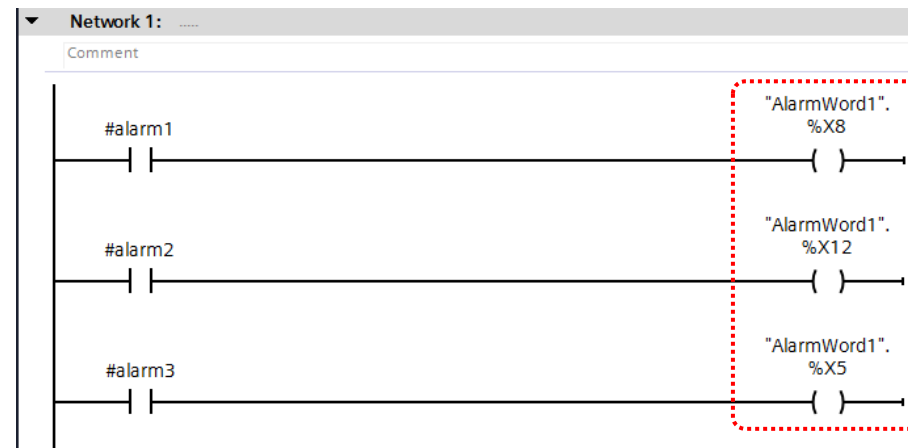
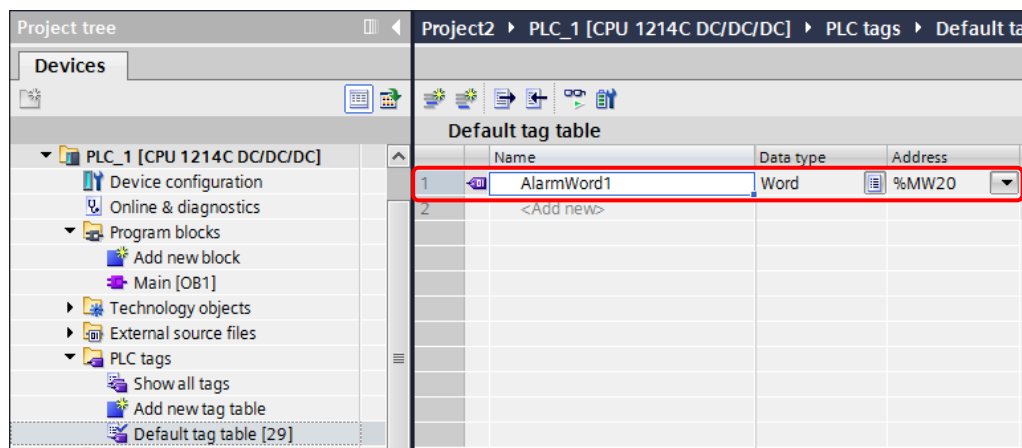
- ใน PLC สร้าง alarm เป็น %Mx.x
- แต่ HMI บังคับให้ใส่เป็น Word
- ดังนั้นต้องสร้าง tag ของ Word ที่ PLC เพิ่มอีก 1 ตัว
- และ map %Mx.x ให้เป็น %MW.x อีกทีใน HMI

Alarm Registration Discrete Alarm

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วิธีการ

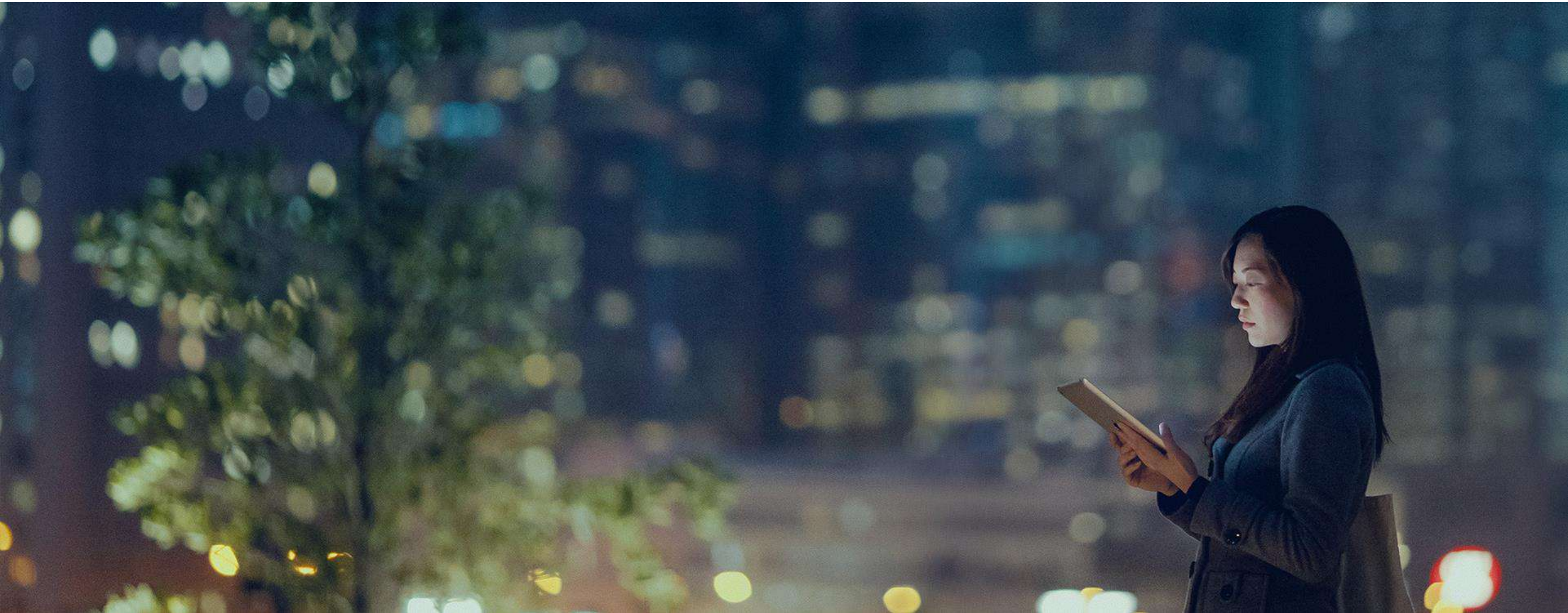
- สร้าง tag ของ Word เพื่อใช้เป็น Alarm ใน PLC
- ใช้รูปแบบ <tag name>.%Xn เพื่อทำ alarm
- ใน HMI จะใส่รูปแบบเดียวกับ PLC ได้เลย

04 - Web browser

Approved
Partner

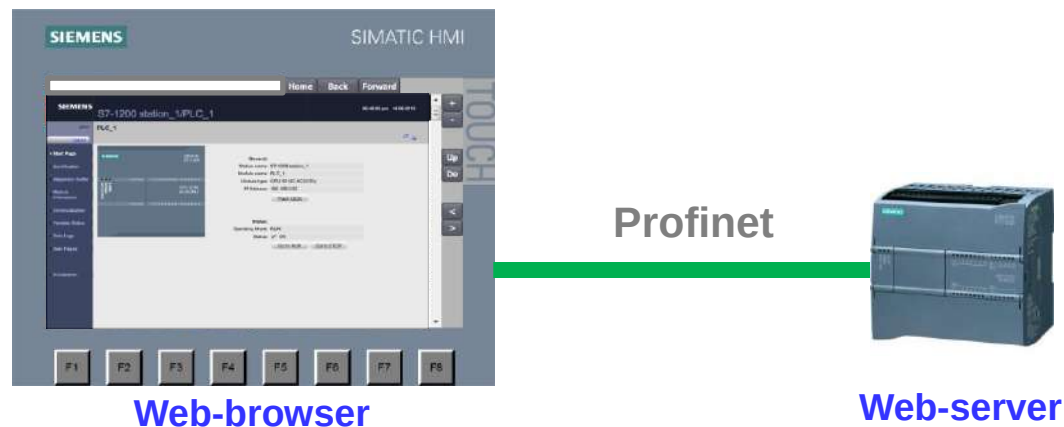
Factory
Automation

SIEMENS

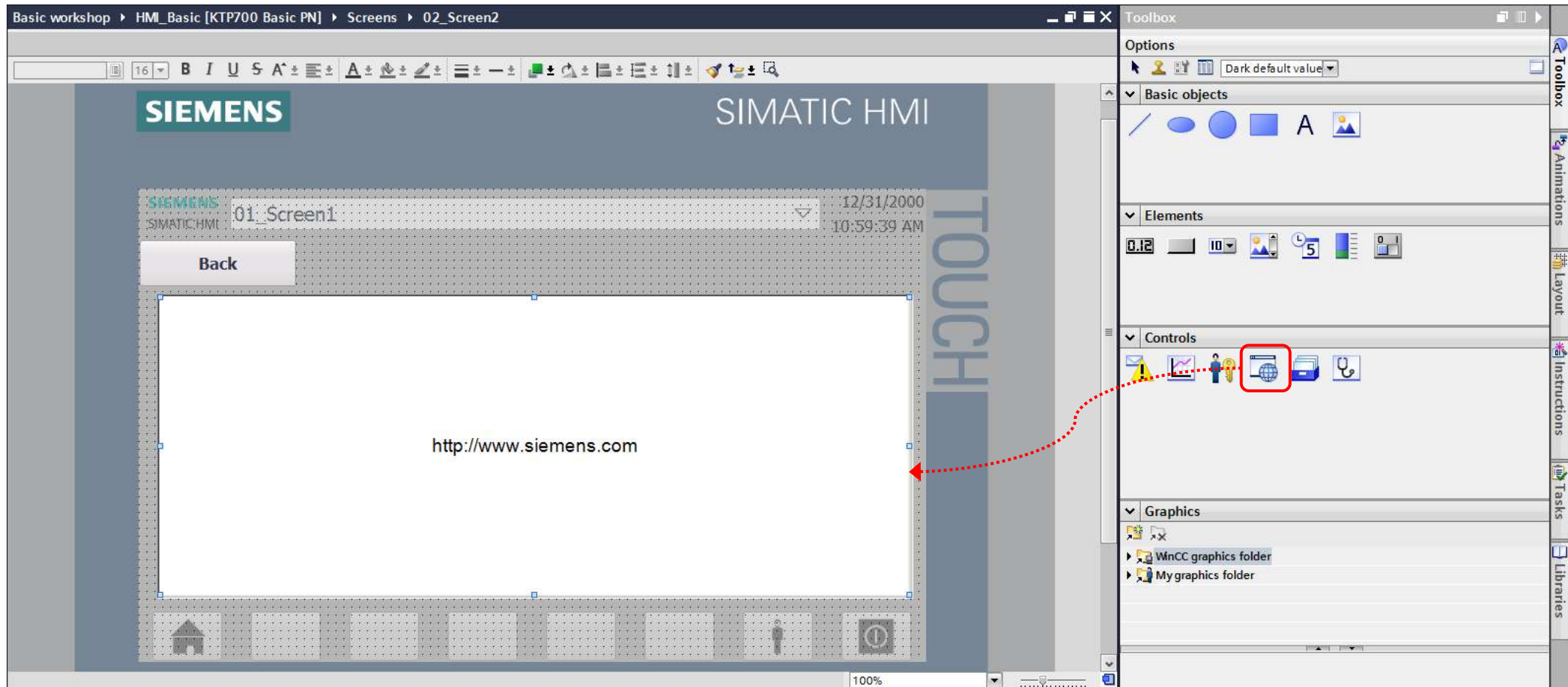


หน้าจอ Basic Panel นั้น สามารถทำหน้าที่เป็น Web browser เสมือนการเปิดหน้าเว็บผ่านทาง Internet Explorer หรือ Chrome ได้

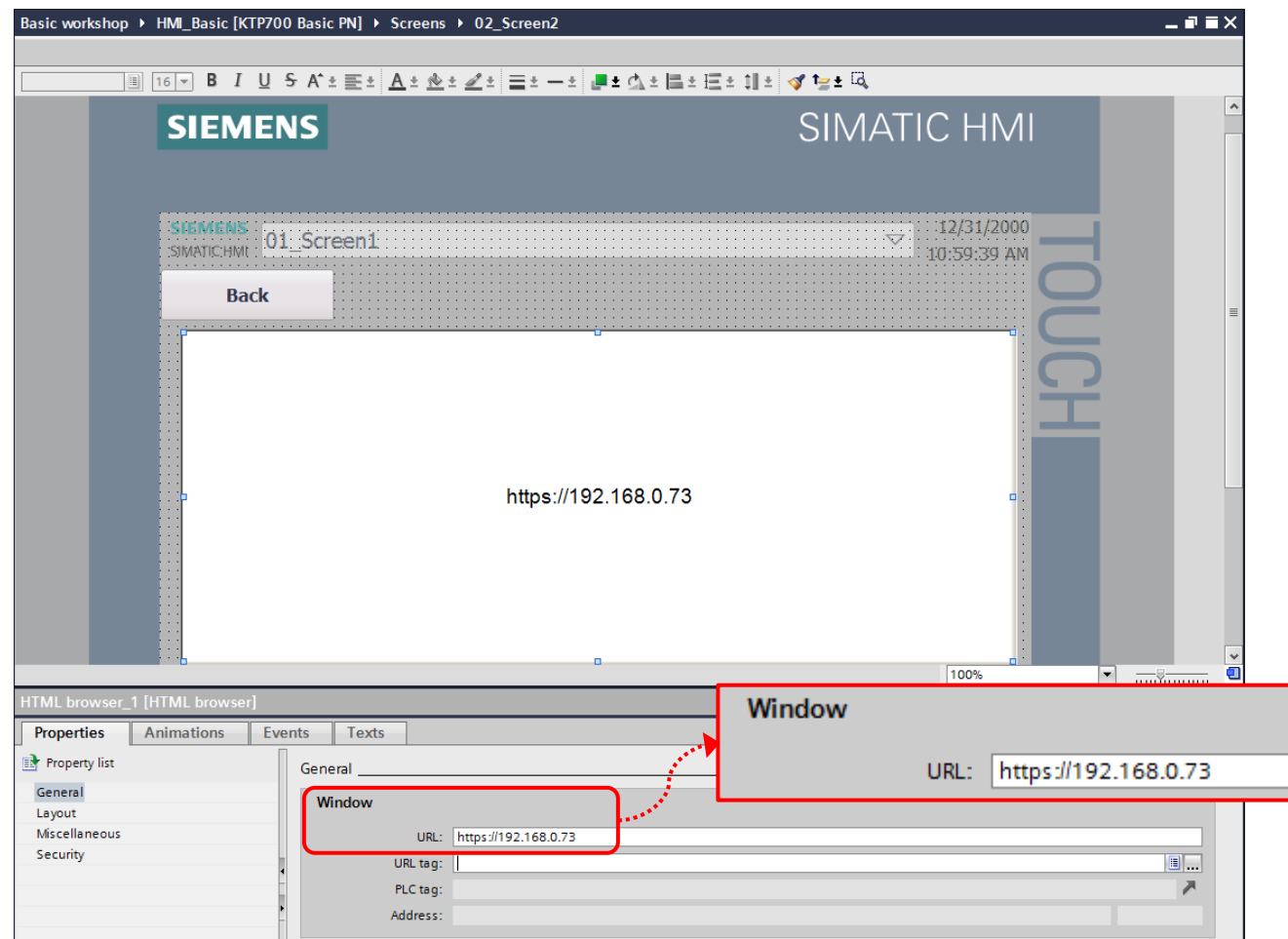
ดังนั้นหากเราประยุกต์ใช้ Web browser นี้ ร่วมกับ function Web-Server ของ PLC S7-1200 แล้ว เราก็จะสามารถใช้หน้าจอ Basic Panel เพื่อดูสถานะต่างๆของ S7-1200 ได้ทันที โดยที่ไม่ต้องใช้ computer มาดูสถานะเลย



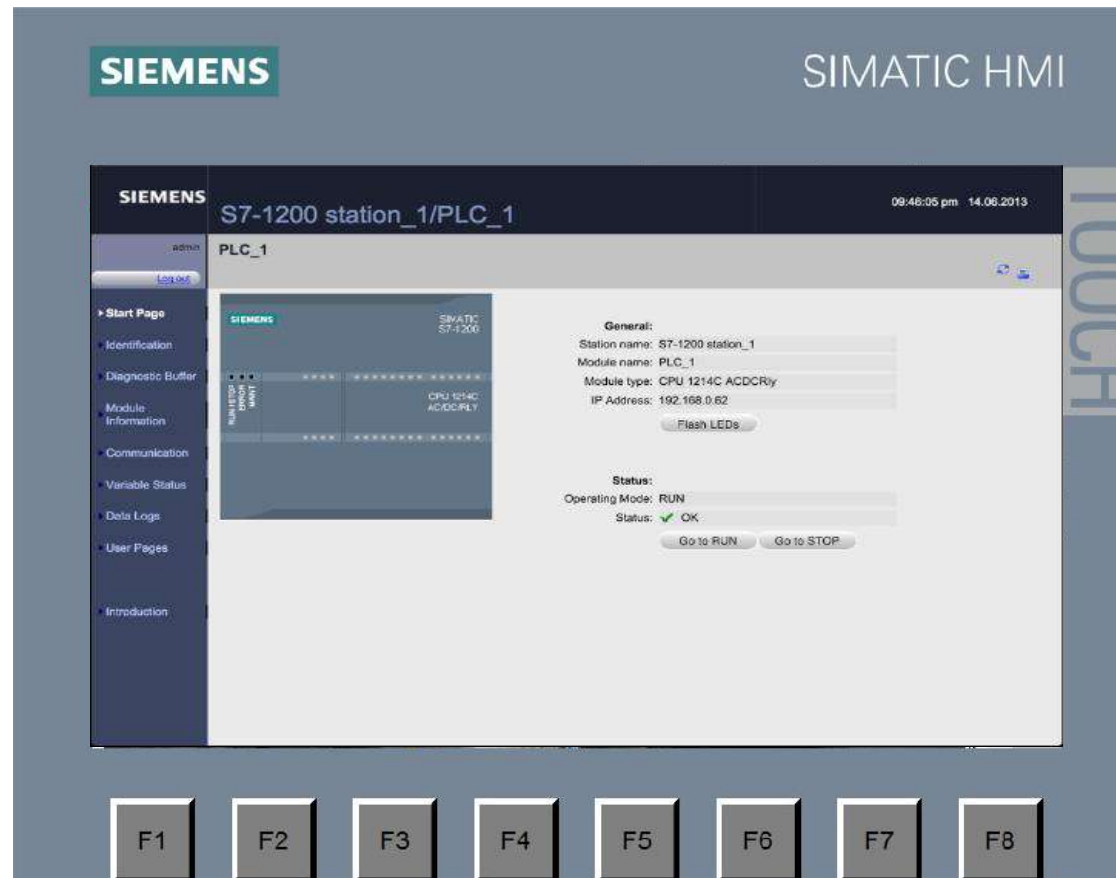
1. วางพาร์ท HTML browser จากในส่วนของ Toolbox -> Control มาวางที่หน้าจอ



2. ตั้งค่า Properties ของ HTML browser โดยใส่ URL เป็น https:// ตามด้วย IP address ของตัว PLC ที่เชื่อมต่อกับ HMI (และ PLC ได้ทำการเปิดใช้งาน Web-Server ไว้แล้ว)



3. เมื่อทำการโหลด project ลง HMI หน้าจอจะแสดงหน้า Web-Server ของ PLC เราสามารถใช้งาน Web-Server จาก HMI ได้เหมือนการเปิด web browser จาก computer ได้เลย

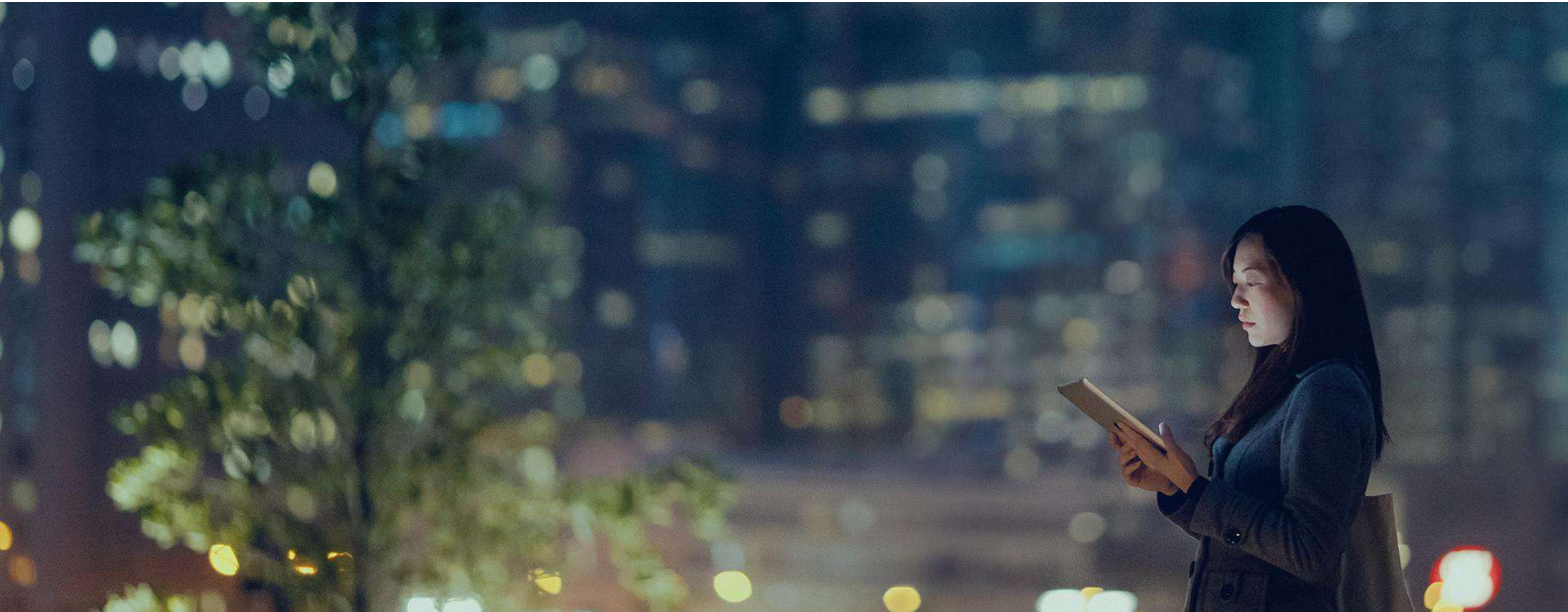


Sm@rt Server

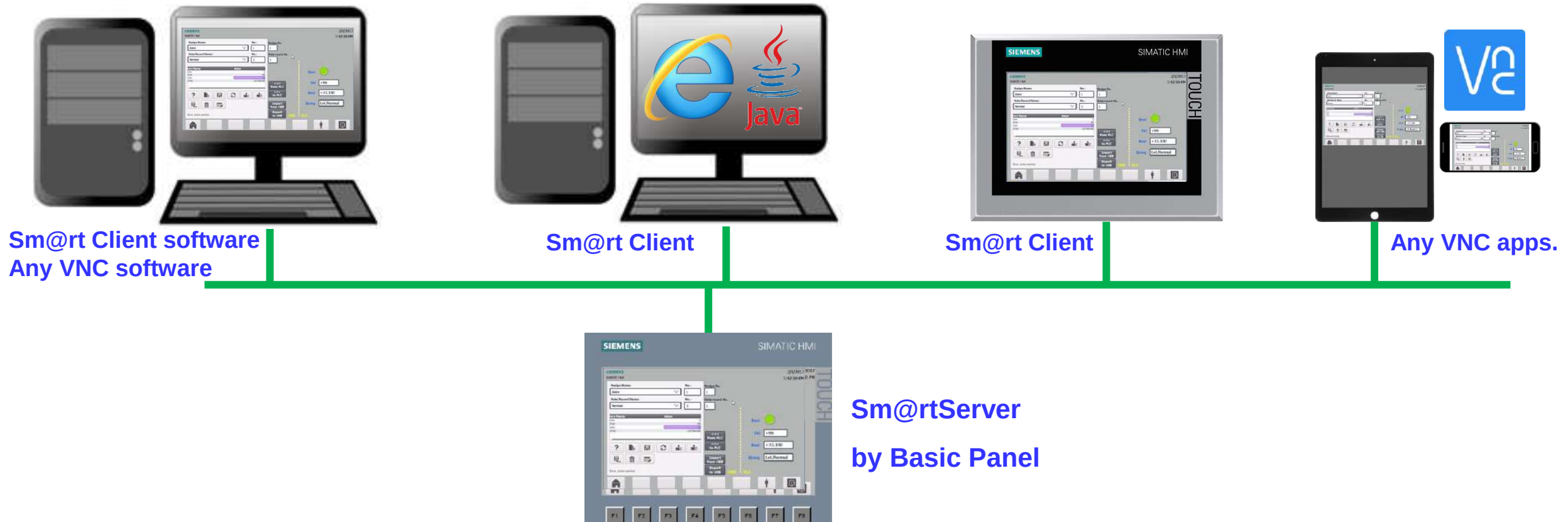
Approved
Partner

Factory
Automation

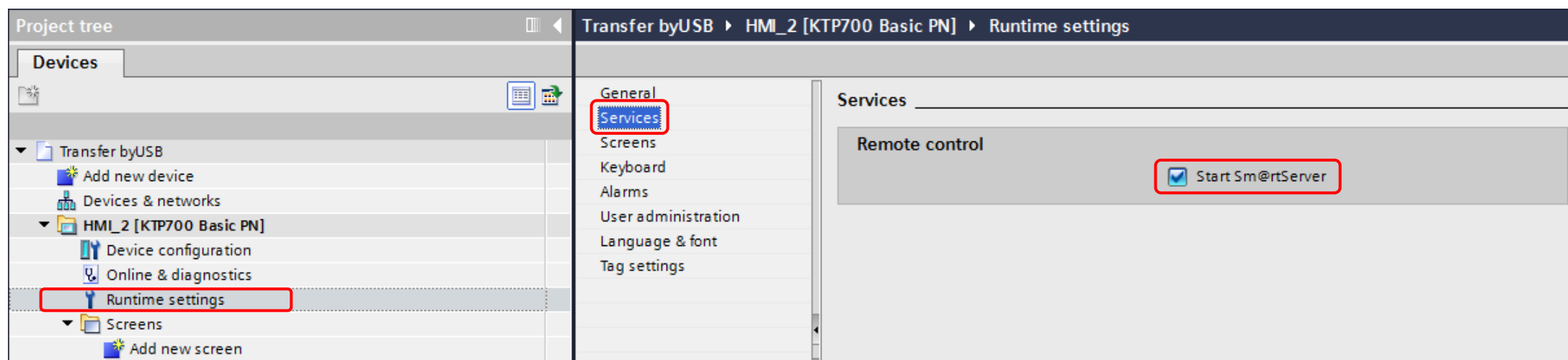
SIEMENS



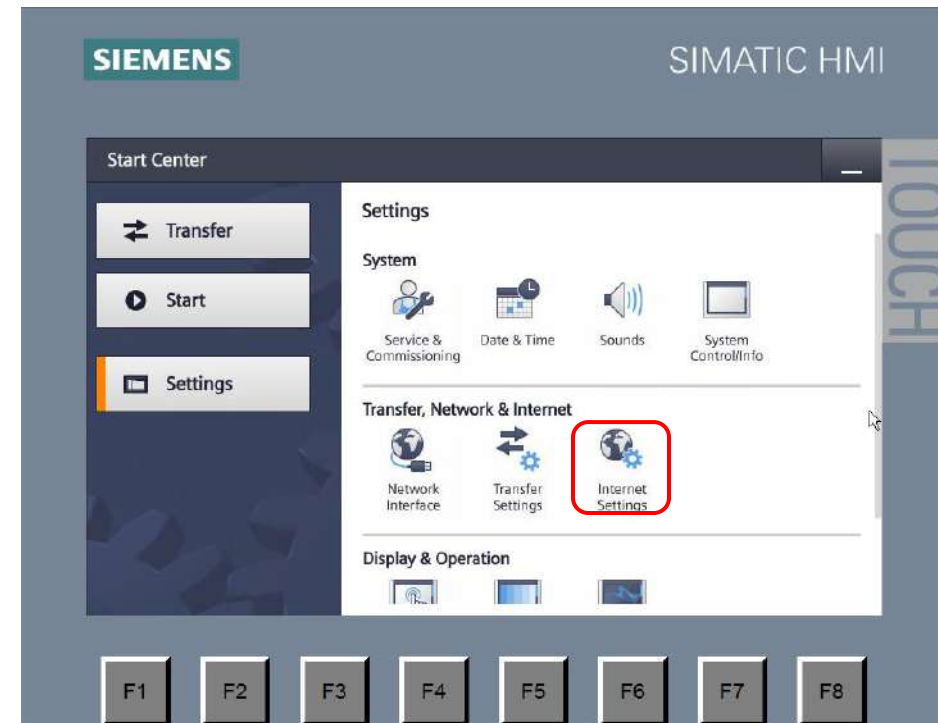
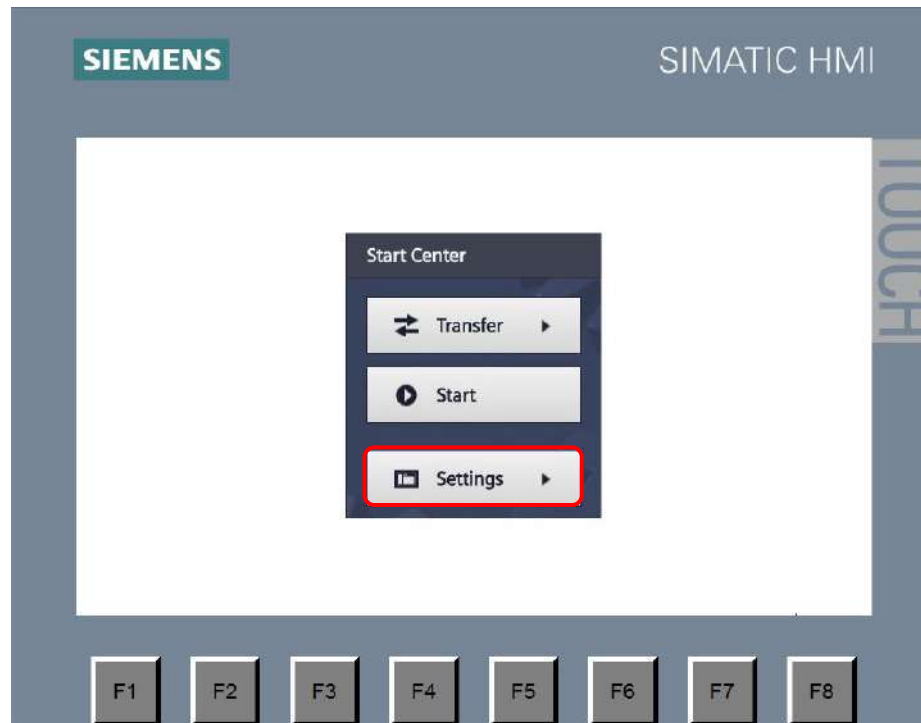
- ตั้งแต่ TIA Portal V14 เป็นต้นไป เราสามารถใช้ Basic Panel เป็น Sm@rt Server ได้แล้ว
แต่ยังคงต้องใช้ license “**SIMATIC WinCC Sm@rtServer for SIMATIC Basic Panels**” เพื่อ transfer เข้าจอ Basic Panel ด้วย
ในขณะที่ถ้าเป็นจอ Comfort Panel จะสามารถเปิดใช้งานได้ฟรีเลย



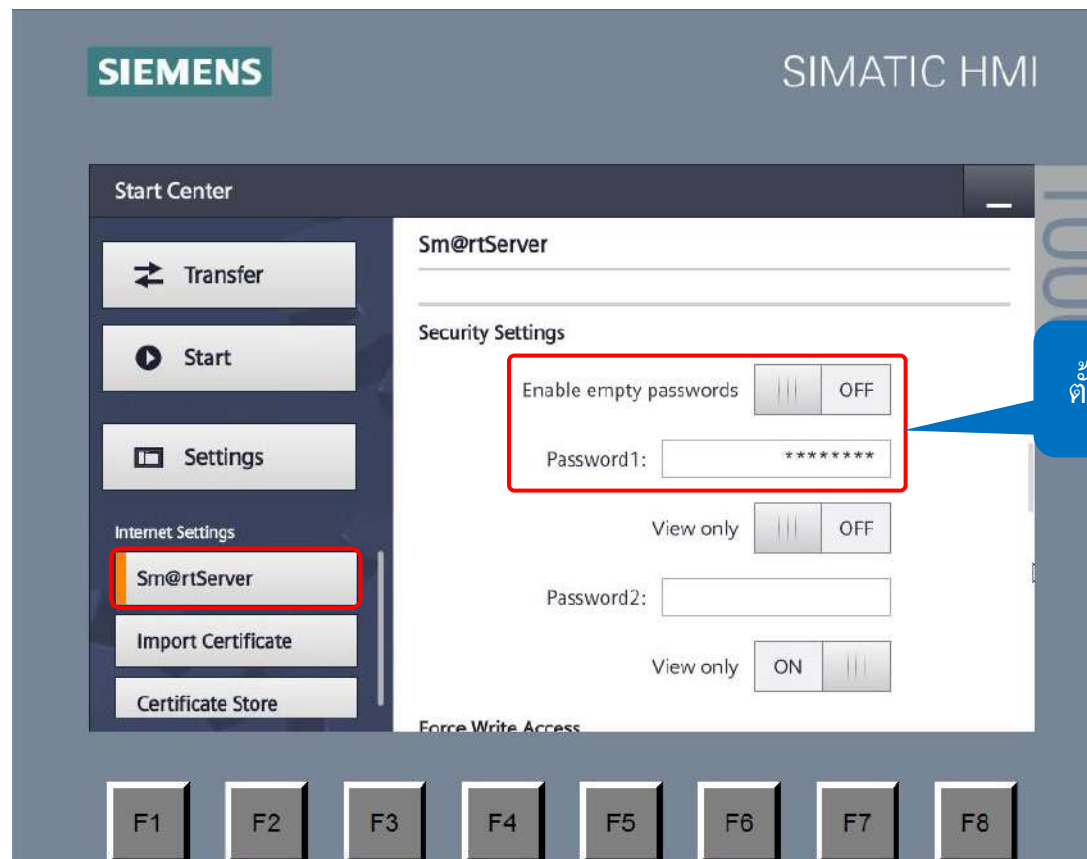
- ตั้งค่าเพื่อ Enable Sm@rtServer จากนั้นทำการ download ของจอ



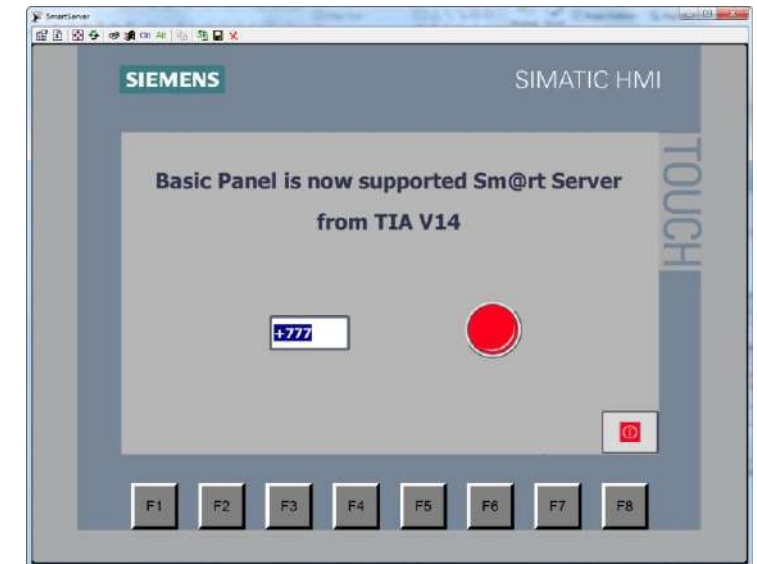
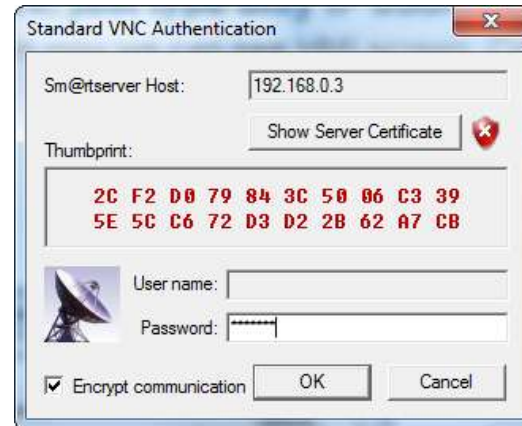
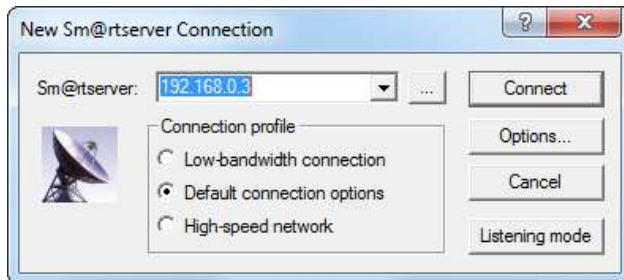
- ก่อนใช้งาน เราต้องเข้าไปตั้ง Password ของ Sm@rtServer ด้วย โดยเข้าไปตั้งใน Settings -> Internet Settings



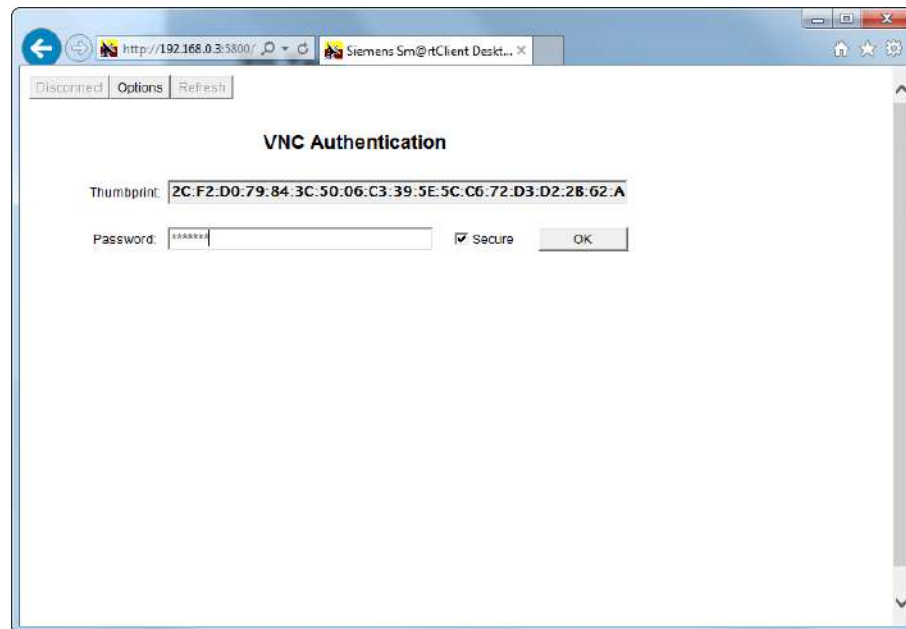
- ในส่วนของ Sm@rtServer ให้ตั้งค่า Password1 เพื่อให้สามารถควบคุมได้ด้วย (หากต้องการให้ดูได้อย่างเดียวให้ใช้ Password2)



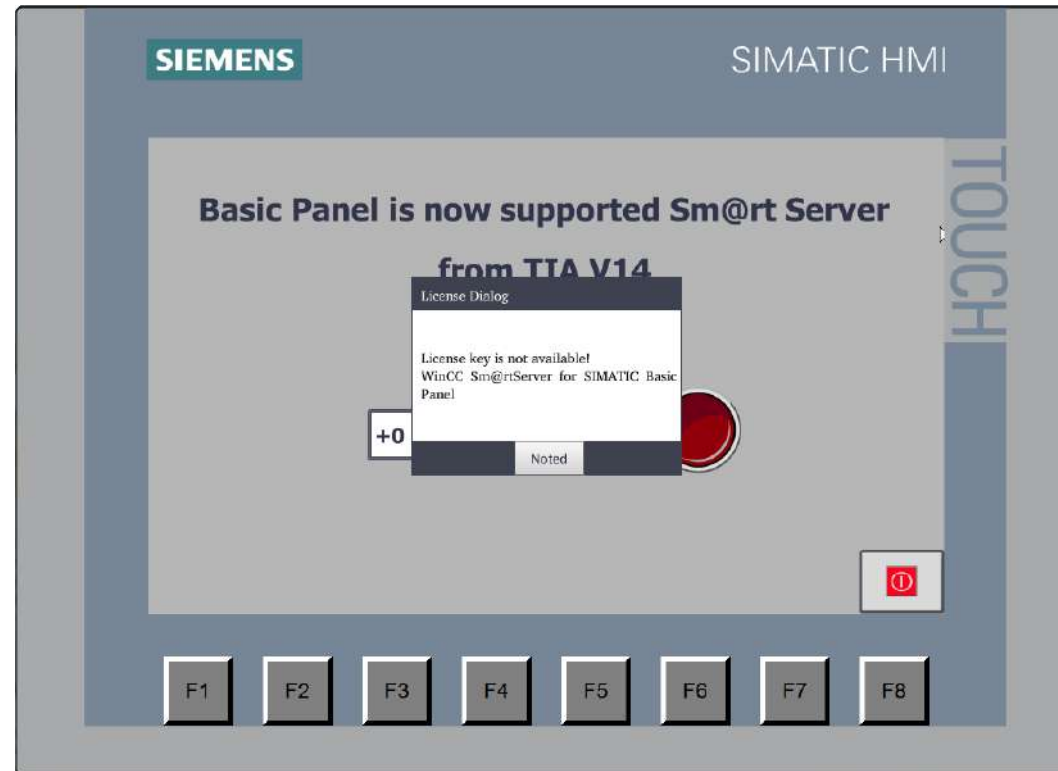
- หากใช้โปรแกรม Sm@rtClient ให้เราพิมพ์ IP ของจอ Basic Panel และให้ใส่ Password ที่เราได้ตั้งเอาไว้ใน Password1 ในขั้นตอนก่อนหน้านี้



- หากใช้ Internet Explorer ให้เราพิมพ์รูปแบบ “**http://IP of HMI:5800**” โดยต้องติดตั้ง Java ให้กับ Internet Explorer ด้วย



- หากที่จอ Basic Panel ไม่ได้ลง license “SIMATIC WinCC Sm@rtServer for SIMATIC Basic Panels” เอาไว้ จะมีข้อความแจ้งเตือนว่า License key is not available ! ขึ้นมาเป็นระยะๆ

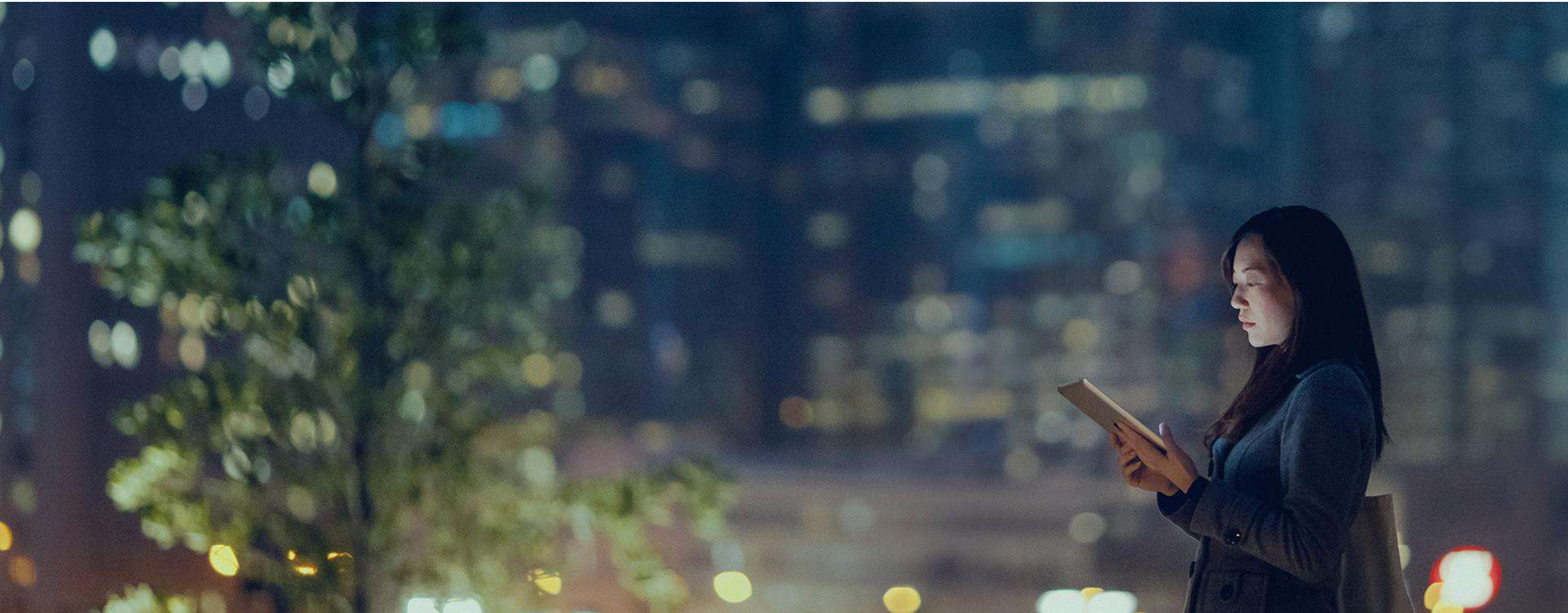


05 - Others

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Automation

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1. Set IP address in Settings mode

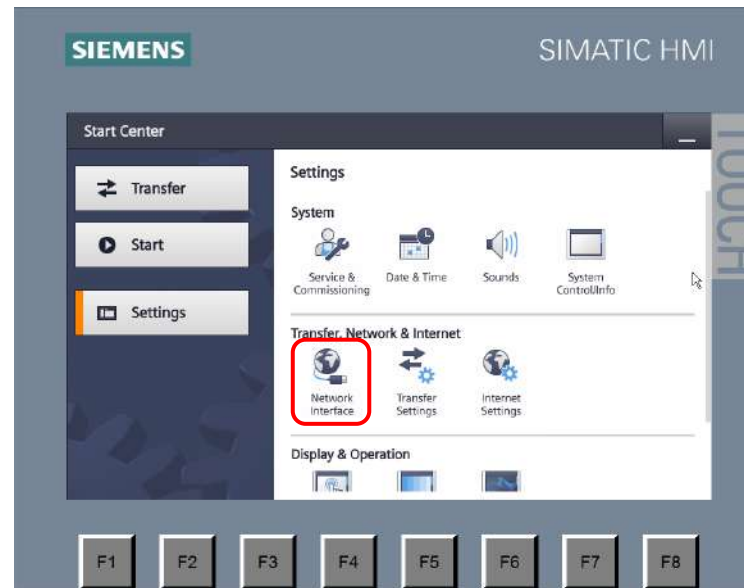
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Partner

Factory
Automation

SIEMENS



Settings



Network Interface



2. Set IP address in TIA Portal

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Factory
Automation

SIEMENS

The screenshot displays the Siemens TIA Portal interface for configuring an HMI_IP3 [KTP700 Basic PN] device. The Project tree on the left shows the hierarchy: Transfer byUSB > HMI_IP3 [KTP700 Basic PN] > Device configuration. The central workspace shows a 3D model of the device. The Properties window on the right is open to the 'Ethernet addresses' tab. The 'Interface networked with' dropdown is set to 'PN/IE_1'. The 'IP protocol' section has 'Set IP address in the project' selected, with the IP address field set to 192.168.0.3 and the Subnet mask set to 255.255.255.0. The 'Use router' checkbox is unchecked, and the 'Router address' field is set to 0.0.0.0. The 'IP address is set directly at the device' option is also visible but not selected.

Module	Index	Type
HMI_RT_2	1	KTP700 ...
	2	
	3	
	4	
HMI_IP3.IE_CP_1	5	PROFIBUS...
PROFINET Interface_1	5 X1	PROFIBUS...
	6	
	7	
	8	
	9	
	10	
	11	
	12	
	13	
	14	
	15	

3. Sync clock from PLC

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SIEMENS

HMI สามารถที่จะ sync เวลาจาก PLC ได้ ทำให้การทำงานใน function ต่างๆทั้งจอและ PLC อยู่บนฐานเวลาเดียวกัน โดยการ sync เวลาจะทำตอนที่ HMI เปิดไฟขึ้นมาและทำอีกอัตโนมัติทุกๆ 10 นาที

The screenshot displays the Siemens TIA Portal V14 interface. The title bar indicates the project path: C:\Users\Admin\Documents\Automation\Transfer byUSB\Transfer byUSB. The main window is divided into several panes:

- Project tree (left):** Shows the project structure. The 'Connections' folder under 'HMI_IP3 [KTP700 Basic PN]' is highlighted with a red box.
- Connections table (center):** Titled 'Connections to S7 PLCs in Devices & Networks'. It contains a table with the following data:

Name	Communication driver	HMI time synchronization mode	Station	Partner	Node	Online	Co...
S7-1200_1	SIMATIC S7 1200	Slave	S7-1200 station_1	PLC_1	CPU 1212C AC/DC/...	☒	
<Add new>		None					
		Slave					

A red dotted arrow points from the 'Slave' option in the 'HMI time synchronization mode' dropdown to the 'Connections' folder in the project tree.
- Parameter pane (bottom):** Contains configuration details for the 'KTP700 Basic PN' HMI device and the connected 'PLC'.
 - KTP700 Basic PN:** Interface is set to 'PROFINET (X1)'.
 - HMI device:** Address is '192.168.0.3' and Access point is 'S7ONLINE'.
 - PLC:** Address is '192.168.0.90' and Access password is empty.

3. Sync clock from PLC

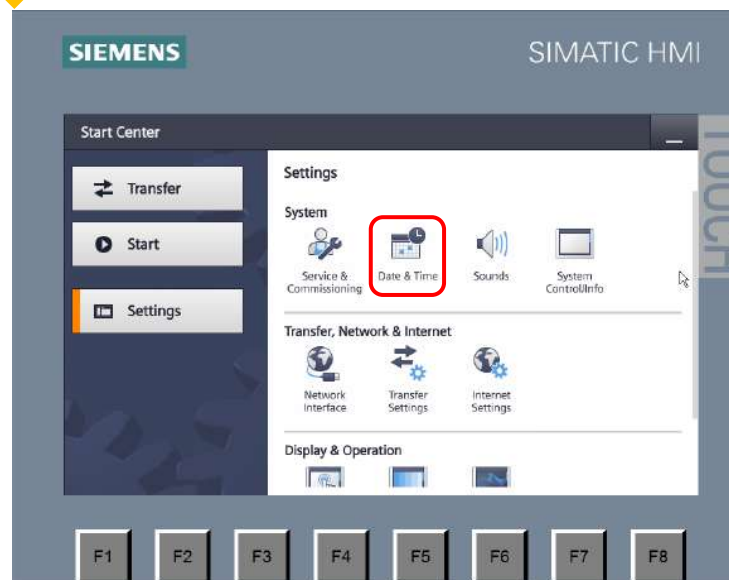
Approved
Partner

Factory
Automation

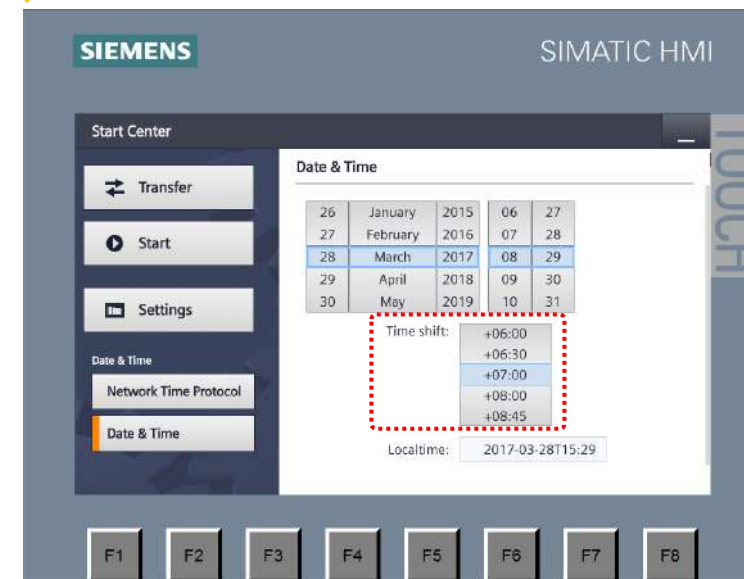
SIEMENS



Settings



Date & Time



อย่าลืมกำหนด Time Zone ของจอให้ตรงกับ PLC ด้วย ไม่เช่นนั้น เวลาที่อ่านอาจจะไม่ตรงกันด้วย Time Zone ที่แตกต่างกัน

3. Sync clock from PLC

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The screenshot displays the Siemens STEP 7 HW Config interface. The top window shows a rack configuration for 'Rack_0' with slots 103, 102, 101, 1, 2, and 3. Slot 1 contains a 'SIEMENS S7-300 CPU 1212C DC/DC/DC' module. Slot 2 contains a 'DI 8xDC24V' module. The bottom window shows the 'Properties' tab for the selected CPU module, specifically the 'Time of day' section. This section includes settings for 'Local time' (Time zone: UTC +07:00 Bangkok, Hanoi, Jakarta), 'Daylight saving time' (Activate daylight saving time: unchecked, Difference between standard and daylight saving time: 60 mins), and 'Start of daylight saving time' (First, Sunday, of: January, at: Midnight).

Training Mar_28 ▸ PLC_MASS_99 [CPU 1212C DC/DC/DC]

Topology view Network view Device view

PLC_MASS_99 [CPU 1212C]

Rack_0

103 102 101 1 2 3

SIEMENS S7-300 CPU 1212C DC/DC/DC

DI 8xDC24V

100%

PLC_MASS_99 [CPU 1212C DC/DC/DC]

Properties Info Diagnostics

General IO tags System constants Texts

- General
- PROFINET interface [X1]
- DI 8/DQ 6
- AI 2
- CB 1241 (RS485)
- High speed counters (HSC)
- Pulse generators (PTO/PWM)
- Startup
- Cycle
- Communication load
- System and clock memory
- Web server
- User interface languages
- Time of day**
- Protection & Security
- Configuration control
- Connection resources
- Overview of addresses

Time of day

Local time

Time zone: (UTC +07:00) Bangkok, Hanoi, Jakarta

Daylight saving time

☐ Activate daylight saving time

Difference between standard and daylight saving time: 60 mins

Start of daylight saving time

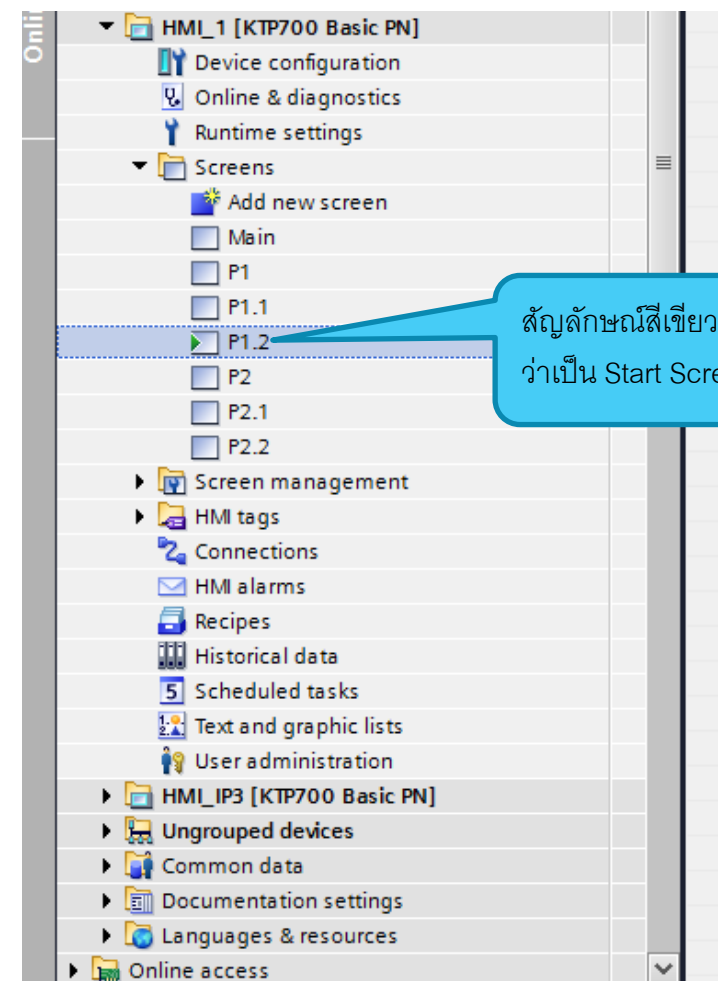
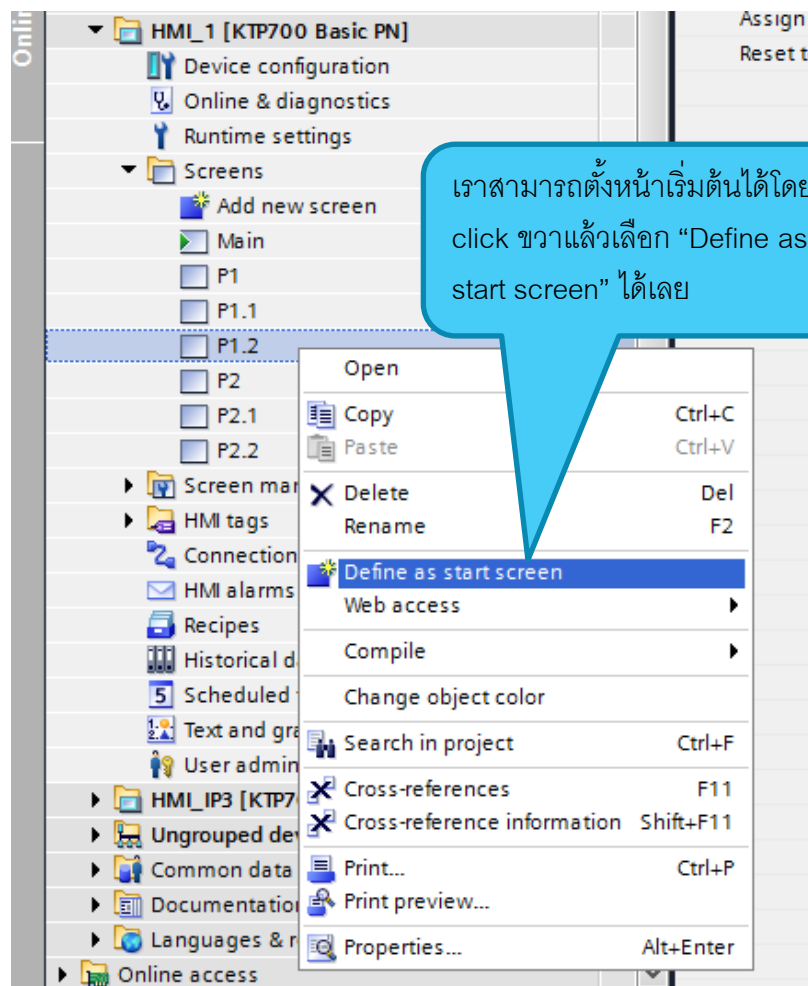
First Sunday of: January at: Midnight

4. Start Screen

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SIEMENS



5. RUN/STOP PLC

หน้าจอ HMI สามารถสั่งงานเปลี่ยน mode ของ PLC ให้ RUN หรือ STOP จากได้ด้วย Event ที่ชื่อว่า SetPLCMode

The image illustrates the configuration of HMI buttons to control the PLC mode (RUN or STOP) using the SetPLCMode event.

Main Screen: The SIMATIC HMI interface shows a main screen with a RUN button and a STOP button. The RUN button is highlighted with a red dashed box, and the STOP button is highlighted with a blue dashed box.

Button_1 [Button] Properties: The Properties window for Button_1 shows the Events tab. The SetPLCMode event is selected, and the Connection is set to HMI_Connection_1. The Mode is set to RUN.

Button_2 [Button] Properties: The Properties window for Button_2 shows the Events tab. The SetPLCMode event is selected, and the Connection is set to HMI_Connection_1. The Mode is set to STOP.

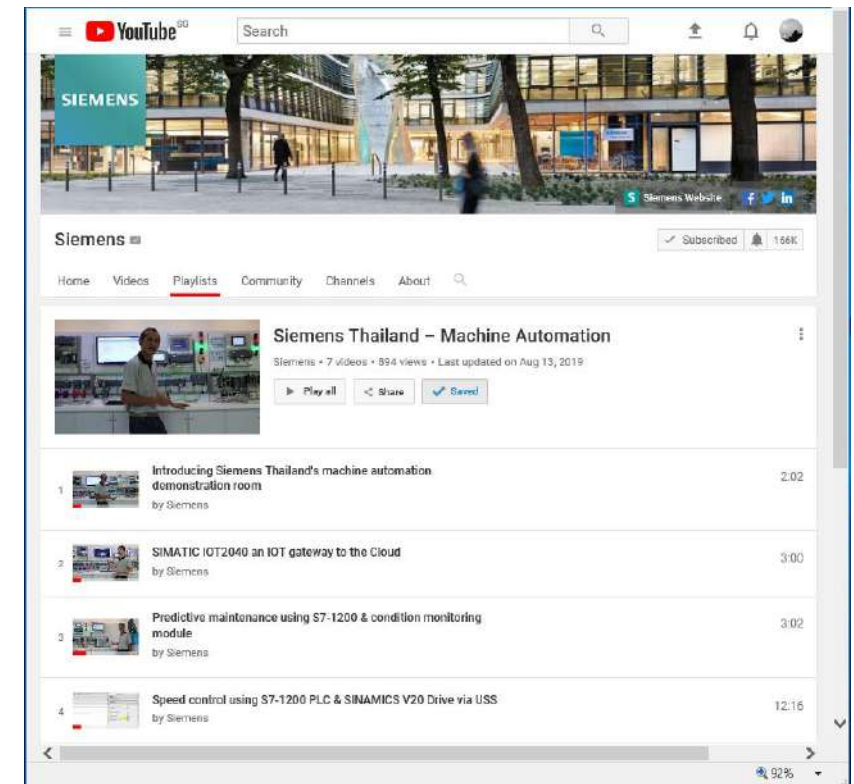
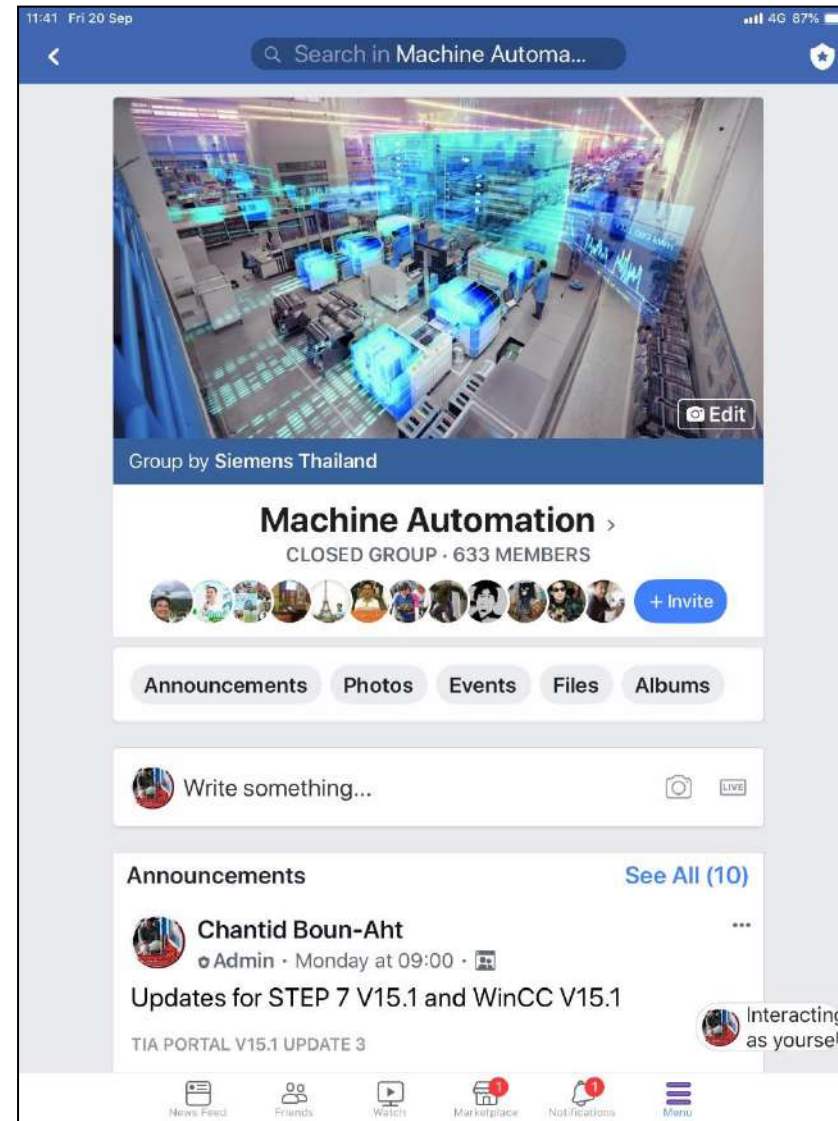
Connections: The Connections window shows the HMI_Connection_1 connection, which is linked to the HMI_Connection_1 event.

Callout: A blue callout box indicates: "เลือก HMI Connection ที่เชื่อมต่อกับ PLC นั้นๆ" (Select the HMI Connection that is connected to that PLC).

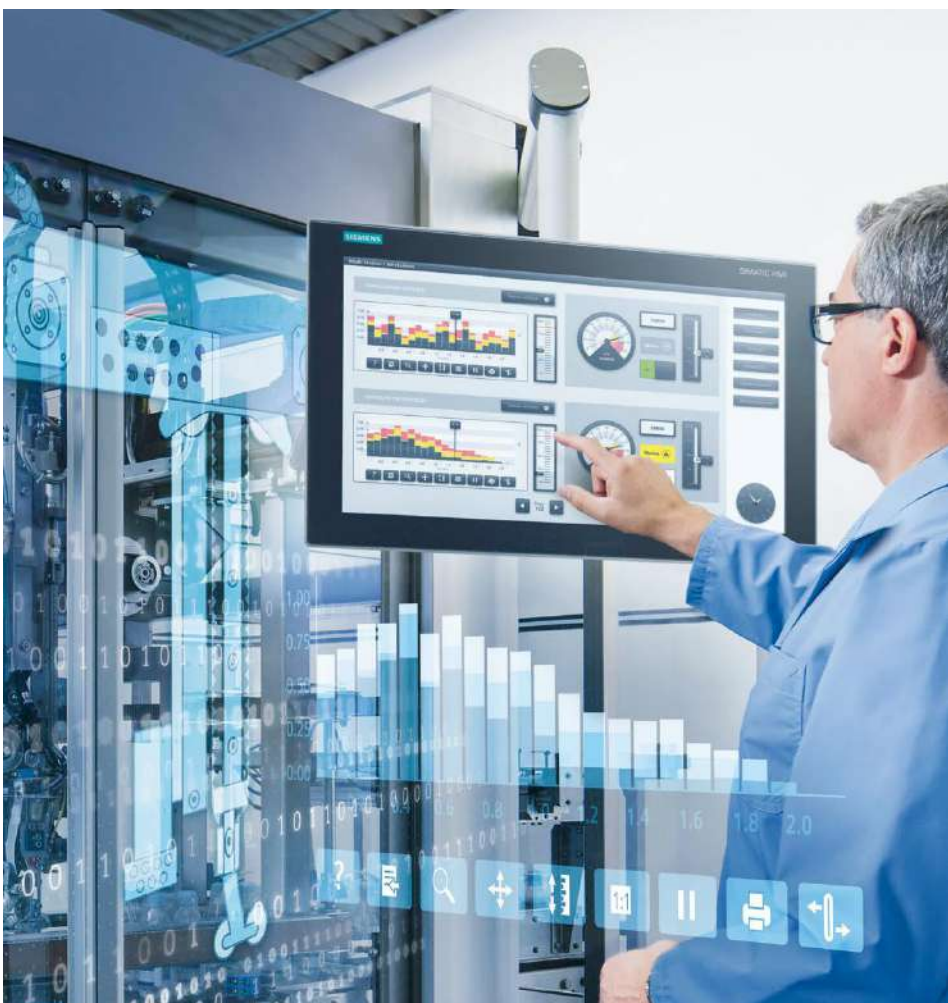
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